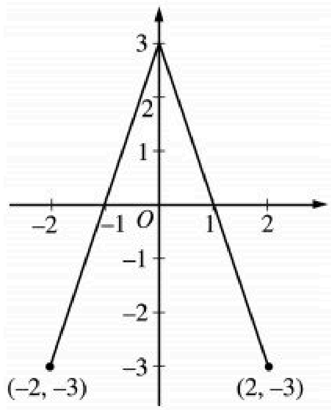
5.3a

1.



Graph of *f*

The graph of the function *f* shown above consists of two line segments. Let *g* be the function given by $g(x) = \int_0^x f(t)dt$.

(a) Find *g*(-1), *g*′(-1), and *g*″(-1).

(b) For what values of *x* in the open interval (-2,2) is *g* increasing? Explain your reasoning.

(c) For what values of *x* in the open interval (-2,2) is the graph of *g* concave down? Explain your reasoning.

(d) On the axes provided, sketch the graph of *g* on the closed interval [-2,2].

Part A

1 point is earned for *g*(-1)

1 point is earned for *g*′(-1)

1 point is earned for *g*″(-1)

$$g(-1) = \int_0^{-1} f(t)dt = -\int_{-1}^{0} f(t)dt = -\frac{3}{2}$$
$$g'(-1) = f(-1) = 0$$
$$g''(-1) = f'(-1) = 3$$

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for *g*(-1)

1 point is earned for *g*′(-1)

5.3a

1 point is earned for $g''(-1)$

$$g(-1) = \int_0^{-1} f(t)dt = -\int_{-1}^{0} f(t)dt = -\frac{3}{2}$$
$$g'(-1) = f(-1) = 0$$
$$g''(-1) = f'(-1) = 3$$

Part B

1 point is earned for the interval

1 point is earned for the answer

g is increasing on $-1 < x < 1$ because $g'(x) = f(x) > 0$ on this interval.

0	1	2

The student response earns all of the following points:

1 point is earned for the interval

1 point is earned for the answer

g is increasing on $-1 < x < 1$ because $g'(x) = f(x) > 0$ on this interval.

Part C

1 point is earned for the interval

1 point is earned for the reason

The graph of g is concave down on $0 < x < 2$ because $g''(x) = f'(x) < 0$ on this interval.

or

because $g'(x) = f(x)$ is decreasing on this interval.

0	1	2

The student response earns all of the following points:

1 point is earned for the interval

5.3a

1 point is earned for the reason

The graph of g is concave down on $0 < x < 2$ because $g''(x) = f'(x) < 0$ on this interval.

or

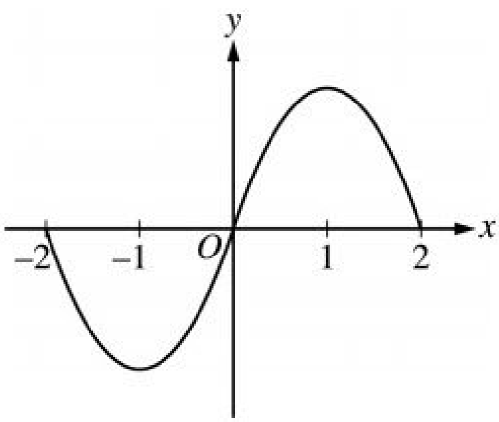
because $g'(x) = f(x)$ is decreasing on this interval.

Part D

1 point is earned for $g(-2) = g(0) = g(2) = 0$

1 point is earned for appropriate increasing/decreasing and concavity behavior

<-1> vertical asymptote



0	1	2

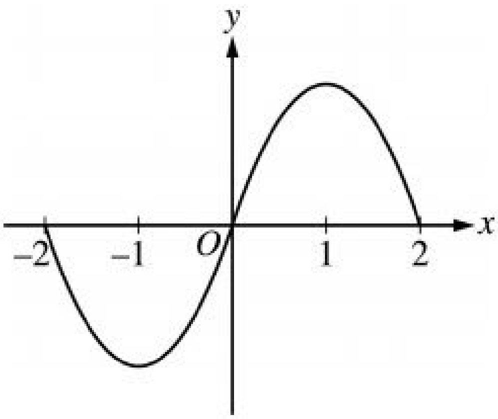
The student response earns all of the following points:

1 point is earned for $g(-2) = g(0) = g(2) = 0$

1 point is earned for appropriate increasing/decreasing and concavity behavior

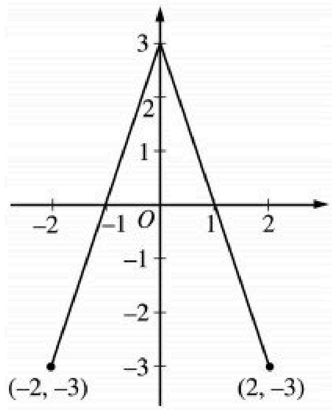
<-1> vertical asymptote

5.3a



5.3a

2.



Graph of f

The graph of the function f shown above consists of two line segments. Let g be the function given by $g(x) = \int_0^x f(t)dt$.

For what values of x in the open interval (-2,2) is g increasing? Explain your reasoning.

Part B

1 point is earned for the interval

1 point is earned for the answer

g is increasing on $-1 < x < 1$ because $g'(x) = f(x) > 0$ on this interval.

0	1	2

The student response earns all of the following points:

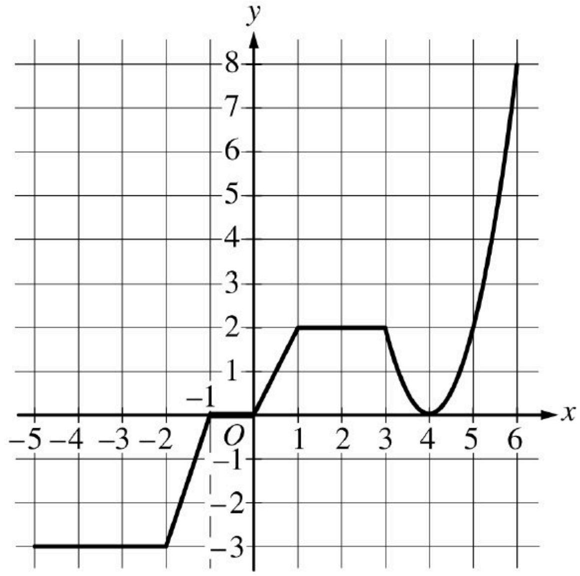
1 point is earned for the interval

1 point is earned for the reason

g is increasing on $-1 < x < 1$ because $g'(x) = f(x) > 0$ on this interval.

5.3a

3.



Graph of g

The graph of the continuous function g, the derivative of the function f, is shown above. The function g is piecewise linear for $-5 \le x < 3$, and $g(x) = 2(x - 4)^2$ for $3 \le x \le 6$.

(a) If $f(1) = 3$, what is the value of $f(-5)$?

(b) Evaluate $\int_1^6 g(x)dx$.

(c) For $-5 < x < 6$, on what open intervals, if any, is the graph of f both increasing and concave up? Give a reason for your answer.

(d) Find the x-coordinate of each point of inflection of the graph of f. Give a reason for your answer.

Part A

1 point is earned for integral

1 point is earned for answer

$$f(-5) = f(1) + \int_1^{-5} g(x)dx = f(1) - \int_{-5}^{1} g(x)dx$$
$$= 3 - \left(-9 - \tfrac{3}{2} + 1\right) = 3 - \left(-\tfrac{19}{2}\right) = \frac{25}{2}$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for integral

5.3a

1 point is earned for answer

$$f(-5) = f(1) + \int_1^{-5} g(x)dx = f(1) - \int_{-5}^{1} g(x)dx$$
$$= 3 - \left(-9 - \tfrac{3}{2} + 1\right) = 3 - \left(-\tfrac{19}{2}\right) = \frac{25}{2}$$

Part B

1 point is earned for split at $x = 3$

1 point is earned for antiderivative $2(x-4)^2$

1 point is earned for answer

$$\int_1^6 g(x)dx = \int_1^3 g(x)dx + \int_3^6 g(x)dx$$
$$= \int_1^3 2dx + \int_3^6 2(x-4)^2 dx$$
$$= 4 + \left[\tfrac{2}{3}(x-4)^3\right]_{x=3}^{x=6} = 4 + \frac{16}{3} - \left(-\tfrac{2}{3}\right) = 10$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for split at $x = 3$

1 point is earned for antiderivative $2(x-4)^2$

1 point is earned for answer

$$\int_1^6 g(x)dx = \int_1^3 g(x)dx + \int_3^6 g(x)dx$$
$$= \int_1^3 2dx + \int_3^6 2(x-4)^2 dx$$
$$= 4 + \left[\tfrac{2}{3}(x-4)^3\right]_{x=3}^{x=6} = 4 + \frac{16}{3} - \left(-\tfrac{2}{3}\right) = 10$$

5.3a

Part C

1 point is earned for intervals

1 point is earned for reason

The graph of f is increasing and concave up on $0 < x < 1$ and $4 < x < 6$ because $f'(x) = g(x) > 0$ and $f'(x) = g(x)$ is increasing on those intervals.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for intervals

1 point is earned for reason

The graph of f is increasing and concave up on $0 < x < 1$ and $4 < x < 6$ because $f'(x) = g(x) > 0$ and $f'(x) = g(x)$ is increasing on those intervals.

Part D

1 point is earned for answer

1 point is earned for reason

The graph of f has a point of inflection at $x = 4$ because $f'(x) = g(x)$ changes from decreasing to increasing at $x = 4$.

0	1	2 ✓

The student response earns all of the following points:

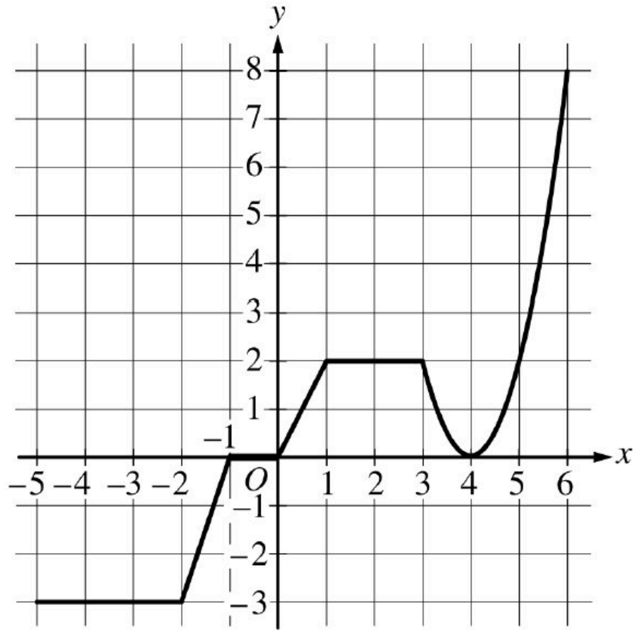
1 point is earned for answer

1 point is earned for reason

The graph of f has a point of inflection at $x = 4$ because $f'(x) = g(x)$ changes from decreasing to increasing at $x = 4$.

5.3a

4.



Graph of g

The graph of the continuous function g, the derivative of the function f, is shown above. The function g is piecewise linear for $-5 \le x < 3$, and $g(x) = 2(x - 4)^2$ for $3 \le x \le 6$.

For $-5 < x < 6$, on what open intervals, if any, is the graph of f both increasing and concave up? Give a reason for your answer.

Part 3C

1 point is earned for: intervals

1 point is earned for: reason

The graph of f is increasing and concave up on $0 < x < 1$ and $4 < x < 6$ because $f'(x) = g(x) > 0$ and $f'(x) = g(x)$ is increasing on those intervals.

0	1	2

The student response earns two of the following points:

1 point is earned for: intervals

1 point is earned for: reason

The graph of f is increasing and concave up on $0 < x < 1$ and $4 < x < 6$ because $f'(x) = g(x) > 0$ and $f'(x) = g(x)$ is increasing on those intervals.

5.3a

5. Let f be the function defined by $f(x) = \sin^2 x - \sin x$ for $0 \leq x \leq 3\pi/2$.

(a) Find the x-intercepts of the graph of f.

(b) Find the intervals on which f is increasing.

(c) Find the absolute maximum value and the absolute minimum value of f. Justify your answer.

Part A

1 point is earned for $\sin^2 x - \sin x = 0$

1 point is earned for three appropriate roots

$\sin^2 x - \sin x = 0$

$\therefore \sin x = 0$ or $\sin x = 1$

$\boxed{x = 0, \frac{\pi}{2}, \pi}$

0	1	2

The student response earns all of the following points:

1 point is earned for $\sin^2 x - \sin x = 0$

1 point is earned for three appropriate roots

$\sin^2 x - \sin x = 0$

$\therefore \sin x = 0$ or $\sin x = 1$

$\boxed{x = 0, \frac{\pi}{2}, \pi}$

Part B

1 point is earned for $f'(x)$

5.3a

1 point is earned for finding 2 interior critical points

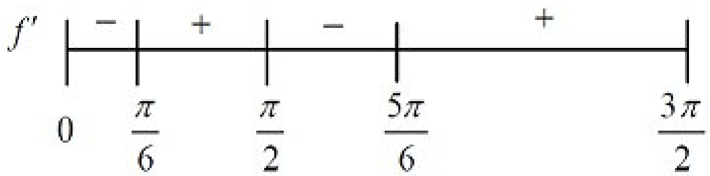
1 point is earned for analysis

1 point is earned for answer

$f'(x) = 2 \sin x \cos x - \cos x$

$= \cos x(2 \sin x - 1)$

$f'(x) = 0$ where $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$



$\boxed{\frac{\pi}{6} \le x \le \frac{\pi}{2} \text{ and } \frac{5\pi}{6} \le x \le \frac{3\pi}{2}}$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for finding 2 interior critical points

1 point is earned for analysis

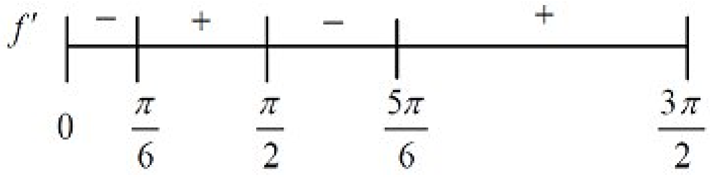
1 point is earned for answer

$f'(x) = 2 \sin x \cos x - \cos x$

$= \cos x(2 \sin x - 1)$

$f'(x) = 0$ where $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

5.3a



$\boxed{\frac{\pi}{6} \leq x \leq \frac{\pi}{2} \text{ and } \frac{5\pi}{6} \leq x \leq \frac{3\pi}{2}}$

Part C

1 point is earned for max value

1 point is earned for min value

1 point is earned for justification

Maximum value: 2
Minimum value : -1/4

x	$f(x)$
0	0
$\frac{\pi}{6}$	$-\frac{1}{4}$
$\frac{\pi}{2}$	0
$\frac{5\pi}{6}$	$-\frac{1}{4}$
$\frac{3\pi}{2}$	2

0	1	2	3

The student response earns all of the following points:

1 point is earned for max value

5.3a

1 point is earned for min value

1 point is earned for justification

Maximum value: 2
Minimum value : -1/4

x	$f(x)$
0	0
$\frac{\pi}{6}$	$-\frac{1}{4}$
$\frac{\pi}{2}$	0
$\frac{5\pi}{6}$	$-\frac{1}{4}$
$\frac{3\pi}{2}$	2

5.3a

6. Let *f* be the function defined by $f(x)=\sin^2 x - \sin x$ for $0 \le x \le 3\pi/2$

Find the intervals on which *f* is increasing.

Part B

1 point is earned for f'(x)

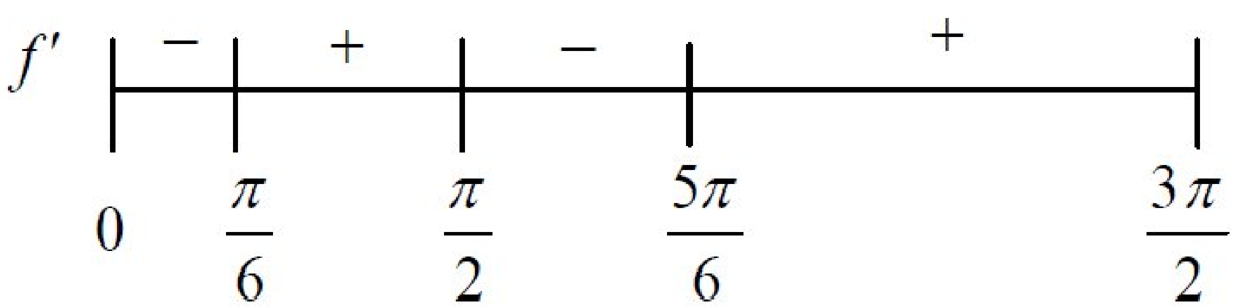
1 point is earned for finding 2 interior critical points

1 point is earned for analysis

1 point is earned for answer

$$f'(x) = 2\sin x\cos x - \cos x$$
$$= \cos x\,(2\sin x - 1)$$

$$f'(x) = 0 \text{ when } x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$$



$\frac{\pi}{6} \le x \le \frac{\pi}{2}$ *and* $\frac{5\pi}{6} \le x \le \frac{3\pi}{2}$

0	1	2	3	4

The student response earns four of the following points:

1 point is earned for f'(x)

1 point is earned for finding 2 interior critical points

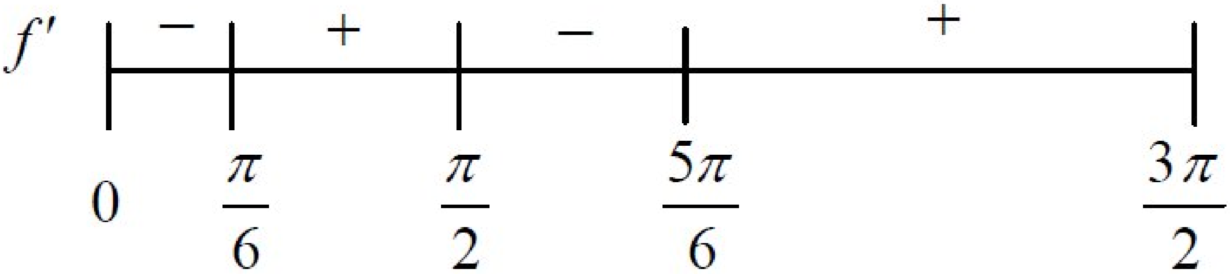
1 point is earned for analysis

1 point is earned for answer

5.3a

$f'(x) = 2\sin x \cos x - \cos x$
$= \cos x\,(2\sin x - 1)$

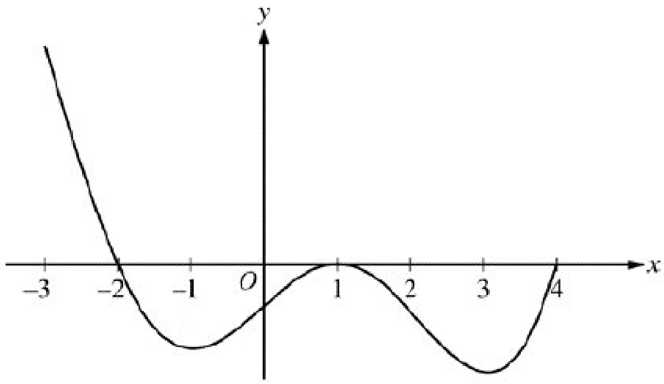
$f'(x) = 0$ when $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$



$\frac{\pi}{6} \leq x \leq \frac{\pi}{2}$ *and* $\frac{5\pi}{6} \leq x \leq \frac{3\pi}{2}$

5.3a

7.



Graph of f'

The figure above shows the graph of f', the derivative of a twice-differentiable function f, on the interval $[-3, 4]$. The graph of f' has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x-axis and the graph of f' on the intervals $[-2, 1]$ and $[1, 4]$ are 9 and 12, respectively.

On what open intervals contained in $-3 < x < 4$ is the graph of f both concave down and decreasing? Give a reason for your answer.

Part B

1 point is earned for the intervals

1 point is earned for the explanation

The graph of f is concave down and decreasing on the intervals $-2 < x < -1$ and $1 < x < 3$ because f' is decreasing and negative on these intervals.

0	1	2

The student response earns all of the following points:

1 point is earned for the intervals

1 point is earned for the explanation

The graph of f is concave down and decreasing on the intervals $-2 < x < -1$ and $1 < x < 3$ because f' is decreasing and negative on these intervals.

5.3a

8. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

The functions f and g are twice differentiable. The table shown gives values of the functions and their first derivatives at selected values of x.

(a) Let h be the function defined by $h(x) = f(g(x))$. Find $h\prime(7)$. Show the work that leads to your answer.

(b) Let k be a differentiable function such that $k'(x) = (f(x))^2 \cdot g(x)$. Is the graph of k concave up or concave down at the point where $x = 4$? Give a reason for your answer.

(c) Let m be the function defined by $m(x) = 5x^3 + \int_0^x f'(t)\,dt$. Find $m(2)$. Show the work that leads to your answer.

(d) Is the function m defined in part (c) increasing, decreasing, or neither at $x = 2$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $h'(x) = f'(g(x)) \cdot g'(x)$ or $h'(7) = f'(g(7)) \cdot g'(7)$.

If the first point is earned, the second point is earned only for an answer of 12 (or equivalent).

If the first point is not earned, the second point can be earned only for a response of either $f'(0) \cdot 8 = 12$ or $\frac{3}{2} \cdot 8$.

A response of 12 with no supporting work does not earn either point.

5.3a

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Chain rule
- [ ] Answer

Solution:

$h'(x) = f'(g(x)) \cdot g'(x)$

$h'(7) = f'(g(7)) \cdot g'(7) = f'(0) \cdot 8 = \frac{3}{2} \cdot 8 = 12$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $k''(x) = 2f(x) \cdot f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $k''(4) = 2f(4) \cdot f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$.

The first point is also earned by any of the following incorrect expressions, each of which has a single error in the application of the product rule or the chain rule:

- $2f(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $2f(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $2f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $2f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$ or $f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
- $2f(x) \cdot f'(x) \cdot g'(x)$ or $2f(4) \cdot f'(4) \cdot g'(4)$
- Note: A response that presents one of these expressions cannot earn the second point.

To earn the second point a response must correctly find $k''(4) = -40$ (or equivalent) with supporting work.

The third point is earned for an answer and reason that are consistent with any declared nonzero value of $k''(4)$.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Product or chain rule

5.3a

- [ ] $k''(4)$
- [ ] Answer with reason

Solution:

$k''(x) = 2f(x) \cdot f'(x) \cdot g(x) + (f(x))^2 \cdot g'(x)$

$k''(4) = 2f(4) \cdot f'(4) \cdot g(4) + (f(4))^2 \cdot g'(4)$
$= 2 \cdot 4 \cdot 3 \cdot (-3) + 4^2 \cdot 2 = -72 + 32 = -40$

The graph of k is concave down at the point where $x = 4$ because $k''(4) < 0$ and k'' is continuous.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned only for an answer of 37 (or equivalent) with supporting work equivalent to $5 \cdot 8 + (f(2) - f(0))$, $40 + (f(2) - f(0))$, $5 \cdot 8 + (7 - 10)$, $40 + (7 - 10)$.

An answer of 37 with no supporting work does not earn the point.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer with supporting work

Solution:

$m(2) = 5 \cdot 8 + \int_0^2 f'(t)dt = 40 + (f(2) - f(0))$
$= 40 + (7 - 10) = 37$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for considering $m'(x)$, $m'(2)$, or m'. This consideration may appear in a justification statement.

The second point is earned for $m'(2) = 15 \cdot 2^2 + f'(2)$, $m'(2) = 60 + f'(2)$, or $m'(2) = 60 - 8$ but is not earned for an unsupported response of $m'(2) = 52$.

The third point is earned for an answer and justification consistent with any declared value of $m'(2)$.

5.3a

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $m'(x)$
- ☐ $m'(2)$ with supporting work
- ☐ Answer with justification

Solution:

$m'(x) = 15x^2 + f'(x)$

$m'(2) = 15 \cdot 4 + f'(2) = 60 + (-8) = 52$

The graph of m is increasing at $x = 2$ because $m'(2) > 0$.

9. Traffic flow is defined as the rate at which cars pass through an intersection, measured in cars per minute. The traffic flow at a particular intersection is modeled by the function F defined by $F(t) = 82 + 4\sin\left(\frac{t}{2}\right)$ for $0 \le t \le 30$, where *F(t)* is measured in cars per minute and *t* is measured in minutes.

Is the traffic flow increasing or decreasing at $t = 7$? Give a reason for your answer.

Part B

1 point is earned for the answer with reason

$F'(7) = -1.872$ or -1.873

Since $F'(7) < 0$, the traffic is decreasing at $t = 7$.

0	1

The student response earns all of the following points:

1 point is earned for the answer with reason

$F'(7) = -1.872$ or -1.873

Since $F'(7) < 0$, the traffic is decreasing at $t = 7$.

5.3a

10. A particle moves along the y-axis so that its velocity at any time $t \geq 0$ is given by $v(t)=t\cos t$. At time $t=0$, the position of the particle is $y=3$.

For what values of t, $0 \leq t \leq 5$, is the particle moving upward?

Part A

1 point is earned for correctly setting $v(t) > 0$

$t\cos t > 0, (\text{ or } v(t) > 0)$

1 point is earned for the first correct interval

1 point is earned for the second correct interval

Particle is moving up for $0 < t < \frac{\pi}{2}$ and for $\frac{3\pi}{2} < t \leq 5$.

0/3 if uses any function other than v(t)

0	1	2	3

The student response earns two of the following points:

1 point is earned for correctly setting $v(t) > 0$

$t\cos t > 0, (\text{ or } v(t) > 0)$

1 point is earned for the first correct interval

1 point is earned for the second correct interval

Particle is moving up for $0 < t < \frac{\pi}{2}$ and for $\frac{3\pi}{2} < t \leq 5$

0/3 if uses any function other than v(t)

5.3a

11. *Let f be the function given by* $f(x) = x^3 - 5x^2 + 3x + k$, where k is a constant.

(a) *On what intervals is f increasing?*

(b) On what intervals is the graph of f concave downward?

(c) Find the value of k for which f has 11 as its relative minimum.

Part A

1 point is earned for the correct $f'(x)$

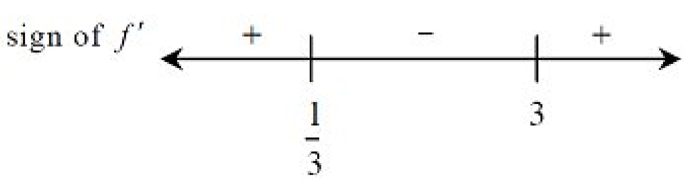
$$\begin{aligned} f'(x) &= 3x^2 - 10x + 3 \\ &= (3x-1)(x-3) \end{aligned}$$

1 point is earned for the correct zeros of $f'(x)$

$f'(x) = 0$ at $x = 1/3$ and $x = 3$

1 point is earned for correctly analyzing the sign of f'

Note: 0/1 if f' has fewer than two zeros or if the zeros are incorrect for f'



1 point is earned for correctly indicating answer is where $f'(x) > 0$

f is increasing on $(-\infty, 1/3]$ and on $[3, \infty)$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for the correct $f'(x)$

$$\begin{aligned} f'(x) &= 3x^2 - 10x + 3 \\ &= (3x-1)(x-3) \end{aligned}$$

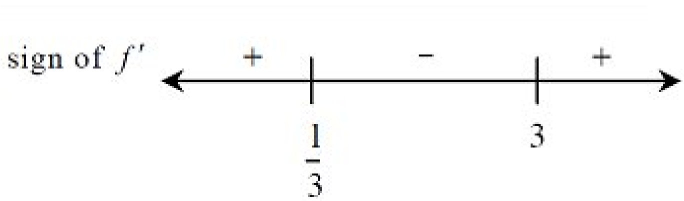
1 point is earned for the correct zeros of $f'(x)$

$f'(x) = 0$ at $x = 1/3$ and $x = 3$

5.3a

1 point is earned for correctly analyzing the sign of f'

Note: 0/1 if f' has fewer than two zeros or if the zeros are incorrect for f'



1 point is earned for correctly indicating answer is where $f'(x) > 0$

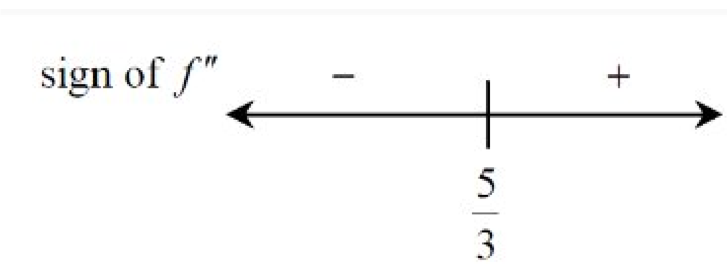
f is increasing on $(-\infty, 1/3]$ and on $[3,\infty)$

Part B

1 point is earned for the correct $f''(x)$

$f''(x) = 6x - 10$

1 point is earned for correctly analyzing sing of f'' OR sets $f''(x) < 0$



1 point is earned for correctly choosing interval where $f''(x) < 0$

graph is concave down on $\left(-\infty, \frac{5}{3}\right)$

0	1	2	3

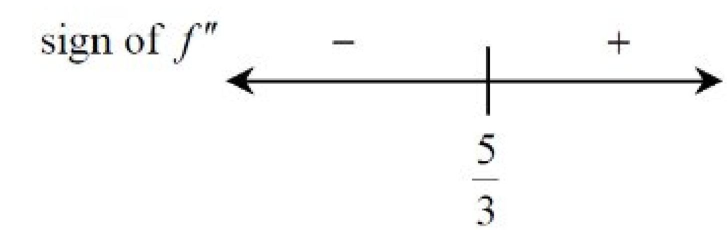
The student response earns all of the following points:

1 point is earned for the correct $f''(x)$

$f''(x) = 6x - 10$

1 point is earned for correctly analyzing sing of f'' OR sets $f''(x) < 0$

5.3a



1 point is earned for correctly choosing interval where $f''(x) < 0$

graph is concave down on $\left(-\infty, \frac{5}{3}\right)$

Part C

1 point is earned for the correct evaluation of the relative minimum

from (a), f has its relative minimum at $x = 3$

1 point is earned for correctly setting relative min. = 11 and solves for k

so,

$$f\,(3) = (3)^3 - 5(3)^2 + 3\,(3) + k$$

$= -9 + k = 11$

$k = 20$

Note: 0/2 if evaluates f at a non-critical number

1/2 if student's work shows no relative min. and student so states

1/2 if evaluates at relative max and correctly solves

0	1	2

The student response earns all of the following points:

1 point is earned for the correct evaluation of the relative minimum

from (a), f has its relative minimum at $x = 3$

1 point is earned for correctly setting relative min. = 11 and solves for k

so,

5.3a

$f(3) = (3)^3 - 5(3)^2 + 3(3) + k$

$= -9 + k = 11$

$k = 20$

Note: 0/2 if evaluates f at a non-critical number

1/2 if student's work shows no relative min. and student so states

1/2 if evaluates at relative max and correctly solves

5.3a

12. Let f be the function given by $f(x)=x^3-5x^2+3x+k$, where k is a constant.
On what intervals is f increasing?

Part A

1 point is earned for the correct f ′ (x)

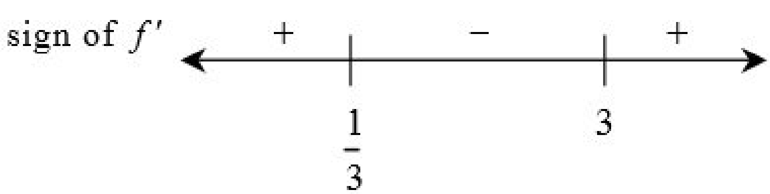
$f'(x) = 3x^2 - 10x + 3 = (3x-1)(x-3)$

1 point is earned for the correct zeros of f ′ (x)

$f'(x)=0$ *at* $x=1/3$ *and* $x=3$

1 point is earned for correctly analyzing the sign of f'

Note: 0/1 if f' has fewer than two zeros or if the zeros are incorrect for f'



1 point is earned for correctly indicating answer is where $f'(x) > 0$

f is increasing on $(-\infty, 1/3]$ and on $[3,\infty)$

0	1	2	3	4

The student response earns four of the following points:

1 point is earned for the correct f ′ (x)

$f'(x) = 3x^2 - 10x + 3 = (3x-1)(x-3)$

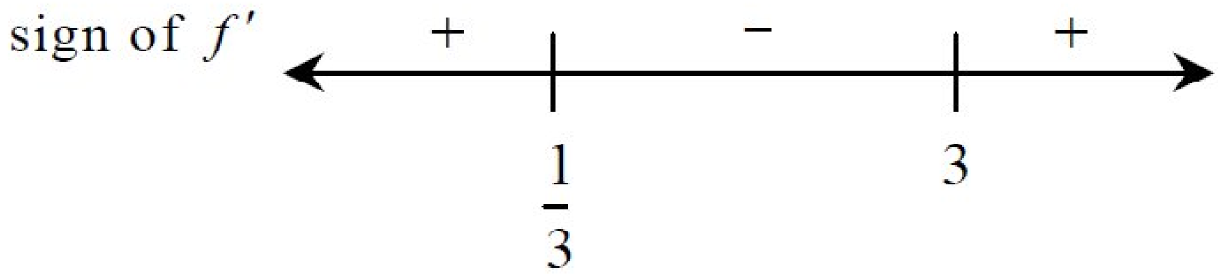
1 point is earned for the correct zeros of f ′ (x)

$f'(x)=0$ *at* $x=1/3$ *and* $x=3$

1 point is earned for correctly analyzing the sign of f'

Note: 0/1 if f' has fewer than two zeros or if the zeros are incorrect for f'

5.3a



1 point is earned for correctly indicating answer is where $f'(x) > 0$

f is increasing on $(-\infty, 1/3\,]$ and on $[3,\infty)$

5.3a

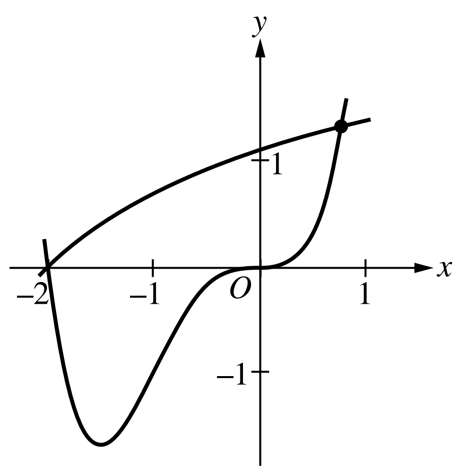
13. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f and g be the functions defined by $f(x) = \ln(x+3)$ and $g(x) = x^4 + 2x^3$. The graphs of f and g, shown in the figure, intersect at $x = -2$ and $x = B$, where $B > 0$.

(a) Find the area of the region enclosed by the graphs of f and g.

(b) For $-2 \le x \le B$, let $h(x)$ be the vertical distance between the graphs of f and g. Is h increasing or decreasing at $x = -0.5$? Give a reason for your answer.

(c) The region enclosed by the graphs of f and g is the base of a solid. Cross sections of the solid taken perpendicular to the x-axis are squares. Find the volume of the solid.

(d) A vertical line in the xy-plane travels from left to right along the base of the solid described in part (c). The vertical line is moving at a constant rate of 7 units per second. Find the rate of change of the area of the cross section above the vertical line with respect to time when the vertical line is at position $x = -0.5$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5.3a

Other forms of the integrand in a definite integral, e.g., $|f(x) - g(x)|$, $|g(x) - f(x)|$, or $g(x) - f(x)$, earn the first point.

To earn the second point, the response must have a lower limit of -2 and an upper limit expressed as either the letter B with no value attached, or a number that is correct to the number of digits presented, with at least one and up to three decimal places.

· Case 1: If the response did not earn the second point because of an incorrect value of B, $0 < B < 1$, but used a lower limit of -2, the response earns the third point only for a consistent answer.

· Case 2: If the response did not earn the second point because the lower limit used was $x = 0$, but the response used a correct upper limit of B the response earns the third point for a consistent answer of 0.708 (or 0.707).

· Case 3: If a response uses any other incorrect limits it does not earn the second or third points.

A response containing the integrand $g(x) - f(x)$ must interpret the value of the resulting integral correctly to earn the third point. For example, the following response earns all 3 points: $\int_{-2}^{B} (g(x) - f(x))\, dx = -3.604$ so the area is 3.604. However, the response "Area $= \int_{-2}^{B} (g(x) - f(x))\, dx = 3.604$" presents an untrue statement and earns the first and second points but not the third point.

A response must earn the first point in order to be eligible for the third point. If the response has earned the second point, then only the correct answer will earn the third point.

Instructions for scoring a response that presents an integrand of $\ln(x+3) - x^4 + 2x^3$:

· The first time a response implicitly presents $f(x) - g(x)$ as $j(x) = \ln(x+3) - x^4 + 2x^3$ (with no explicit connection) in any part of this question, the response loses a point. The response is then eligible for all remaining points for a correct or consistent answer.

· In part (a) the consistent answer using $j(x)$ is negative and will not earn the third point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Integrand
- ☐ Limits of integration
- ☐ Answer

Solution:

$$\ln(x+3) = x^4 + 2x^3 \;\Rightarrow\; x = -2,\;\; x = B = 0.781975$$

5.3a

$$\int_{-2}^{B} (f\ (x) - g\ (x))\ dx = 3.603548$$

The area of the region is 3.604 (or 3.603).

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response need not present the value of $h'\ (-0.5)$. The last line earns both points. However, if a value is presented it must be correct for the digits reported up to three decimal places.

A response that reports an incorrect value of $h'\ (-0.5)$ earns only the first point.

A response that presents only $h'\ (x)$ does not earn either point.

A response that compares the values of $f'\ (x)$ and $g'\ (x)$ at $x = -0.5$ earns the first point and is eligible for the second point. This comparison can be made symbolically or verbally; for example, the response "the rate of change of $f\ (x)$ is less than the rate of change of $g\ (x)$ at $x = -0.5$" earns the first point.

Instructions for scoring a response that presents $h(x)$ as $\ln(x+3) - x^4 + 2x^3$:

· If the first time the response presents $f\ (x) - g(x)$ as $j\ (x) = \ln(x+3) - x^4 + 2x^3$ occurs in part (b), the response loses a point. If the response already lost a point for this error in part (a), then the response is eligible for all remaining points for a correct or consistent answer.

· In part (b), the consistent answer using $j\ (x)$ is that $j'\ (-0.5) = 2.4 > 0$, so h is increasing at $x = -0.5$.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'\ (-0.5)$ – OR – $f'\ (x) - g'\ (x)$
- ☐ Answer with reason

Solution:

$h\ (x) = f\ (x) - g\ (x)$

$h'\ (x) = f'\ (x) - g'\ (x)$

$h'\ (-0.5) = f'\ (-0.5) - g'\ (-0.5) = -0.6$ (or -0.599)

Since $h'\ (-0.5) < 0$, h is decreasing at $x = -0.5$.

Part C

5.3a

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an integrand of $k(f(x) - g(x))^2$ or its equivalent with $k \neq 0$ in any definite integral. If $k \neq 1$, then the response is not eligible for the second point.

A response that does not earn the first point is ineligible to earn the second point, with the following exceptions:

· A response which has a presentation error in the integrand (for example, mismatched or missing parentheses, misplaced exponent) does not earn the first point but would earn the second point for the correct answer. A response which has a presentation error in the integrand and which reports an incorrect answer earns no points.

· A response that presents an integrand of $\left(\ln(x+3) - x^4 + 2x^3\right)^2$.

If the first time the response presents $f(x) - g(x)$ as $j(x) = \ln(x+3) - x^4 + 2x^3$ occurs in part (c), the response loses a point. If the response already lost a point for this error in part (a) or part (b), then the response is eligible for all remaining points for a correct or consistent answer.

In part (c), the consistent answer using $j(x)$ is 252.187 252.187 (or 252.188).

A response that uses incorrect limits is only eligible for the second point, provided the limits are imported from part (a) in Case 1 or Case 2. In both of these situations, the second point is earned only for answers consistent with the imported limits.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Integrand
- ☐ Answer

Solution:

$$\int_{-2}^{B} (f(x) - g(x))^2 \, dx = 5.340102$$

The volume of the solid is 5.340.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point may be earned by presenting $\frac{dA}{dx} \cdot \frac{dx}{dt}$, $A' \cdot \frac{dx}{dt}$, $A'(x) \cdot \frac{dx}{dt}$, $A'(x) \cdot x'$, $A'(-0.5) \cdot 7$, or $k \cdot 7$, where k is a declared value of $A'(-0.5)$, or any equivalent expression, including

5.3a

$2\left(f\left(x\right)-g\left(x\right)\right)\left(f'\left(x\right)-g'\left(x\right)\right)\frac{dx}{dt}$.

If a response defines $f\left(x\right)-g\left(x\right)$ as a function in parts (b) or (c) (for example, $h\left(x\right)=f\left(x\right)-g\left(x\right)$), then a correct expression for $\frac{dA}{dt}$ (for example, $2h\frac{dh}{dt}$) earns the first point.

A response that imports a function $A\left(x\right)$ declared in part (c) is eligible for both points (the answer must be consistent with the imported function $A\left(x\right)$).

A response that presents an incorrect function for $A\left(x\right)$ that is not imported from part (c) is eligible only for the first point.

Except when $A\left(x\right)$ is imported from part (c), the second point is earned only for the correct answer.

A response that does not earn the first point is ineligible to earn the second point except in the special case noted below.

· If the first time the response presents $f\left(x\right)-g(x)$ as $j\left(x\right)=\ln\left(x+3\right)-x^4+2x^3$ occurs in part (d), the response loses a point. If the response already lost a point for this error in part (a), part (b), or part (c), then the response is eligible for all remaining points for a correct or consistent answer. In part (d) the consistent answer using $j\left(x\right)$ is 20.287.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $\frac{dA}{dx}\cdot\frac{dx}{dt}$
- ☐ Answer

Solution:

The cross section has area $A\left(x\right)=\left(f\left(x\right)-g\left(x\right)\right)^2$.

$$\frac{d}{dt}\left[A\left(x\right)\right]=\frac{dA}{dx}\cdot\frac{dx}{dt}$$

$$\frac{d}{dt}\left[A\left(x\right)\right]\Big|_{x=-0.5}=A'\left(-0.5\right)\cdot 7=-9.271842$$

At $x=-0.5$, the area of the cross section above the line is changing at a rate of -9.272 (or -9.271) square units per second.

5.3a

14. When a certain grocery store opens, it has 50 pounds of bananas on a display table. Customers remove bananas from the display table at a rate modeled by

$$f(t) = 10 + (0.8t)\sin\left(\frac{t^3}{100}\right) \text{ for } 0 < t \leq 12,$$

where $f(t)$ is measured in pounds per hour and t is the number of hours after the store opened. After the store has been open for three hours, store employees add bananas to the display table at a rate modeled by

$$g(t) = 3 + 2.4\ln\left(t^2 + 2t\right) \text{ for } 3 < t \leq 12,$$

where $g(t)$ is measured in pounds per hour and t is the number of hours after the store opened.

(a) How many pounds of bananas are removed from the display table during the first 2 hours the store is open?

(b) Find $f'(7)$. Using correct units, explain the meaning of $f'(7)$ in the context of the problem.

(c) Is the number of pounds of bananas on the display table increasing or decreasing at time $t = 5$? Give a reason for your answer.

(d) How many pounds of bananas are on the display table at time $t = 8$?

Part A

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 f(t)dt = 20.051175$$

20.051 pounds of bananas are removed from the display table during the first 2 hours the store is open.

0	1	2

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 f(t)dt = 20.051175$$

20.051 pounds of bananas are removed from the display table during the first 2 hours the store is open.

Part B

1 point is earned for value

5.3a

1 point is earned for meaning

$f'(7) = -8.120$ (or -8.119)

After the store has been open 7 hours, the rate at which bananas are being removed from the display table is decreasing by 8.120 (or 8.119) pounds per hour per hour.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for value

1 point is earned for meaning

$f'(7) = -8.120$ (or -8.119)

After the store has been open 7 hours, the rate at which bananas are being removed from the display table is decreasing by 8.120 (or 8.119) pounds per hour per hour.

Part C

1 point is earned for considers $f(5)$ and $g(5)$

1 point is earned for answer with reason

$g(5) - f(5) = -2.263103 < 0$

Because $g(5) - f(5) < 0$, the number of pounds of bananas on the display table is decreasing at time $t = 5$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for considers $f(5)$ and $g(5)$

1 point is earned for answer with reason

$g(5) - f(5) = -2.263103 < 0$

Because $g(5) - f(5) < 0$, the number of pounds of bananas on the display table is decreasing at time $t = 5$.

Part D

5.3a

2 points are earned for integrals

1 point is earned for answer

$$50+\int_3^8 g(t)dt-\int_0^8 f(t)dt=23.347396$$

23.347 pounds of bananas are on the display table at time $t = 8$.

0	1	2	3

The student response earns all of the following points:

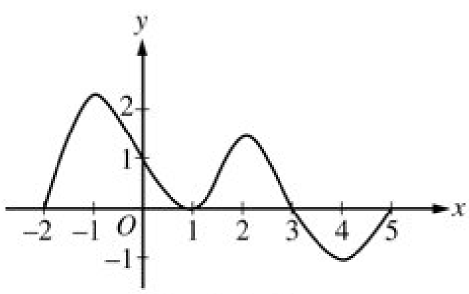
2 points are earned for integrals

1 point is earned for answer

$$50+\int_3^8 g(t)dt-\int_0^8 f(t)dt=23.347396$$

23.347 pounds of bananas are on the display table at time $t = 8$.

15.



Graph of f'

The graph of f' the derivative of f, is shown above for $-2 \le x \le 5$. On what intervals is f increasing?

(A) [−2,1] only

(B) [−2,3]

(C) [3,5] only

(D) [0,1.5] and[3,5]

(E) [−2,−1] ,[1,2] and[4,5]

5.3a

16. The first derivative of the function f is defined by $f'(x) = \sin(x^3 - x)$ for $0 \le x \le 2$. On what intervals is f increasing?

(A) $1 \le x \le 1.445$ only

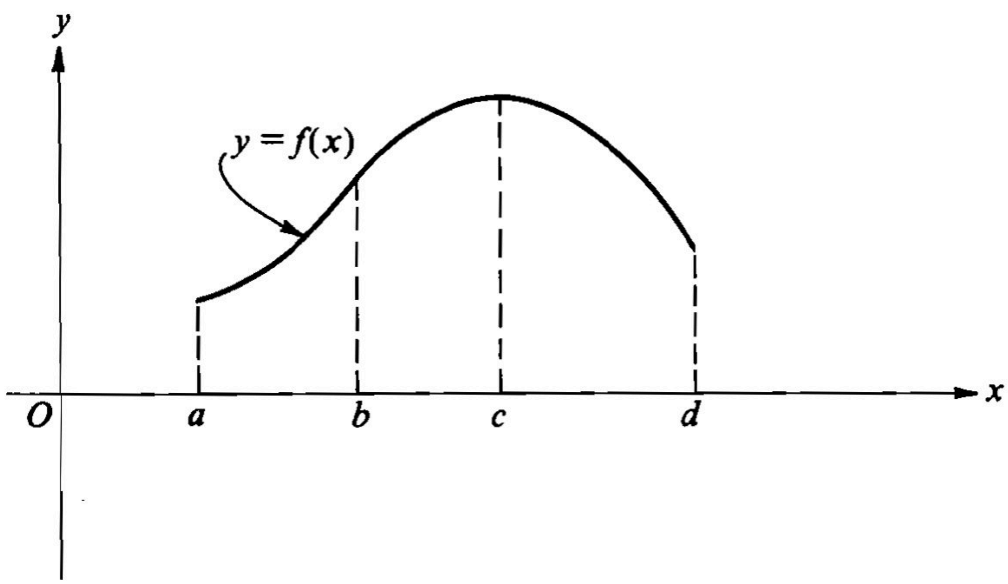
(B) $1 \le x \le 1.691$ ✓

(C) $1.445 \le x \le 1.875$

(D) $0.577 \le x \le 1.445$ and $1.875 \le x \le 2$

(E) $0 \le x \le 1$ and $1.691 \le x \le 2$

17.



The graph of $y = f(x)$ is shown in the figure above. On which of the following intervals are $\frac{dy}{dx} > 0$ and $\frac{d^2y}{dx^2} < 0$?

I. $a<x<b$

II. $b<x<c$

III. $c<x<d$

(A) I only

(B) II only ✓

(C) III only

(D) I and II

(E) II and III

5.3a

18. Let $F(x) = \int_0^x \sin(t^2)\,dt$ for $0 \le x \le 3$.

On what intervals is F increasing?

Part B

1 point is earned for correctly using $F'(x) = \sin(x^2)$

$F'(x) = \sin(x^2)$
$F'(x) = 0$ when $x^2 = 0, \pi, 2\pi, \ldots$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$x = 0, \sqrt{\pi}, \sqrt{2\pi}$

1 point is earned for correctly analyzing sign of F' within [0,3]

1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $[0, \sqrt{\pi}]$ and on $[\sqrt{2\pi}, 3]$

0	1	2	3	4

The student response earns four of the following points:

1 point is earned for correctly using $F'(x) = \sin(x^2)$

$F'(x) = \sin(x^2)$
$F'(x) = 0$ when $x^2 = 0, \pi, 2\pi, \ldots$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$x = 0, \sqrt{\pi}, \sqrt{2\pi}$

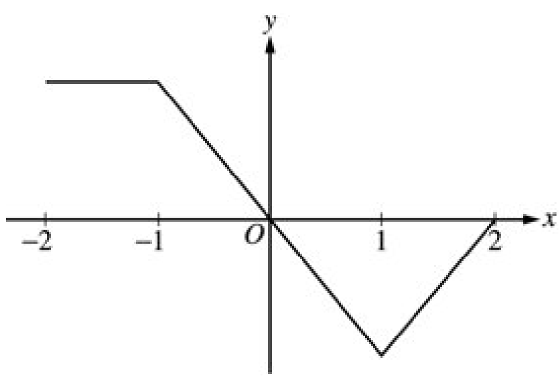
5.3a

1 point is earned for correctly analyzing sign of F' within [0,3]

1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $\left[0, \sqrt{\pi}\right]$ and on $\left[\sqrt{2\pi}, 3\right]$

19.



Graph of f'

The graph of f', the derivative of the function f, is shown above. Which of the following statements is true about f?

(A) f is decreasing for $-1 \le x \le 1$.

(B) f is increasing for $-2 \le x \le 0$. ✓

(C) f is increasing for $1 \le x \le 2$.

(D) f has a local minimum at $x = 0$.

(E) f is not differentiable at $x = -1$ and $x = 1$.

20. Let g be a twice-differentiable function with $g'(x) > 0$ and $g''(x) > 0$ for all real numbers x, such that $g(4) = 12$ and $g(5) = 18$ Of the following, which is a possible value for $g(6)$?

(A) 15

(B) 18

(C) 21

(D) 24

(E) 27 ✓

5.3a

21.

t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \le t \le 2.4$, as shown in the table above, For $0 < t \le 2.4, v(t) > 0$.

According to the model, $g(t) = \dfrac{24t+5\sin(6t)}{t+0.7}$, is Ruth's speed increasing or decreasing at time t=1.3 ? Give a reason reason for your answer.

Part D

The response can earn up to 2 points:

2 points: For correct conclusion with reason

Velocity $= g(1.3) = 18.096358 > 0$

Acceleration $= g'(1.3) = 3.761152 > 0$

Ruth's speed is increasing at time t=1.3 since velocity and acceleration have the same sign at this time.

0	1	2

The response earns both of the following points:

2 points: For correct conclusion with reason

Velocity $= g(1.3) = 18.096358 > 0$

Acceleration $= g'(1.3) = 3.761152 > 0$

Ruth's speed is increasing at time t=1.3 since velocity and acceleration have the same sign at this time.

5.3a

22.

t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \le t \le 2.4$ hours, as shown in the table above. For $0 < t \le 2.4$, $v(t) > 0$.

(a) Use the data in the table to approximate Ruth's acceleration at time $t = 1.4$ hours. Show the computations that lead to your answer. Indicate units of measure.

(b) Using correct units, interpret the meaning of $\int_0^{2.4} v(t)\,dt$ in the context of the problem. Approximate $\int_0^{2.4} v(t)\,dt$ using a midpoint Riemann sum with three subintervals of equal length and values from the table.

(c) For $0 \le t \le 2.4$ hours, Ruth's velocity can be modeled by the function g given by $g(t) = \frac{24t+5\sin(6t)}{t+0.7}$. According to the model, what was Ruth's average velocity during the time interval $0 \le t \le 2.4$?

(d) According to the model, $g(t) = \frac{24t+5\sin(6t)}{t+0.7}$, is Ruth's speed increasing or decreasing at time $t = 1.3$? Give a reason for your answer.

Part A

The response can earn up to 2 points:

1 point is earned for correct approximation

1 point is earned for correct units

$$a(1.4) \approx \frac{v(1.6)-v(1.2)}{1.6-1.2} = \frac{16.3-17.2}{1.6-1.2} = -2.25 \text{ miles/hr}^2$$

0	1	2

The response earns all of the following points:

1 point is earned for correct approximation

1 point is earned for correct units

$$a(1.4) \approx \frac{v(1.6)-v(1.2)}{1.6-1.2} = \frac{16.3-17.2}{1.6-1.2} = -2.25 \text{ miles/hr}^2$$

Part B

5.3a

The response can earn up to 3 points:

1 point is earned for the correct interpretation

$\int_0^{2.4} v(t)dt$ is the total distance Ruth traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

1 point is earned for use of the midpoint Riemann sum

1 point is earned for the correct approximation

$$\int_0^{2.4} v(t)dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8)$$
$$= 36.64 \text{ miles}$$

0	1	2	3

The response earns all of the following points:

1 point is earned for the correct interpretation

$\int_0^{2.4} v(t)dt$ is the total distance Ruth traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

1 point is earned for use of the midpoint Riemann sum

1 point is earned for the correct approximation

$$\int_0^{2.4} v(t)dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8)$$
$$= 36.64 \text{ miles}$$

Part C

The response can earn up to 2 points:

1 point is earned for the correct integral

$$\text{Average velocity} = \frac{1}{2.4}\int_0^{2.4} g(t)dt$$

1 point is earned for the correct answer

Average velocity = 14.064 miles/hr

5.3a

0	1	2

The response earns all of the following points:

1 point is earned for the correct integral

Average velocity $= \frac{1}{2.4}\int_0^{2.4} g(t)dt$

1 point is earned for the correct answer

Average velocity = 14.064 miles/hr

Part D

The response can earn up to 2 points:

2 points are earned for correct conclusion with reason

Velocity $= g(1.3) = 18.096358 > 0$
Acceleration $= g'(1.3) = 3.761152 > 0$

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.

0	1	2

The response earns all of the following points:

2 points are earned for correct conclusion with reason

Velocity $= g(1.3) = 18.096358 > 0$
Acceleration $= g'(1.3) = 3.761152 > 0$

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.

5.3a

23. Let f be a function defined by $f(x) = \begin{cases} 2x - x^2 & \text{for } x \le 1, \\ x^2 + kx + p & \text{for } x > 1. \end{cases}$

(a) For what values of k and p will f be continuous and differentiable at $x = 1$?

(b) For the values of k and p found in part (a), on what interval or intervals is f increasing?

(c) Using the values of k and p found in part (a), find all points of inflection of the graph of f. Support your conclusion.

Part A

1 point is earned for equation of $f(1)$

1 point is earned for $1 = 1 + k + p$

1 point is earned for $0 = 2 + k$

1 point is earned for solution of system

For continuity at $x = 1$,

$$\lim_{x \to 1^-} \left(2x - x^2\right) = f(1) = \lim_{x \to 1^+} \left(x^2 + kx + p\right)$$

Therefore $1 = 1 + k + p$

Since f is continuous at $x = 1$ and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k$$

For differentiability at $x = 1$, $0 = 2 + k$.

Therefore $k = -2$, $p = 2$.

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for equation of $f(1)$

1 point is earned for $1 = 1 + k + p$

1 point is earned for $0 = 2 + k$

1 point is earned for solution of system

5.3a

For continuity at $x = 1$,

$$\lim_{x \to 1^-} \left(2x - x^2\right) = f(1) = \lim_{x \to 1^+} \left(x^2 + kx + p\right)$$

Therefore $1 = 1 + k + p$

Since f is continuous at $x = 1$ and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k$$

For differentiability at $x = 1$, $0 = 2 + k$.

Therefore $k = -2$, $p = 2$.

Part B

1 point is earned for correct f' and analysis of sign of f' on domain

1 point is earned for $(-\infty,\infty)$

$f'(x) = 2 - 2x,\ x \le 1$

$2 - 2x > 0$

$x < 1$

$f'(x) = 2x - 2,\ x > 1$

$2x - 2 > 0$

$x > 1$

Since f increases on each of $(-\infty,1)$ and $(1,\infty)$ and is continuous at $x = 1$, f is increasing on $(-\infty,\infty)$.

0	1	2

The student response earns all of the following points:

1 point is earned for correct f' and analysis of sign of f' on domain

1 point is earned for $(-\infty,\infty)$

$f'(x) = 2 - 2x,\ x \le 1$

5.3a

$2 - 2x > 0$

$x < 1$

$f'(x) = 2x - 2,\ x > 1$

$2x - 2 > 0$

$x > 1$

Since f increases on each of $(-\infty,1)$ and $(1,\infty)$ and is continuous at $x = 1$, f is increasing on $(-\infty,\infty)$.

Part C

1 point is earned for f''

1 point is earned for noting sign changes in f'' or change of concavity in f

1 point is earned for ordered pair (or equivalent)

$f''(x) = -2,\ x < 1$

$f''(x) = 2,\ x > 1$

Since $f''(x) < 0$ on $(-\infty,1)$ and

$f''(x) > 0$ on $(1,\infty)$ and

$f(1)$ is defined,

$(1, f(1)) = (1,\ 1)$ is a point of inflection.

0	1	2	3

The student response earns all of the following points:

1 point is earned for f''

1 point is earned for noting sign changes in f'' or change of concavity in f

1 point is earned for ordered pair (or equivalent)

$f''(x) = -2,\ x < 1$

$f''(x) = 2,\ x > 1$

Since $f''(x) < 0$ on $(-\infty,1)$ and

5.3a

$f''(x) > 0$ on $(1,\infty)$ and

$f(1)$ is defined,

$(1, f(1)) = (1, 1)$ is a point of inflection.

24. Let f be a function defined by $f(x)=\{2x-x^2$ for $x \le 1,\ x^2+kx+p$ for $x>1$.
For the values of k and p found in part (a), on what interval or intervals is f increasing?

Part B

1 point is earned for correct f' and analysis of sign of f' on domain

1 point is earned for $(-\infty, \infty)$

$f'(x) = 2 - 2x, x \le 1$
$2 - 2x > 0$
$x < 1$
$f'(x) = 2x - 2, x > 1$
$2x - 2 > 0$
$x > 1$

Since f increases on each of $(-\infty, 1)$ and $(1, \infty)$ and is continuous at $x = 1$, f is increasing on $(-\infty, \infty)$

0	1	2

The student response earns two of the following points:

1 point is earned for correct f' and analysis of sign of f' on domain

1 point is earned for $(-\infty, \infty)$

$f'(x) = 2 - 2x, x \le 1$
$2 - 2x > 0$
$x < 1$
$f'(x) = 2x - 2, x > 1$
$2x - 2 > 0$
$x > 1$

Since f increases on each of $(-\infty, 1)$ and $(1, \infty)$ and is continuous at $x = 1$, f is increasing on $(-\infty, \infty)$

5.3a

25. A water tank at Camp Newton holds 1200 gallons of water at time $t = 0$. During the time interval $0 \le t \le$ 18 hours, water is pumped into the tank at the rate

$$W(t) = 95\sqrt{t}\sin^2\left(\frac{t}{6}\right) \text{ gallons per hour.}$$

During the same time interval, water is removed from the tank at the rate

$$R(t) = 275\sin^2\left(\frac{t}{3}\right) \text{ gallons per hour.}$$

(a) Is the amount of water in the tank increasing at time $t = 15$? Why or why not?

(b) To the nearest whole number, how many gallons of water are in the tank at time $t = 18$?

(c) At what time t, $0 \le t \le 18$, is the amount of water in the tank at an absolute minimum? Show the work that leads to your conclusion.

(d) For $t > 18$, no water is pumped into the tank, but water continues to be removed at the rate $R(t)$ until the tank becomes empty. Let k be the time at which the tank becomes empty. Write, but do not solve, an equation involving an integral expression that can be used to find the value of k.

Part A

1 point is earned for the answer with reason

No; the amount of water is not increasing at $t = 15$ since $W(15) - R(15) = -121.09 < 0$.

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer with reason

No; the amount of water is not increasing at $t = 15$ since $W(15) - R(15) = -121.09 < 0$.

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$1200 + \int_0^{18} (W(t) - R(t))dt = 1309.788$$

1310 gallons

5.3a

0	1	2	3

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$1200 + \int_0^{18} (W(t) - R(t))dt = 1309.788$$

1310 gallons

Part C

1 point is earned for the interior critical points

1 point is earned for the amount of water is least at $t = 6.494$ or 6.495

1 point is earned for the analysis for absolute minimum

$W(t) - R(t) = 0$

$t = 0, 6.4948, 12.9748$

t (hours)	gallons of water
0	1200
6.495	525
12.975	1697
18	1310

The values at the endpoints and the critical points show that the absolute minimum occurs when $t = 6.494$ or 6.495.

0	1	2	3

The student response earns all of the following points:

1 point is earned for the interior critical points

1 point is earned for the amount of water is least at $t = 6.494$ or 6.495

5.3a

1 point is earned for the analysis for absolute minimum

$W(t)$ - $R(t) = 0$

$t = 0, 6.4948, 12.9748$

t (hours)	gallons of water
0	1200
6.495	525
12.975	1697
18	1310

The values at the endpoints and the critical points show that the absolute minimum occurs when $t = 6.494$ or 6.495.

Part D

1 point is earned for the limits

1 point is earned for the equation

$$\int_{18}^{k} R(t)dt = 1310$$

0	1	2

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the equation

$$\int_{18}^{k} R(t)dt = 1310$$

5.3a

26. A water tank at Camp Newton holds 1200 gallons of water at time t = 0. During the time interval $0 \le t \le 18$ hours, water is pumped into the tank at the rate

$$W(t) = 95\sqrt{t}\sin^2\left(\frac{t}{6}\right) \text{ gallons per hour.}$$

During the same time interval, water is removed from the tank at the rate

$$R(t) = 275\sin^2\left(\frac{t}{3}\right) \text{ gallons per hour.}$$

Is the amount of water in the tank increasing at time $t=15$? Why or why not?

Part A

1 point is earned for the answer with reason

No; the amount of water is not increasing at $t=15$ since $W(15)-R(15)=-121.09<0$.

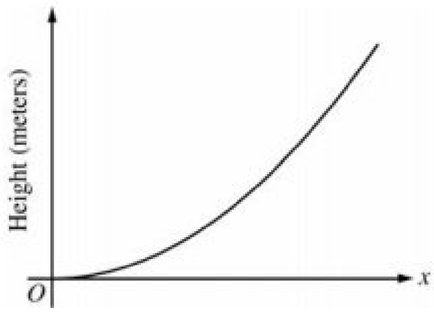
0	1

The student response earns one of the following points:

1 point is earned for the answer with reason

No; the amount of water is not increasing at $t=15$ since $W(15)-R(15)=-121.09<0$.

5.3a



The figure above is the graph of a function of x, which models the height of a skateboard ramp. The function meets the following requirements.

(i) At x=0, the value of the function is 0, and the slope of the graph of the function is 0.

(ii) At x=4, the value of the function is 1, and the slope of the graph of the function is 1.

(iii) Between x=0 and x=4, the function is increasing.

27. Let $h(x) = x^n / k$, where k is a nonzero constant and n is a positive integer. Find the values of k and n so that h meets requirement (ii) above. Show that h also meets requirements (i) and (iii) above.

Part D

1 point is earned for $\frac{4^n}{k} = 1$

1 point is earned for $\frac{n4^{n-1}}{k} = 1$

1 point is earned for values for k and n

1 point is earned for verification

$h(4) = \frac{4^n}{k} = 1$ implies that $4^n = k$.

$h\prime(4) = \frac{n4^{n-1}}{k} = \frac{n4^{n-1}}{4^n} = \frac{n}{4} = 1$ gives n=4 and $k = 4^4 = 256$.

$h(x) = \frac{x^4}{256} \Rightarrow h(0) = 0$.

$h\prime(x) = \frac{4x^3}{256} \Rightarrow h\prime(0) = 0$ and h ′ (x)>0 for 0<x<4.

0	1	2	3	4 ✓

The student response earns all of the following points:

5.3a

1 point is earned for $\frac{4^n}{k} = 1$

1 point is earned for $\frac{n4^{n-1}}{k} = 1$

1 point is earned for values for k and n

1 point is earned for verification

$h\left(4\right) = \frac{4^n}{k} = 1$ implies that $4^n = k$.

$h\prime(4) = \frac{n4^{n-1}}{k} = \frac{n4^{n-1}}{4^n} = \frac{n}{4} = 1$ gives n=4 and $k = 4^4 = 256$.

$h\left(x\right) = \frac{x^4}{256} \Rightarrow h\left(0\right) = 0$.

$h\prime(x) = \frac{4x^3}{256} \Rightarrow h\prime(0) = 0$ and h ′ (x)>0 for 0<x<4.

5.3a

28. Using the function g and your value of c from part (b), show that g does not meet requirement (iii) above.

Part C

1 point is earned for $g'(x)$

1 point is earned for the explanation

$$g\prime(x) = \frac{3}{32}x^2 - \frac{x}{8} = \frac{1}{32}x\,(3x - 4)$$

$g'(x)<0$ for $0 < x < \dfrac{4}{3}$, so g does not satisfy (iii).

0	1	2

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for the explanation

$$g\prime(x) = \frac{3}{32}x^2 - \frac{x}{8} = \frac{1}{32}x\,(3x - 4)$$

$g'(x)<0$ for $0<x<\dfrac{4}{3}$, so g does not satisfy (iii).

5.3a

29. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

t (minutes)	0	1	5	6	8
$g(t)$ (cubic feet per minute)	12.8	15.1	20.5	18.3	22.7

Grain is being added to a silo. At time $t = 0$, the silo is empty. The rate at which grain is being added is modeled by the differentiable function g, where $g(t)$ is measured in cubic feet per minute for $0 \le t \le 8$ minutes. Selected values of $g(t)$ are given in the table above.

(a) Using the data in the table, approximate $g'(3)$. Using correct units, interpret the meaning of $g'(3)$ in the context of the problem.

(b) Write an integral expression that represents the total amount of grain added to the silo from time $t = 0$ to time $t = 8$. Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate the integral.

(c) The grain in the silo is spoiling at a rate modeled by $w(t) = 32 \cdot \sqrt{\sin\left(\frac{\pi t}{74}\right)}$, where $w(t)$ is measured in cubic feet per minute for $0 \le t \le 8$ minutes. Using the result from part (b), approximate the amount of unspoiled grain remaining in the silo at time $t = 8$.

(d) Based on the model in part (c), is the amount of unspoiled grain in the silo increasing or decreasing at time $t = 6$? Show the work that leads to your answer.

Part A

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5)-g(1)}{5-1} = \frac{20.5-15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.

0	1	2

The student response earns all of the following points:

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5)-g(1)}{5-1} = \frac{20.5-15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.

5.3a

Part B

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t)\,dt$ cubic feet.

$$\int_0^8 g(t)\,dt \approx g(1)\cdot(1-0) + g(5)\cdot(5-1) + g(6)\cdot(6-5) + g(8)\cdot(8-6)$$
$$= 15.1\cdot 1 + 20.5\cdot 4 + 18.3\cdot 1 + 22.7\cdot 2 = 160.8$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t)\,dt$ cubic feet.

$$\int_0^8 g(t)\,dt \approx g(1)\cdot(1-0) + g(5)\cdot(5-1) + g(6)\cdot(6-5) + g(8)\cdot(8-6)$$
$$= 15.1\cdot 1 + 20.5\cdot 4 + 18.3\cdot 1 + 22.7\cdot 2 = 160.8$$

Part C

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t)\,dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t)\,dt = 61.749$ (or 61.748) cubic feet.

5.3a

0	1	2

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t)\,dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t)\,dt = 61.749$ (or 61.748) cubic feet.

Part D

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.

0	1	2

The student response earns all of the following points:

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.

5.3a

A particle moves along the y-axis with velocity given by $v(t) = t\sin(t^2)$ for $t \geq 0$.

30. In what direction (up or down) is the particle moving at time $t = 1.5$?

Part A

1 point is earned for the answer and reason

$v(1.5) = 1.5\sin(1.5^2) = 1.167$

Up, because $v(1.5) > 0$

0	1

The student response earns all of the following points:

1 point is earned for the answer and reason

$v(1.5) = 1.5\sin(1.5^2) = 1.167$

Up, because $v(1.5) > 0$

5.3a

31. Find the acceleration of the particle at time $t = 1.5$. Is the velocity of the particle increasing at $t = 1.5$? Why or why not?

Part B

1 point is earned for: $a(1.5)$

1 point is earned for the conclusion and reasons

$a(t) = v\prime 9(t) = \sin t^2 + 2t^2 \cos t^2$
$a(1.5) = v\prime(1.5) = -2.048$ or -2.049

No; v is decreasing at 1.5 because $v'(1.5) < 0$

0	1	2

The student response earns all of the following points:

1 point is earned for: $a(1.5)$

1 point is earned for the conclusion and reasons

$a(t) = v\prime 9(t) = \sin t^2 + 2t^2 \cos t^2$
$a(1.5) = v\prime(1.5) = -2.048$ or -2.049

No; v is decreasing at 1.5 because $v'(1.5) < 0$

5.3a

For $0 \leq t \leq 31$, the rate of change of the number of mosquitoes on Tropical Island at time *t* days is modeled by $R(t) = 5\sqrt{t}\cos\left(\frac{t}{5}\right)$ mosquitoes per day. There are 1000 mosquitoes on Tropical Island at time t=0.

32. Show that the number of mosquitoes is increasing at time t=6.

Part A

1 point is earned for showing that $R(6) > 0$

Since $R(6)$=4.438>0, the number of mosquitoes is increasing at t=6.

0	1 ✓

The student response earns one of the following points:

1 point is earned for showing that $R(6) > 0$

Since $R(6)$=4.438>0, the number of mosquitoes is increasing at t=6.

33. At time t=6, is the number of mosquitoes increasing at an increasing rate, or is the number of mosquitoes increasing at a decreasing rate? Give a reason for your answer.

Part B

1 point is earned for considering R'(6)

1 point is earned for the answer with reason

$R'(6)$=-1.913 Since $R'(6)$<0, the number of mosquitoes is increasing at a decreasing rate at t=6.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for considering R'(6)

1 point is earned for the answer with reason

$R'(6)$=-1.913 Since $R'(6)$<0, the number of mosquitoes is increasing at a decreasing rate at t=6.

5.3a

34. An object moves along the x-axis with initial position $x(0) = 2$. The velocity of the object at time $t \geq 0$ is given by $v(t) = \sin\left(\frac{\pi}{3}t\right)$.

Consider the following two statements.

Statement I: For $3 < t < 4.5$, the velocity of the object is decreasing.

Statement II: For $3 < t < 4.5$, the speed of the object is increasing.

Are either or both of these statements correct? For each statement provide a reason why it is correct or not correct.

Part B

1 point is earned for I correct, with reason

1 point is earned for II correct

1 point is earned for the reason for II

On $3 < t < 4.5$:

$$a(t) = v'(t) = \frac{\pi}{3}\cos\left(\frac{\pi}{3}t\right) < 0$$

Statement I is correct since $a(t) < 0$.

Statement II is correct since $v(t) < 0$ and $a(t) < 0$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for I correct, with reason

1 point is earned for II correct

1 point is earned for the reason for II

On $3 < t < 4.5$:

$$a(t) = v'(t) = \frac{\pi}{3}\cos\left(\frac{\pi}{3}t\right) < 0$$

Statement I is correct since $a(t) < 0$.

Statement II is correct since $v(t) < 0$ and $a(t) < 0$.

5.3a

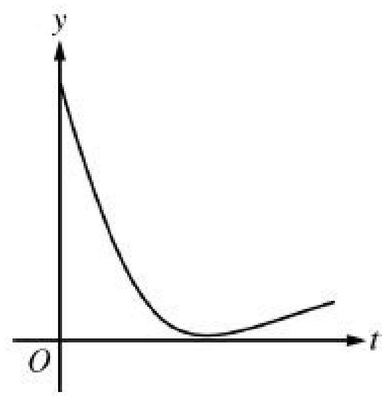
5.3a

35. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

During a chemical reaction, the function $y = f(t)$ models the amount of a substance present, in grams, at time t seconds. At the start of the reaction $(t = 0)$, there are 10 grams of the substance present. The function $y = f(t)$ satisfies the differential equation $\frac{dy}{dt} = -0.02y^2$.

(a) Use the line tangent to the graph of $y = f(t)$ at $t = 0$ to approximate the amount of the substance remaining at time $t = 2$ seconds.

(b) Using the given differential equation, determine whether the graph of f could resemble the following graph. Give a reason for your answer.



(c) Find an expression for $y = f(t)$ by solving the differential equation $\frac{dy}{dt} = -0.02y^2$ with the initial condition $f(0) = 10$.

(d) Determine whether the amount of the substance is changing at an increasing or a decreasing rate. Explain your reasoning.

Part A

1 point is earned for: $y'(0)$

1 point is earned for: approximation

$y'(0) = -0.02\left(10^2\right) = -2$

An equation for the line tangent to the graph of $y = f(t)$ at $t = 0$ is $y = 10 - 2t$.

$y(2) \approx 10 - 2(2) = 6$ grams

0	1	2

The student response earns all of the following points:

1 point is earned for: $y'(0)$

1 point is earned for: approximation

5.3a

$y'(0) = -0.02\left(10^2\right) = -2$

An equation for the line tangent to the graph of $y = f(t)$ at $t = 0$ is $y = 10 - 2t$.

$y(2) \approx 10 - 2(2) = 6$ grams

Part B

1 point is earned for: answer with reason

$\frac{dy}{dt} = -0.02y^2 \leq 0$, so the graph of *f* is nonincreasing.

The graph of *f* cannot resemble the given graph because the given graph is increasing on a portion of its domain.

0	1

The student response earns all of the following points:

1 point is earned for: answer with reason

$\frac{dy}{dt} = -0.02y^2 \leq 0$, so the graph of *f* is nonincreasing.

The graph of *f* cannot resemble the given graph because the given graph is increasing on a portion of its domain.

Part C

1 point is earned for: separation of variables

1 point is earned for: antiderivatives

1 point is earned for: constant of integration and uses initial condition

1 point is earned for: answer

$$\int \left(-\frac{1}{y^2}\right) dy = \int 0.02\, dt$$

$$\frac{1}{y} = 0.02t + C$$

$$\frac{1}{10} = 0.02(0) + C \Rightarrow C = 0.1$$

$$\frac{1}{y} = 0.02t + 0.1 \Rightarrow y = \frac{1}{0.02t+0.1} = \frac{50}{t+5}$$

Note: this solution is valid for $t > -5$.

Note: max $2/4$ $[1\text{-}1\text{-}0\text{-}0]$ if no constant of integration

5.3a

Note: $0/4$ if no separation of variables

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for: separation of variables

1 point is earned for: antiderivatives

1 point is earned for: constant of integration and uses initial condition

1 point is earned for: answer

$$\int\left(-\frac{1}{y^2}\right)\,dy=\int 0.02\,dt$$

$$\frac{1}{y}=0.02t+C$$

$$\frac{1}{10}=0.02(0)+C\Rightarrow C=0.1$$

$$\frac{1}{y}=0.02t+0.1\Rightarrow y=\frac{1}{0.02t+0.1}=\frac{50}{t+5}$$

Note: this solution is valid for $t>-5$.

Note: max $2/4$ $[1\text{-}1\text{-}0\text{-}0]$ if no constant of integration

Note: $0/4$ if no separation of variables

Part D

1 point is earned for: $\frac{d^2y}{dt^2}$

1 point is earned for: answer with reason

$$\begin{aligned}\frac{d^2y}{dt^2}&=-0.04y\frac{dy}{dt}\\&=-0.04y(-0.02y^2)\\&=0.0008y^3\end{aligned}$$

Because $y>0$, $0.0008y^3>0$.

The amount of substance is changing at an increasing rate.

— OR —

From part (c), $f(t)=\frac{50}{t+5}$, and from context, $t\geq 0$.

5.3a

$f'(t) = \frac{-50}{(t+5)^2}$ and $f''(t) = \frac{100}{(t+5)^3} > 0$ for $t \geq 0$.

The amount of substance is changing at an increasing rate.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\frac{d^2y}{dt^2}$

1 point is earned for: answer with reason

$$\begin{aligned}\frac{d^2y}{dt^2} &= -0.04y\frac{dy}{dt}\\ &= -\,0.04y(-0.02y^2)\\ &= 0.0008y^3\end{aligned}$$

Because $y > 0$, $0.0008y^3 > 0$.

The amount of substance is changing at an increasing rate.

— OR —

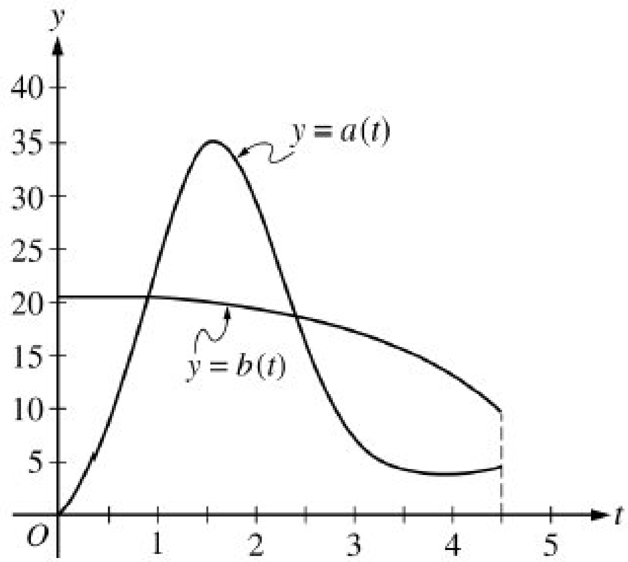
From part (c), $f(t) = \frac{50}{t+5}$, and from context, $t \geq 0$.

$f'(t) = \frac{-50}{(t+5)^2}$ and $f''(t) = \frac{100}{(t+5)^3} > 0$ for $t \geq 0$.

The amount of substance is changing at an increasing rate.

5.3a

36. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.



During the time interval $0 \leq t \leq 4.5$ hours, water flows into tank A at a rate of $a(t) = (2t - 5) + 5e^{2\sin t}$ liters per hour. During the same time interval, water flows into tank B at a rate of $b(t)$ liters per hour. Both tanks are empty at time $t = 0$. The graphs of $y = a(t)$ and $y = b(t)$, shown in the figure above, intersect at $t = k$ and $t = 2.416$.

(a) How much water will be in tank A at time $t = 4.5$?

(b) During the time interval $0 \leq t \leq k$ hours, water flows into tank B at a constant rate of 20.5 liters per hour. What is the difference between the amount of water in tank A and the amount of water in tank B at time $t = k$?

(c) The area of the region bounded by the graphs of $y = a(t)$ and $y = b(t)$ for $k \leq t \leq 2.416$ is 14.470. How much water is in tank B at time $t = 2.416$?

(d) During the time interval $2.7 \leq t \leq 4.5$ hours, the rate at which water flows into tank B is modeled by $w(t) = 21 - \frac{30t}{(t-8)^2}$ liters per hour. Is the difference $w(t) - a(t)$ increasing or decreasing at time $t = 3.5$? Show the work that leads to your answer.

Part A

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^{4.5} a(t)\, dt = 66.532128$$

At time $t = 4.5$, tank A contains 66.532 liters of water.

0	1	2

5.3a

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^{4.5} a(t)\,dt = 66.532128$$

At time $t = 4.5$, tank A contains 66.532 liters of water.

Part B

1 point is earned for: sets $a(k) = 20.5$

1 point is earned for: integral

1 point is earned for: answer

$$a(k) = 20.5 \Rightarrow k = 0.892040$$
$$\int_0^{k} (20.5 - a(t))\,dt = 10.599191$$

At time $t = k$, the difference in the amounts of water in the tanks is 10.599 liters.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: sets $a(k) = 20.5$

1 point is earned for: integral

1 point is earned for: answer

$$a(k) = 20.5 \Rightarrow k = 0.892040$$
$$\int_0^{k} (20.5 - a(t))\,dt = 10.599191$$

At time $t = k$, the difference in the amounts of water in the tanks is 10.599 liters.

Part C

1 point is earned for: $\int_k^{2.416} a\,(t)\,dt$

5.3a

1 point is earned for: answer

$$\int_0^{2.416} b(t)\,dt = \int_0^k b(t)\,dt + \int_k^{2.416} b(t)\,dt$$

$$\int_0^k b(t)\,dt = 20.5 \cdot k = 18.286826$$

On $k < t < 2.416$, tank A receives $\int_k^{2.416} a(t)\,dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B.

Therefore, $\int_k^{2.416} b(t)\,dt = \int_k^{2.416} a(t)\,dt - 14.470 = 30.027051$.

$$\int_0^k b(t)\,dt + \int_k^{2.416} b(t)\,dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\int_k^{2.416} a(t)\,dt$

1 point is earned for: answer

$$\int_0^{2.416} b(t)\,dt = \int_0^k b(t)\,dt + \int_k^{2.416} b(t)\,dt$$

$$\int_0^k b(t)\,dt = 20.5 \cdot k = 18.286826$$

On $k < t < 2.416$, tank A receives $\int_k^{2.416} a(t)\,dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B.

Therefore, $\int_k^{2.416} b(t)\,dt = \int_k^{2.416} a(t)\,dt - 14.470 = 30.027051$.

$$\int_0^k b(t)\,dt + \int_k^{2.416} b(t)\,dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

Part D

5.3a

1 point is earned for: $w'(3.5) - a'(3.5) < 0$

1 point is earned for: conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $w'(3.5) - a'(3.5) < 0$

1 point is earned for: conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

5.3a

37. Let $x = ky^2 + 2$, where $k > 0$.

Let R be the region in the first quadrant bounded by the x-axis, the graph of $x = ky^2 + 2$, and the line $x = 4$. Write an integral expression for the area of the region R and show that this area decreases as k increases.

Part C

3 points are earned for correctly expressing the area as a definite integral

$$A = \int_0^{\sqrt{2/k}} (4 - (ky^2 + 2))dy$$

or

$$A = \frac{1}{\sqrt{k}} \int_2^4 \sqrt{x - 2}dx$$

< -1 > error in limits

< -1 > considers twice the region

< -2 > error in integrand

1 point is earned for correctly showing that A decreases using algebra, geometry, or calculus

$$A = \frac{4\sqrt{2}}{3}k^{-0.5}$$

$$\frac{dA}{dk} = -\frac{2\sqrt{2}}{3}k^{-1.5} < 0$$

for all $k > 0$

thus the area decreases as k increases

0	1	2	3	4

The student response earns four of the following points:

3 points are earned for correctly expressing the area as a definite integral

$$A = \int_0^{\sqrt{2/k}} (4 - (ky^2 + 2))dy$$

or

$$A = \frac{1}{\sqrt{k}} \int_2^4 \sqrt{x - 2}dx$$

< -1 > error in limits

< -1 > considers twice the region

5.3a

< -2 > error in integrand

1 point is earned for correctly showing that A decreases using algebra, geometry, or calculus

$$A = \frac{4\sqrt{2}}{3}k^{-0.5}$$

$$\frac{dA}{dk} = -\frac{2\sqrt{2}}{3}k^{-1.5} < 0$$

for all $k > 0$

thus the area decreases as k increases

38. If g is the function given by $g(x) = \frac{1}{3}x^3 + \frac{3}{2}x^2 - 70x + 5$, on which of the following intervals is g decreasing?

(A) $(-\infty, -10)$ and $(7, \infty)$

(B) $(-\infty, -7)$ and $(10, \infty)$

(C) $(-\infty, 10)$

(D) $(-10, 7)$

(E) $(-7, 10)$

5.3a

39. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

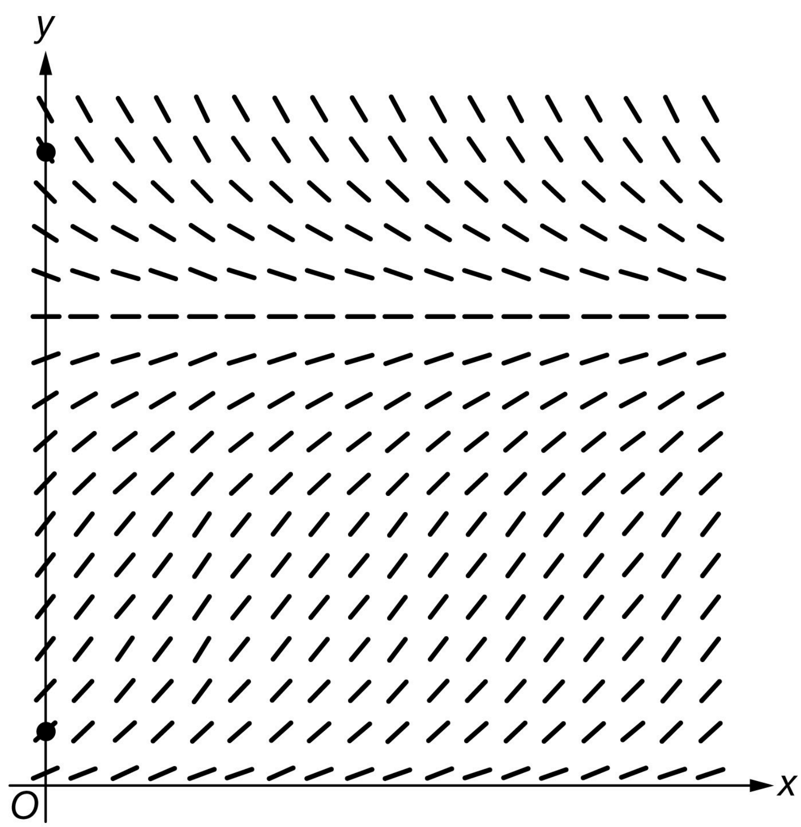
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{-y}{4}\ln\left(\frac{y}{50}\right)$.

(a) A portion of the slope field for the differential equation is given below. Sketch the solution curves through the two indicated points for $x \geq 0$.



(b) Find $\lim\limits_{y \to 50} \frac{dy}{dx}$.

(c) Let f be the particular solution to the differential equation with initial condition $f(0) = 20$. Find $f'(0)$, and explain why f is increasing for all x.

(d) For $0 < y < 40$, at what value of y does $\frac{dy}{dx}$ attain its maximum value? Justify your answer.

Part A

5.3a

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Each solution curve must pass through the appropriate point, extend reasonably close to the right edge of the square, and have no obvious conflicts with the given slope lines.

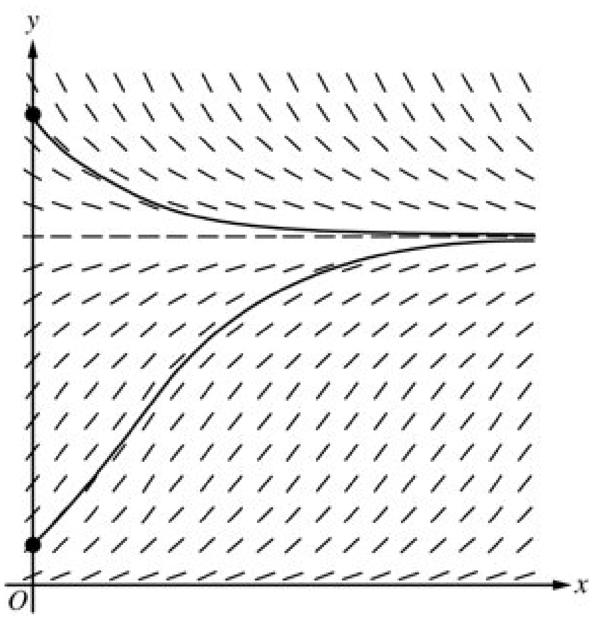
Ignore any portion of a solution curve that extends beyond the square.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Top solution curve
- ☐ Bottom solution curve

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An answer of 0 earns the point.

The answer of $\frac{-50}{4}\ln\left(\frac{50}{50}\right)$ need not be simplified, but any subsequent simplification must be correct.

Limit notation is not required.

5.3a

Expressions that equate the limit with the derivative such as $\lim\limits_{y \to 50} \frac{dy}{dx} = \frac{-y}{4} \ln\left(\frac{y}{50}\right) = \frac{-50}{4} \ln\left(\frac{50}{50}\right)$, or other linkage errors, will not earn the point, even in the presence of the correct evaluation.

Expressions with incomplete limit notation, such as $\lim\limits_{y \to 50} = \frac{-50}{4} \ln\left(\frac{50}{50}\right)$, are eligible for the point.

The expression $\lim\limits_{y \to 50} \frac{-50}{4} \ln\left(\frac{50}{50}\right)$ is not incorrect, but the limit must be resolved to earn this point.

0	1

The student response accurately includes the criteria below.

☐ Limit

Solution:

$$\lim_{y \to 50} \frac{dy}{dx} = \lim_{y \to 50} \frac{-y}{4} \ln\left(\frac{y}{50}\right) = \frac{-50}{4} \ln\left(\frac{50}{50}\right) = 0$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $-5 \ln\left(\frac{2}{5}\right)$ or the equivalent; no supporting work is necessary.

The evaluation does not need to be simplified, and incorrect attempts at simplifying a correct expression will not be read.

For the second point to be earned, the response must:

· Argue that $\frac{dy}{dx}$ is positive for $0 < y < 50$, thus f is increasing for $0 < y < 50$.

· Argue that the particular solution cannot cross the line $y = 50$. (This may be done implicitly by restricting the domain of consideration to $0 < y < 50$.)

Explanations may be graphical but must address both of the above points.

Explanations that appeal only to the sign of $f'(0)$ will not earn this point.

0	1	2

The student response accurately includes **both** of the criteria below.

5.3a

- [ ] $f'(0)$
- [ ] Explanation

Solution:

$$f'(0) = \frac{dy}{dx}\bigg|_{(x,\,y)=(0,\,20)} = \frac{-20}{4}\ln\left(\frac{20}{50}\right) = -5\ln\left(\frac{2}{5}\right)$$

Because $\frac{dy}{dx} = \frac{-y}{4}\ln\left(\frac{y}{50}\right) > 0$ for $0 < y < 50$, f is increasing provided $f(x) < 50$.

Note that $y = 50$ is also a solution to the differential equation, so the graph of the particular solution $y = f(x)$ with $f(0) = 20$ is below the line $y = 50$ and cannot cross the line $y = 50$.

Thus, f is increasing for all x.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by correctly applying the product rule in finding $\frac{d^2y}{dx^2}$, with or without using the chain rule. For example, $\frac{d^2y}{dx^2} = -\frac{1}{4} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{1}{y}$ would earn the first point.

The second point is earned for correctly applying the chain rule (twice) in finding $\frac{d^2y}{dx^2}$. This point can be earned even if an error was made in applying the product rule that resulted in the first point not being earned. In general, expressions of the form $\frac{d^2y}{dx^2} = c \cdot \ln\left(\frac{y}{50}\right) \cdot \frac{dy}{dx} + g(y) \cdot \frac{dy}{dx}$, where $g(y)$ may be a nonzero constant and c is a nonzero constant, earn this point.

For the first two points, if both the factors 50 and $\frac{1}{50}$ are not present in the response, assume that these factors were canceled. Furthermore, in the presence of the factor 50, assume that a missing factor $\frac{1}{50}$ is an error in the application of the chain rule. Thus, $\frac{d^2y}{dx^2} = -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{50}{y} \cdot \frac{dy}{dx}$ would earn the first point but not the second point.

For the first two points, each point is earned with the correct expression and any subsequent errors in simplification come off of the fourth point.

The third point is earned when the declared $\frac{d^2y}{dx^2}$ is set equal to zero and one initial (correct or otherwise) algebraic step towards solving this equation for y is made. However, the declared expression for $\frac{d^2y}{dx^2}$ must contain the variable y and must be distinct from $\frac{dy}{dx} = \frac{-y}{4}\ln\left(\frac{y}{50}\right)$. (This means a score of 0-0-1-0 is possible in this part.)

The fourth point cannot be earned without having earned the first three points.

5.3a

The fourth point requires a discussion of the behavior of $\frac{d^2y}{dx^2}$ both to the left and right of $y = 50e^{-1}$.

Sign charts, labeled or otherwise, are not read for the fourth point.

0	1	2	3	4

The student response accurately includes **all four** of the criteria below.

- [ ] Product rule
- [ ] Chain rule
- [ ] Attempts to solve $\frac{d^2y}{dx^2} = 0$ for y
- [ ] Answer with justification

Solution:

$$\frac{d^2y}{dx^2} = -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{50}{y} \frac{1}{50} \frac{dy}{dx}$$

$$= -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{1}{y} \frac{dy}{dx} = -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \left(\ln\left(\frac{y}{50}\right) + 1\right)$$

$$\frac{d^2y}{dx^2} = 0 \Rightarrow \left(\ln\left(\frac{y}{50}\right) + 1\right) = 0 \Rightarrow y = 50e^{-1}$$

Because $\frac{d^2y}{dx^2} > 0$ for $0 < y < 50e^{-1}$ and $\frac{d^2y}{dx^2} < 0$ for $50e^{-1} < y < 40$, $\frac{dy}{dx}$ attains its maximum value at $y = 50e^{-1}$.

5.3a

40. The rate at which rainwater flows into a drainpipe is modeled by the function R , where $R(t) = 20\sin\left(\frac{t^2}{35}\right)$ cubic feet per hour, t is measured in hours, and $0 \le t \le 8$. The pipe is partially blocked, allowing water to drain out the other end of the pipe at a rate modeled by $D(t) = -0.04t^3 + 0.4t^2 + 0.96t$ cubic feet per hour, for $0 \le t \le 8$. There are 30 cubic feet of water in the pipe at time $t = 0$.

(a) How many cubic feet of rainwater flow into the pipe during the 8-hour time interval $0 \le t \le 8$?

(b) Is the amount of water in the pipe increasing or decreasing at time $t = 3$ hours? Give a reason for your answer.

(c) At what time t, $0 \le t \le 8$, is the amount of water in the pipe at a minimum? Justify your answer.

(d) The pipe can hold 50 cubic feet of water before overflowing. For $t > 8$, water continues to flow into and out of the pipe at the given rates until the pipe begins to overflow. Write, but do not solve, an equation involving one or more integrals that gives the time w when the pipe will begin to overflow.

Part A

1 point is earned for the Integrand

1 point is earned for the answer

$$\int_0^8 R(t)dt = 76.570$$

0	1	2

The student response earns all of the following points:

1 point is earned for the Integrand

1 point is earned for the answer

$$\int_0^8 R(t)dt = 76.570$$

Part B

1 point is earned for considering $R(3)$ and $D(3)$

1 point is earned for the answer and reason

$R(3) - D(3) = -0.313632 < 0$

5.3a

Since $R(3) < D(3)$, the amount of water in the pipe is decreasing at time $t = 3$ hours.

0	1	2

The student response earns all of the following points:

1 point is earned for considering $R(3)$ and $D(3)$

1 point is earned for the answer and reason

$R(3) - D(3) = -0.313632 < 0$

Since $R(3) < D(3)$, the amount of water in the pipe is decreasing at time $t = 3$ hours.

Part C

1 point is earned for considering $R(t) - D(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of water in the pipe at time t, $0 \le t \le 8$, is $30 + \int_0^t [R(x) - D(x)]dx$.

$R(t) - D(t) = 0 \Rightarrow t = 0, 3.271658$

t	Amount of water in the pipe
0	30
3.271658	27.964561
8	48.543686

The amount of water in the pipe is a minimum at time $t = 3.272$ (or 3.271) hours.

0	1	2	3

The student response earns all of the following points:

1 point is earned for considering $R(t) - D(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

5.3a

The amount of water in the pipe at time t, $0 \le t \le 8$, is $30 + \int_0^t [R(x) - D(x)]dx$.

$R(t) - D(t) = 0 \Rightarrow t = 0, 3.271658$

t	Amount of water in the pipe
0	30
3.271658	27.964561
8	48.543686

The amount of water in the pipe is a minimum at time $t = 3.272$ (or 3.271) hours.

Part D

1 point is earned for the integral

1 point is earned for the equation

$$30 + \int_0^w [R(t) - D(t)]dt = 50$$

0	1	2

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the equation

$$30 + \int_0^w [R(t) - D(t)]dt = 50$$

5.3a

41. The rate at which rainwater flows into a drainpipe is modeled by the function R, where $R(t) = 20 \sin\left(\frac{t^2}{35}\right)$ cubic feet per hour, t is measured in hours, and $0 \le t \le 8$. The pipe is partially blocked, allowing water to drain out the other end of the pipe at a rate modeled by $D(t) = -0.04t^3 + 0.4t^2 + 0.96t$ or $0 \le t \le 8$. There are 30 cubic feet of water in the pipe at time t=0.

Is the amount of water in the pipe increasing or decreasing at time t=3 hours? Give a reason for your answer.

Part B

1 point is earned for considering $R(3)$ and $D(3)$

1 point is earned for the answer and reason

$R(3)-D(3) = -0.313632 < 0$

Since $R(3) < D(3)$, the amount of water in the pipe is decreasing at time t=3 hours.

0	1	2

The student response earns all of the following points:

1 point is earned for considering $R(3)$ and $D(3)$

1 point is earned for the answer and reason

$R(3)-D(3) = -0.313632 < 0$

Since $R(3) < D(3)$, the amount of water in the pipe is decreasing at time t=3 hours.

5.3a

42. If $f'(x) > 0$ for all x, and $f''(x) < 0$ for all x, which of the following could be a table of values for f?

(A)

x	$f(x)$
–1	4
0	3
1	1

(B)

x	$f(x)$
–1	4
0	4
1	4

(C)

x	$f(x)$
–1	4
0	5
1	6

(D)

x	$f(x)$
–1	4
0	5
1	7

(E)

x	$f(x)$
–1	4
0	6
1	7

5.3a

43. Let $x = ky^2 + 2$, where $k > 0$.

(a) Show that for all $k > 0$, the point $\left(4, \sqrt{\frac{2}{k}}\right)$ is on the graph of $x = ky^2 + 2$.

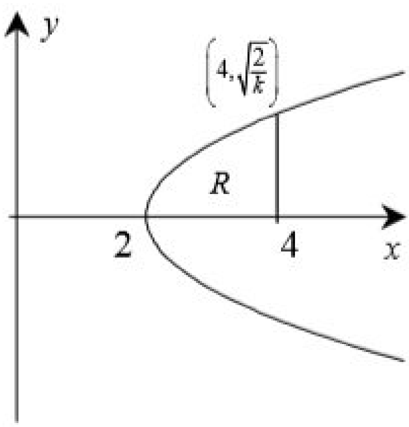
(b) Show that for all $k > 0$, the tangent line to the graph of $x = ky^2 + 2$ at the point $\left(4, \sqrt{\frac{2}{k}}\right)$ passes through the origin.

(c) Let R be the region in the first quadrant bounded by the x-axis, the graph of $x = ky^2 + 2$, and the line $x = 4$. Write an integral expression for the area of the region R and show that this area decreases as k increases.

Part A

1 point is earned for correctly substituting $x = 4$ and $y = \sqrt{\frac{2}{k}}$

$4 = k\left(\frac{2}{k}\right) + 2$
$4 = 4$



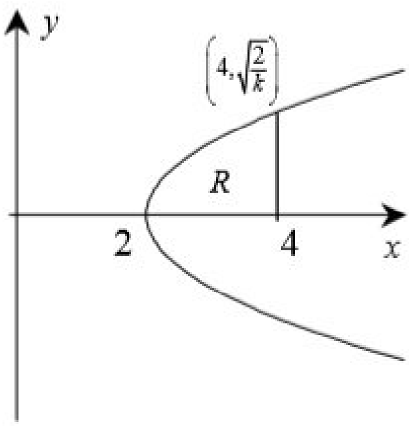
0	1

The student response earns all of the following points:

1 point is earned for correctly substituting $x = 4$ and $y = \sqrt{\frac{2}{k}}$

$4 = k\left(\frac{2}{k}\right) + 2$
$4 = 4$

5.3a



Part B

1 point is earned for the correct equation with $\frac{dy}{dx}$ or $\frac{dx}{dy}$

$x = ky^2 + 2$

$1 = 2ky\frac{dy}{dx}$

1 point is earned for the correct evaluation of $\frac{dy}{dx}$ or $\frac{dx}{dy}$ at $\left(4, \sqrt{\frac{2}{k}}\right)$

$\left.\frac{dy}{dx}\right|_{y=\sqrt{2/k}} = \frac{1}{2\sqrt{2k}}$

1 point is earned for the correct equation of tangent line or slope of line using $\left(4, \sqrt{2/k}\right)$ and (0,0)

the tangent line is

$y - \sqrt{\frac{2}{k}} = \frac{1}{2\sqrt{2k}}(x - 4)$

$y = \frac{1}{2\sqrt{2k}}x$ which contains (0,0)

or

slope of line through (0,0) and $\left(4, \sqrt{2/k}\right)$ is $\frac{\sqrt{2/k}}{4} = \frac{1}{2\sqrt{2k}}$

which is the same as the slope of the tangent line

1 point is earned for correctly observing (0,0) is a solution or observing the slopes are equal

5.3a

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for the correct equation with $\frac{dy}{dx}$ or $\frac{dx}{dy}$

$x = ky^2 + 2$
$1 = 2ky\frac{dy}{dx}$

1 point is earned for the correct evaluation of $\frac{dy}{dx}$ or $\frac{dx}{dy}$ at $\left(4, \sqrt{\frac{2}{k}}\right)$

$$\frac{dy}{dx}\Big|_{y=\sqrt{2/k}} = \frac{1}{2\sqrt{2k}}$$

1 point is earned for the correct equation of tangent line or slope of line using $\left(4, \sqrt{2/k}\right)$ and (0,0)

the tangent line is

$$y - \sqrt{\frac{2}{k}} = \frac{1}{2\sqrt{2k}}(x-4)$$

$y = \frac{1}{2\sqrt{2k}}x$ which contains (0,0)

or

slope of line through (0,0) and $\left(4, \sqrt{2/k}\right)$ is $\frac{\sqrt{2/k}}{4} = \frac{1}{2\sqrt{2k}}$

which is the same as the slope of the tangent line

1 point is earned for correctly observing (0,0) is a solution or observing the slopes are equal

Part C

3 points are earned for correctly expressing the area as a definite integral

$$A = \int_0^{\sqrt{2/k}} (4 - (ky^2 + 2))dy$$

or

$$A = \frac{1}{\sqrt{k}} \int_2^4 \sqrt{x-2}dx$$

5.3a

<-1> error in limits

<-1> considers twice the region

<-2> error in integrand

1 point is earned for correctly showing that A decreases using algebra, geometry, or calculus

$$A = \frac{4\sqrt{2}}{3}k^{-0.5}$$

$$\frac{dA}{dk} = -\frac{2\sqrt{2}}{3}k^{-1.5} < 0 \text{ for all } k > 0$$

thus the area decreases as k increases

0	1	2	3	4

The student response earns all of the following points:

3 points are earned for correctly expressing the area as a definite integral

$$A = \int_0^{\sqrt{2/k}} (4 - (ky^2 + 2))dy$$

or

$$A = \frac{1}{\sqrt{k}} \int_2^4 \sqrt{x-2}dx$$

<-1> error in limits

<-1> considers twice the region

<-2> error in integrand

1 point is earned for correctly showing that A decreases using algebra, geometry, or calculus

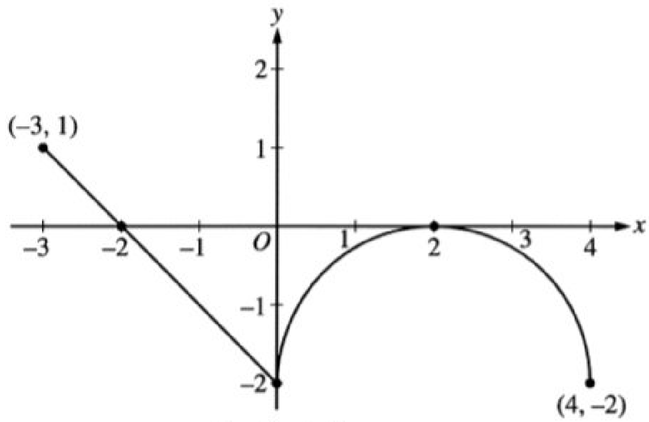
$$A = \frac{4\sqrt{2}}{3}k^{-0.5}$$

$$\frac{dA}{dk} = -\frac{2\sqrt{2}}{3}k^{-1.5} < 0 \text{ for all } k > 0$$

thus the area decreases as k increases

5.3a

44.



Graph of f'

Let f be a function defined on the closed interval $-3 \le x \le 4$ with $f(0) = 3$. The graph of f', the derivative of f, consists of one line segment and a semicircle, as shown above.

On what intervals, if any, is f increasing? Justify your answer.

Part A

1 point is earned for the correct interval

1 point is earned for the correct reason

The function f is increasing on $[-3, -2]$ since $f' > 0$ for $-3 \le x < -2$

0	1	2

The student response earns two of the following points:

1 point is earned for the correct interval

1 point is earned for the correct reason

The function f is increasing on $[-3, -2]$ since $f' > 0$ for $-3 \le x < -2$

45. What are all values of x for which the function f defined by $f(x) = (x^2 - 3)e^{-x}$ is increasing?

(A) There are no such values of x.

(B) $x < -1$ and $x > 3$

(C) $-3 < x < 1$

(D) $-1 < x < 3$

(E) All values of x

5.3a

46. The first derivative of the function g is given by $g'(x) = \cos(\pi x^2)$ for $-0.5 < x < 1.5$. On which of the following intervals is g decreasing?

(A) $-0.5 < x < 0$

(B) $0 < x < 1$

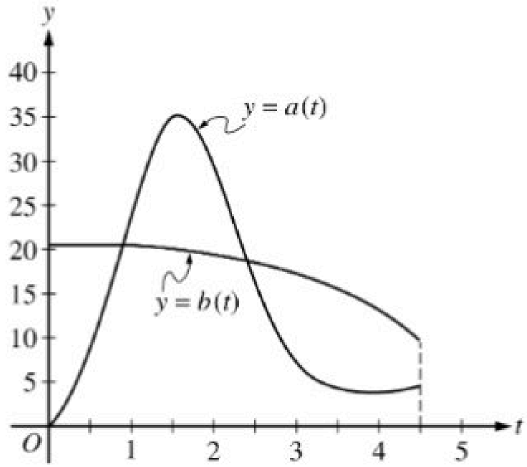
(C) $0.707 < x < 1.225$

(D) $1.225 < x < 1.414$

(E) $1.414 < x < 1.5$

5.3a

47.



During the time interval $0 \leq t \leq 4.5$ hours, water flows into tank A at a rate of $a\left(t\right) = \left(2t - 5\right) + 5e^{2\sin t}$ liters per hour. During the same time interval, water flows into tank B at a rate of $b(t)$ liters per hour. Both tanks are empty at time $t = 0$. The graphs of $y = a(t)$ and $y = b(t)$, shown in the figure above, intersect at $t = k$ and $t = 2.416$.

During the time interval $2.7 \leq t \leq 4.5$ hours, the rate at which water flows into tank B is modeled by $w\left(t\right) = 21 - \frac{30t}{(t-8)^2}$ liters per hour. Is the difference $w(t) - a(t)$ increasing or decreasing at time $t = 3.5$? Show the work that leads to your answer.

General

1 point is earned for: $w'(3.5) - a'(3.5) < 0$

1 point is earned for: conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $w'(3.5) - a'(3.5) < 0$

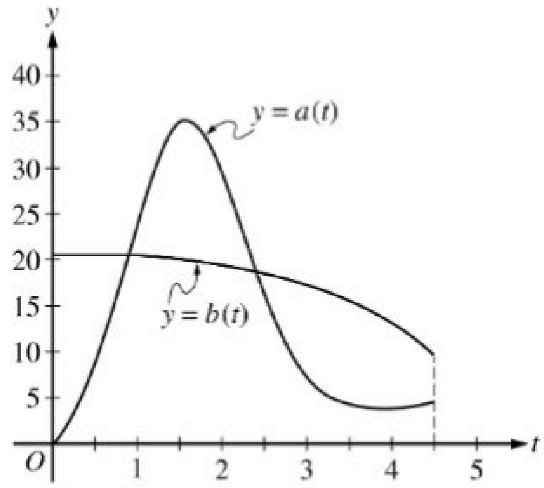
1 point is earned for: conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$

5.3a

48.



During the time interval $0 \leq t \leq 4.5$ hours, water flows into tank *A* at a rate of $a\left(t\right)=\left(2t-5\right)+5e^{2\sin t}$ liters per hour. During the same time interval, water flows into tank *B* at a rate of $b(t)$ liters per hour. Both tanks are empty at time $t = 0$. The graphs of $y = a(t)$ and $y = b(t)$, shown in the figure above, intersect at $t = k$ and $t = 2.416$.

During the time interval $2.7 \leq t \leq 4.5$ hours, the rate at which water flows into tank *B* is modeled by $w\left(t\right)=21-\frac{30t}{\left(t-8\right)^{2}}$ liters per hour. Is the difference $w(t) - a(t)$ increasing or decreasing at time $t = 3.5$? Show the work that leads to your answer.

General

1 point is earned for: $w'(3.5) - a'(3.5) < 0$

1 point is earned for: conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $w'(3.5) - a'(3.5) < 0$

1 point is earned for: conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

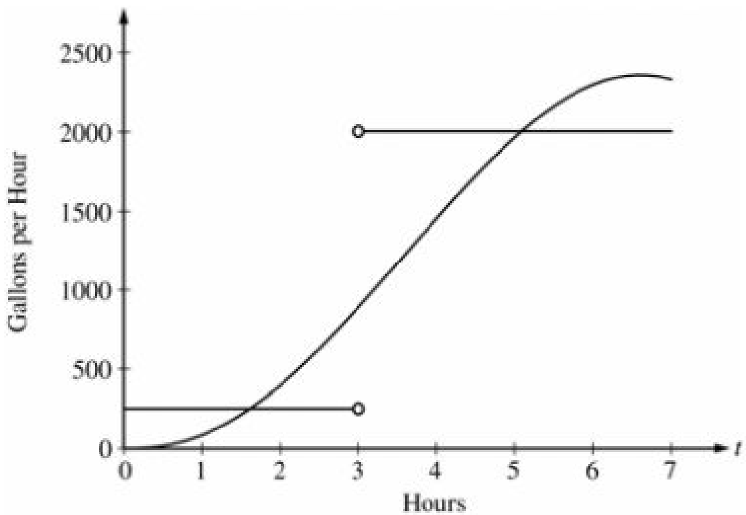
The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

5.3a

49. The amount of water in a storage tank, in gallons, is modeled by a continuous function on the time interval $0 \leq t \leq 7$, where t is measured in hours. In this model, rates are given as follows:

i. The rate at which water enters the tank is $f(t) = 100t^2 \sin\left(\sqrt{t}\right)$ gallons per hour for $0 \leq t \leq 7$.

ii. The rate at which water enters the tank is $g(t) = \begin{cases} 250 & \text{for } 0 \leq t < 3 \\ 2000 & \text{for } 3 < t \leq 7 \end{cases}$ gallons per hour.



The graphs of *f* and *g*, which intersect at *t* = *1.617* and *t* = *5.076*, are shown in the figure above. At time *t* = *0*, the amount of water in the tank is 5000 gallons.

For $0 \leq t \leq 7$, find the time intervals during which the amount of water in the tank is decreasing. Give a reason for each answer.

Part B

1 point is earned for the intervals

1 point is earned for the reason

The amount of water in the tank is decreasing on the intervals $0 \leq t \leq 1.617$ and $3 \leq t \leq 5.076$ because *f(t)* < *g(t)* for $0 \leq t < 1.617$ and $3 < t < 5.076$.

0	1	2

The student response earns all of the following points:

1 point is earned for the intervals

1 point is earned for the reason

5.3a

The amount of water in the tank is decreasing on the intervals $0 \leq t \leq 1.617$ and $3 \leq t \leq 5.076$ because *f(t)* < *g(t)* for $0 \leq t < 1.617$ and $3 < t < 5.076$.

50. At $x = 0$, which of the following is true of the function *f* defined by $f(x) = x^2 + e^{-2x}$?

(A) *f* is increasing.

(B) *f* is decreasing. ✓

(C) *f* is discontinuous.

(D) *f* has a relative minimum.

(E) *f* has a relative maximum.

51. Let *f* be the function with derivative given by $f'(x) = x^2 - \frac{2}{x}$. On which of the following intervals is *f* decreasing?

(A) $(-\infty, -1]$ only

(B) $(-\infty, 0)$

(C) $[-1, 0)$ only

(D) $\left(0, \sqrt[3]{2}\right]$ ✓

(E) $\left[\sqrt[3]{2}, \infty\right)$

52. Let *f* be the function given by $f(x) = \frac{kx}{x^2+1}$, where *k* is a constant. For what values of *k*, if any, is *f* strictly decreasing on the interval (-1, 1)?

(A) $k < 0$ ✓

(B) $k = 0$

(C) $k > 0$

(D) $k > 1$ only

(E) There are no such values of *k*.

53. If $f(x) = x + \frac{1}{x}$, then the set of values for which *f* increases is

(A) $(-\infty, -1] \cup [1, \infty)$ ✓

(B) $[-1, 1]$

(C) $(-\infty, \infty)$

(D) $(0, \infty)$

(E) $(-\infty, 0) \cup (0, \infty)$

5.3a

54. The first derivative of the function *f* is given by $f'(x) = sin(x^2)$. At which of the following values of *x* does *f* have a local minimum?

(A) 2.507 ✓

(B) 2.171

(C) 1.772

(D) 1.253

(E) 0

55. If $f(x) = \ln x/x$, for all $x > 0$, which of the following is true?

(A) *f* is increasing for all *x* greater than 0.

(B) *f* is increasing for all *x* greater than 1.

(C) *f* is decreasing for all *x* between 0 and 1.

(D) *f* is decreasing for all *x* between 1 and e.

(E) *f* is decreasing for all *x* greater than e. ✓

56. Let *f* be the function given by $f(x) = 3 - 2x$. If *g* is a function with derivative given by $g'(x) = f(x)f'(x)(x - 3)$, on what intervals is *g* increasing?

(A) $\left(-\infty, \frac{3}{2}\right]$ and $[3, \infty)$ ✓

(B) $\left(-\infty, \frac{3}{2}\right]$ only

(C) $\left[\frac{3}{2}, 3\right]$ only

(D) $\left[\frac{3}{2}, \infty\right)$

(E) $[3, \infty)$ only

Answer A

57. The function *f* given by $f(x) = x^3 + 12x - 24$ is

(A) increasing for $x<-2$, decreasing for $-2<x<2$, increasing for $x>2$

(B) decreasing for $x<0$, increasing for $x>0$

(C) increasing for all *x* ✓

(D) decreasing for all *x*

(E) decreasing for $x<-2$, increasing for $-2<x<2$, decreasing for x>2

5.3a

58. The number of gallons, $P(t)$, of a pollutant in a lake changes at the rte $P'(t) = 1 - 3e^{-0.2\sqrt{t}}$ gallons per day, where t is measured in days. There are 50 gallons of the pollutant in the lake at time $t = 0$. The lake is considered to be safe when it contains 40 gallons or less of pollutant.

Is the amount of pollutant increasing at time $t = 9$? Why or why not?

Part A

1 point is earned for the answer with reason

$p\prime(9) = 1 - 3e^{-0.6} = -0.646 < 0$ so the amount is not increasing at this time.

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer with reason

$p\prime(9) = 1 - 3e^{-0.6} = -0.646 < 0$ so the amount is not increasing at this time.

59. If $f(x) = x^2e^x$, then the graph of f is decreasing for all x such that

(A) $x<-2$

(B) $-2<x<0$ ✓

(C) $x>-2$

(D) $x<0$

(E) $x>0$

60. Let f be the function given by $f(x) = 300x - x^3$. On which of the following intervals is the function f increasing?

(A) $(-\infty, 10]$ and $[10, \infty)$

(B) $[-10, 10]$ ✓

(C) [0, 10] only

(D) $[0, 10\sqrt{3}]$ only

(E) $[0, \infty)$

5.3a

61. The function *f* is given by *f(x)*=x^4+x^2 -2. On which of the following intervals is *f* increasing?

(A) $(-\frac{1}{\sqrt{2}}, \infty)$

(B) $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

(C) $(0, \infty)$ ✓

(D) $(-\infty, 0)$

(E) $(-\infty, -\frac{1}{\sqrt{2}})$

62. The function $y = g(x)$ is differentiable and increasing for all real numbers. On what intervals is the function $y = g\left(x^3 - 6x^2\right)$ increasing?

(A) $(-\infty, 0]$ and $[4, \infty)$ only ✓

(B) $[0, 4]$ only

(C) $[2, \infty)$ only

(D) $[6, \infty)$ only

(E) $(-\infty, \infty)$

5.3a

63. Consider the function f defined by $f(x)=e^x \cos x$ with domain $[0,2\pi]$.

Find the intervals on which f is increasing.

Part B

1 point is earned for positive intervals

1 point is earned for negative intervals

1 point is **deducted** for intervals outside of domain

Increasing on $\left[0, \frac{\pi}{4}\right], \left[\frac{5\pi}{4}, 2\pi\right]$

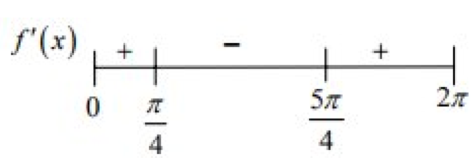
0	1	2

The student response earns two of the following points:

1 point is earned for positive intervals

1 point is earned for negative intervals

1 point is **deducted** for intervals outside of domain



Increasing on $\left[0, \frac{\pi}{4}\right], \left[\frac{5\pi}{4}, 2\pi\right]$

5.3a

64. What are all values of x for which the function f defined by $f(x) = x^3 + 3x^2 - 9x + 7$ is increasing?

(A) $-3<x<1$

(B) $-1<x<1$

(C) $x<-3$ or $x>1$

(D) $x<-1$ or $x>3$

(E) All real numbers

Answer C

65. Let f be the function defined by $f(x)=xe^{1-x}$ for all real numbers x.

Find each interval on which f is increasing.

Part A

1 point is earned for derivative

1 point is earned for answer

$f'(x) = xe^{1-x}(-1) + e^{1-x} = (1-x)e^{1-x}$

f increases on $(-\infty, 1]$

0	1	2

The student response earns two of the following points:

1 point is earned for derivative

1 point is earned for answer

$f'(x) = xe^{1-x}(-1) + e^{1-x} = (1-x)e^{1-x}$

f increases on $(-\infty, 1]$

5.3a

66. Let f be the function with derivative given by $f'(x) = \frac{-2x}{(1+x^2)^2}$. On what interval is f decreasing?

(A) [0, ∞) only
(B) (- ∞, 0] only
(C) [-1/√3 , 1/√3] only
(D) (- ∞, ∞)
(E) There is no such interval

5.3a

67. Let f be the function defined by $f(x)=3x^5-5x^3+2$.
On what intervals is f increasing?

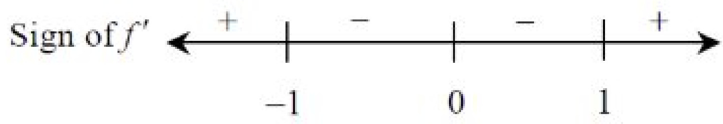
Part A

1 point is earned for f'(x)

1 point is earned for analyze sign of f'(x) or explicitly setting students f'(x)>0

1 point is earned for answer

$f\,\prime(\,x\,) = 15\,x^4 - 15\,x^2 = 15\,x^2(x^2-1)$



Answer: f is increasing on the intervals $(-\infty, 1]$ and $[1, \infty)$

0	1	2	3

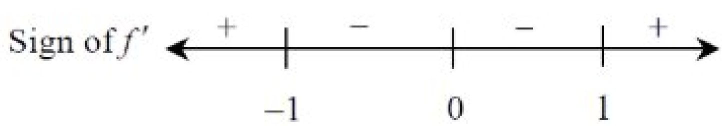
The student response earns three of the following points:

1 point is earned for f'(x)

1 point is earned for analyze sign of f'(x) or explicitly setting students f'(x)>0

1 point is earned for answer

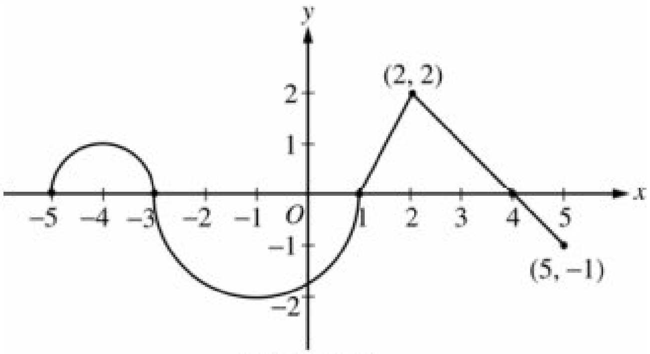
$f\,\prime(\,x\,) = 15\,x^4 - 15\,x^2 = 15\,x^2(x^2-1)$



Answer: f is increasing on the intervals $(-\infty, 1]$ and $[1, \infty)$

5.3a

68.



Graph of f'

Let f be a function defined on the closed interval $-5 \le x \le 5$ with $f(1)=3$. The graph of f' the derivative of f, consists of two semicircles and two line segments, as shown above.

Find all intervals on which the graph of f is concave up and also has positive slope. Explain your reasoning.

Part C

1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of f is concave up with positive slope where f' is increasing and positive: $-5<x<-4$ and $1<x<2$.

0	1	2

The student response earns two of the following points:

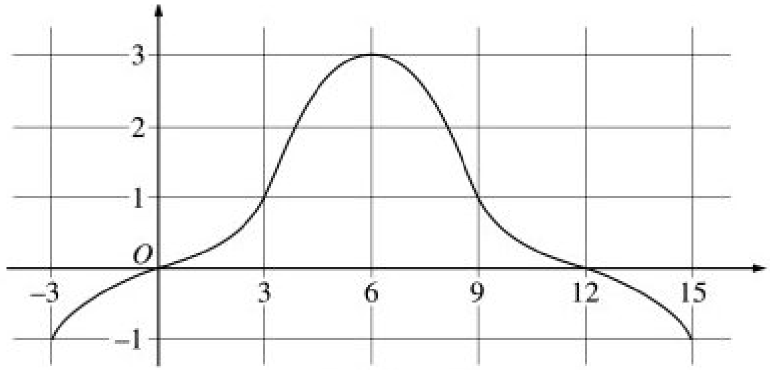
1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of f is concave up with positive slope where f' is increasing and positive: $-5<x<-4$ and $1<x<2$.

5.3a

69.



Graph of f

The graph of a differentiable function f on the closed interval [-3, 15] is shown in the figure above. The graph of f has a horizontal tangent line at $x = 6$. Let $g(x) = 5 + \int_6^x f(t)dt$ for $-3 \leq x \leq 15$.

On what intervals is g decreasing? Justify your answer.

Part B

1 point is earned for: [-3, 0]

1 point is earned for: [12, 15]

1 point is earned for the justification

g is decreasing on [-3, 0] and [12, 15] since $g'(x) = f(x) < 0$ for $x < 0$ and $x > 12$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: [-3, 0]

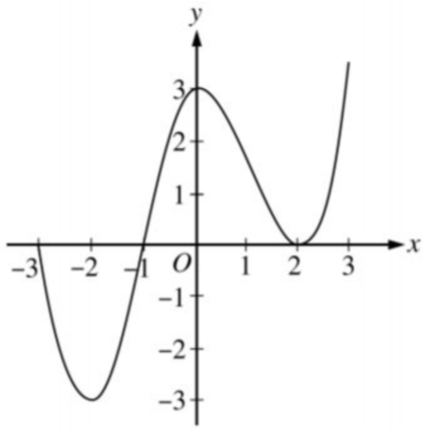
1 point is earned for: [12, 15]

1 point is earned for the justification

g is decreasing on [-3, 0] and [12, 15] since $g'(x) = f(x) < 0$ for $x < 0$ and $x > 12$.

5.3a

70.



Graph of f'

The graph of f', the derivative of the function f, is shown above for $-3 \le x \le 3$.

On what intervals is f increasing?

(A) $[-3, -1]$ only

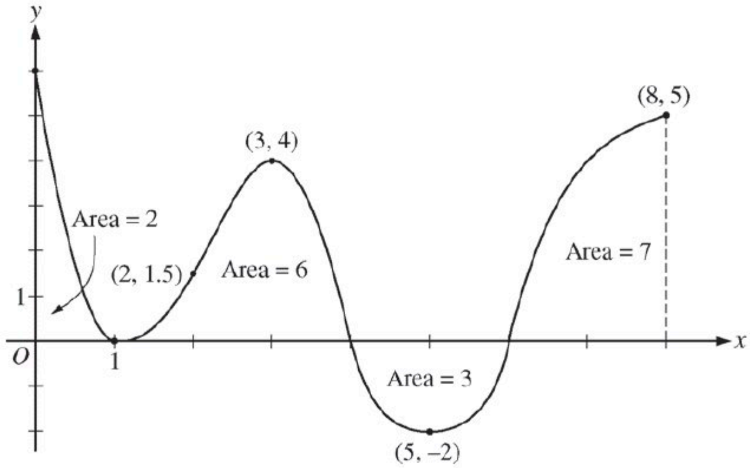
(B) $[-1, 3]$

(C) $[-2, 0]$ and $[2, 3]$

(D) $[-3, -1]$ and $[1, 3]$

5.3a

71.



Graph of f'

The figure above shows the graph of f' the derivative of a twice-differentiable function f, on the closed interval $0 \leq x \leq 8$. The graph of f' has horizontal tangent lines at x=1, x=3, and x=5. The areas of the regions between the graph of f' and the x-axis are labeled in the figure. The function f is defined for all real numbers and satisfied $f(8)=4$.

On what open intervals contained in $0<x<8$ is the graph of f both concave down and increasing? Explain your reasoning.

Part C

1 point is earned for the answer

1 point is earned for the explanation

The graph of f is concave down and increasing on $0<x<1$ and $3<x<4$, because f' is decreasing and positive on these intervals.

0	1	2

The student response earns all of the following points:

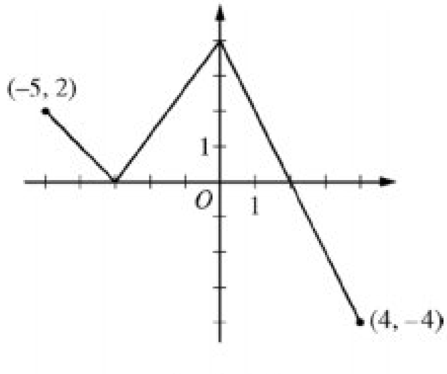
1 point is earned for the answer

1 point is earned for the explanation

The graph of f is concave down and increasing on $0<x<1$ and $3<x<4$, because f' is decreasing and positive on these intervals.

5.3a

72.



Graph of f

The function f is defined on the closed interval [-5, 4]. The graph of f consists of three line segments and is shown in the figure above. Let g be the function defined by $g(x) = \int_{-3}^{x} f(t)dt$.

On what open intervals contained in $-5<x<4$ is the graph of g both increasing and concave down? Give a reason for your answer.

Part B

1 point is earned for the answer

1 point is earned for the reason

$g'(x)=f(x)$ The graph of g is increasing and concave down on the intervals $-5<x<-3$ and $0<x<2$ because $g'=f$ is positive and decreasing on these intervals.

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

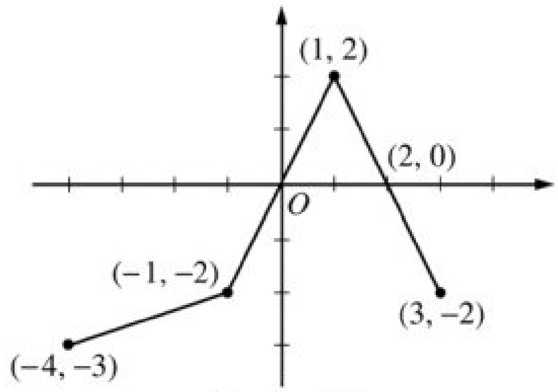
1 point is earned for the reason

$g'(x)=f(x)$

The graph of g is increasing and concave down on the intervals $-5<x<-3$ and $0<x<2$ because $g'=f$ is positive and decreasing on these intervals.

5.3a

73.



Graph of f

The graph of the function f above consists of three line segments.

For the function h defined in part (c), find all intervals on which h is decreasing. Explain your reasoning.

Part D

1 point is earned for interval

1 point is earned for reason

h is decreasing on [0,2]

h' = -f<0 when f>0

0	1	2

The student response earns all of the following points:

1 point is earned for interval

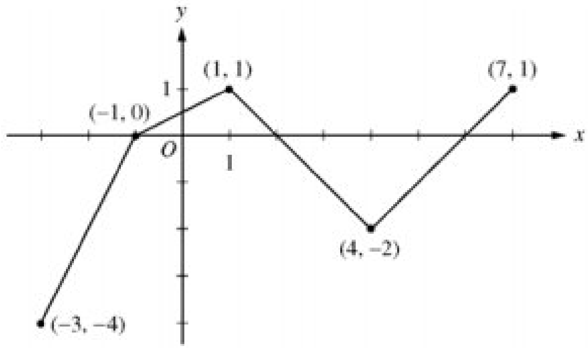
1 point is earned for reason

h is decreasing on [0,2]

h' = -f<0 when f>0

5.3a

74.



Graph of g'

Let g be a continuous function with $g(2) = 5$. The graph of the piecewise-linear function g', the derivative of g, is shown above for $-3 \le x \le 7$.

Find the x-coordinate of all points of inflection of the graph of $y = g(x)$ for $-3 < x < 7$. Justify your answer.

Part A

1 point is earned for x=values

1 point is earned for the justification

g' changes from increasing to decreasing at x=1;

g' changes from decreasing to increasing at x=4. Points of inflection for the graph of y=g(x) occur at x=1 and x=4.

0	1	2

The student response earns all of the following points:

1 point is earned for x=values

1 point is earned for the justification

g' changes from increasing to decreasing at x=1;

g' changes from decreasing to increasing at x=4.

Points of inflection for the graph of y=g(x) occur at x=1 and x=4.

5.3a

75.

x	-4	-3	-2	-1	0	1	2	3	4
$g'(x)$	2	3	0	-3	-2	-1	0	3	2

The derivative g' of a function g is continuous and has exactly two zeros. Selected values of g' are given in the table above. If the domain of g is the set of all real numbers, then g is decreasing on which of the following intervals?

(A) $-2 \leq x \leq 2$ only ✓

(B) $-1 \leq x \leq 1$ only

(C) $x \geq -2$

(D) $x \geq 2$ only

(E) $x \leq -2$ or $x \geq 2$

76. If $f'(x)$ and $g'(x)$ exist and $f'(x) > g'(x)$ for all real x, then the graph of $y = f(x)$ and the graph of $y = g(x)$

(A) intersect exactly once.

(B) intersect no more than once. ✓

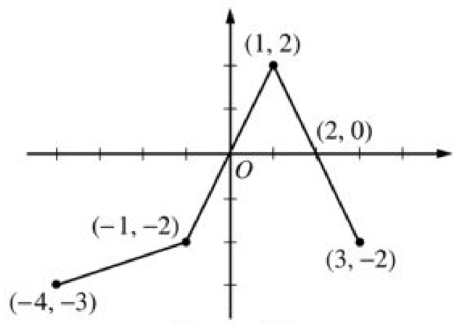
(C) do not intersect.

(D) could intersect more than once.

(E) have a common tangent at each point of intersection.

5.3a

77.



Graph of f

The graph of the function f above consists of three line segments.

(a) Let g be the function given by $g(x) = \int_{-4}^{x} f(t)dt$. For each of $g(-1)$, $g'(-1)$, and $g''(-1)$, find the value or state that it does not exist.

(b) For the function g defined in part (a), find the x-coordinate of each point of inflection of the graph of g on the open interval $-4 < x < 3$. Explain your reasoning.

(c) Let h be the function given by $h(x) = \int_{x}^{3} f(t)dt$. Find all values of x in the closed interval $-4 \le x \le 3$ for which $h(x) = 0$.

(d) For the function h defined in part (c), find all intervals on which h is decreasing. Explain your reasoning.

Part A

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

1 point is earned for $g''(-1)$

$$g(-1) = \int_{-4}^{-1} f(t)dt = -\tfrac{1}{2}(3)(5) = -\frac{15}{2}$$

$g'(-1) = f(-1) = -2$

$g''(-1)$ does not exist because f is not differentiable at $x = -1$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

5.3a

1 point is earned for $g''(-1)$

$$g(-1)=\int_{-4}^{-1} f(t)dt=-\frac{1}{2}(3)(5)=-\frac{15}{2}$$

$g'(-1)=f(-1)=-2$

$g''(-1)$ does not exist because f is not differentiable at $x=-1$.

Part B

1 point is earned for $x = 1$ (only)

1 point is earned for the reason

$x = 1$

$g'=f$ changes from increasing to decreasing at $x = 1$.

0	1	2

The student response earns all of the following points:

1 point is earned for $x = 1$ (only)

1 point is earned for the reason

$x = 1$

$g'=f$ changes from increasing to decreasing at $x = 1$.

Part C

2 points are earned for the correct values

<-1> each missing or extra value

$x = -1, 1, 3$

0	1	2

The student response earns all of the following points:

2 points are earned for the correct values

5.3a

<-1> each missing or extra value

$x = -1, 1, 3$

Part D

1 point is earned for interval

1 point is earned for reason

h is decreasing on [0, 2]

$h' = -f < 0$ when $f > 0$

0	1	2

The student response earns all of the following points:

1 point is earned for interval

1 point is earned for reason

h is decreasing on [0, 2]

$h' = -f < 0$ when $f > 0$

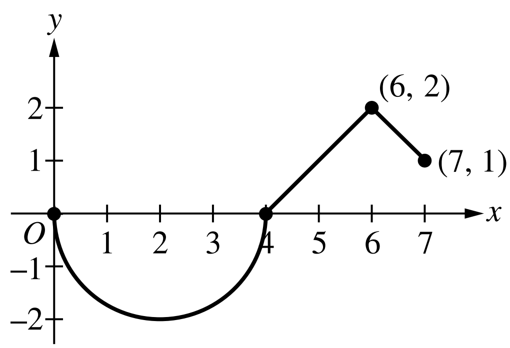
5.3a

78. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f'

Let f be a differentiable function with $f(4) = 3$. On the interval $0 \leq x \leq 7$, the graph of f', the derivative of f, consists of a semicircle and two line segments, as shown in the figure.

(a) Find $f(0)$ and $f(5)$.

(b) Find the x-coordinates of all points of inflection of the graph of f for $0 < x < 7$. Justify your answer.

(c) Let g be the function defined by $g(x) = f(x) - x$. On what intervals, if any, is g decreasing for $0 \leq x \leq 7$? Show the analysis that leads to your answer.

(d) For the function g defined in part (c), find the absolute minimum value on the interval $0 \leq x \leq 7$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with answers of only $f(0) = \pm 2\pi$, or only $f(5) = \frac{1}{2}$, or both earns 1 of the 3 points.

A response displaying $f(5) = \frac{7}{2}$ and a missing or incorrect value for $f(0)$ earns 2 of the 3 points.

The second and third points can be earned in either order.

Read unlabeled values from left to right and from top to bottom as $f(0)$ and $f(5)$. A single value must be labeled as either $f(0)$ or $f(5)$ in order to earn any points.

5.3a

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Area of either region – OR – $\int_0^4 f'(x)\,dx$ – OR – $\int_4^5 f'(x)\,dx$
- [ ] $f(0)$
- [ ] $f(5)$

Solution:

$$f(0) = f(4) + \int_4^0 f'(x)\,dx = 3 - \int_0^4 f'(x)\,dx = 3 + 2\pi$$

$$f(5) = f(4) + \int_4^5 f'(x)\,dx = 3 + \frac{1}{2} = \frac{7}{2}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that gives only one of $x = 2$ or $x = 6$, along with a correct justification, earns 1 of the 2 points.

A response that claims that there is a point of inflection at any value other than $x = 2$ or $x = 6$ earns neither point.

To earn the second point a response must use correct reasoning based on the graph of f'. Examples of correct reasoning include:

· Correctly discussing the signs of the slopes of the graph of f'

· Citing $x = 2$ and $x = 6$ as the locations of local extrema on the graph of f'

Examples of reasoning not (sufficiently) connected to the graph of f' include:

· Reasoning based on sign changes in f'' unless the connection is made between the sign of f'' and the slopes of the graph of f'

· Reasoning based only on the concavity of the graph of f

The second point cannot be earned by use of vague or undefined terms such as "it" or "the function" or "the derivative."

Responses that report inflection points as ordered pairs must report the points $(2,\ 3 + \pi)$ and $(6,\ 5)$ in order to earn the first point. If the y-coordinates are reported incorrectly, the response remains eligible for the second point.

5.3a

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Answer
- [ ] Justification

Solution:

The graph of f has a point of inflection at each of $x = 2$ and $x = 6$, because $f'(x)$ changes from decreasing to increasing at $x = 2$ and from increasing to decreasing at $x = 6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned for $f'(x) \leq 1$ or the equivalent, in words or symbols.

Endpoints do not need to be included in the interval to be eligible for the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $g'(x) = f'(x) - 1$
- [ ] Interval with reason

Solution:

$g'(x) = f'(x) - 1$

$f'(x) - 1 \leq 0 \;\Rightarrow\; f'(x) \leq 1$

The graph of g is decreasing on the interval $0 \leq x \leq 5$ because $g'(x) \leq 0$ on this interval.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A justification that uses a local argument, such as "g' changes from negative to positive (or g changes from decreasing to increasing) at $x = 5$" must also state that $x = 5$ is the only critical point.

5.3a

If $g'(x) = 0$ (or equivalent) is not declared explicitly, a response that isolates $x = 5$ as the only critical number belonging to $(0,\ 7)$ earns the first point.

A response that imports $g'(x) = f'(x)$ from part (c) is eligible for the first point but not the second.

· In this case, consideration of $x = 4$ as the only critical number on $(0,\ 7)$ earns the first point.

Solution using Candidates Test:

$g'(x) = f'(x) - 1 = 0 \ \Rightarrow \ x = 5,\ \ x = 7$

x	$g(x)$
0	$3 + 2\pi$
5	$-\frac{3}{2}$
7	$-\frac{1}{2}$

The absolute minimum value of g on the interval $0 \le x \le 7$ is $-\frac{3}{2}$.

When using a Candidates Test, a response may import an incorrect value of $f(0) = g(0) > -\frac{3}{2}$ from part (a). The second point can only be earned for an answer of $-\frac{3}{2}$.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Considers $g'(x) = 0$
- [ ] Answer with justification

Solution:

g is continuous, $g'(x) < 0$ for $0 < x < 5$, and $g'(x) > 0$ for $5 < x < 7$.

Therefore, the absolute minimum occurs at $x = 5$, and $g(5) = f(5) - 5 = \frac{7}{2} - 5 = -\frac{3}{2}$ is the absolute minimum value of g.

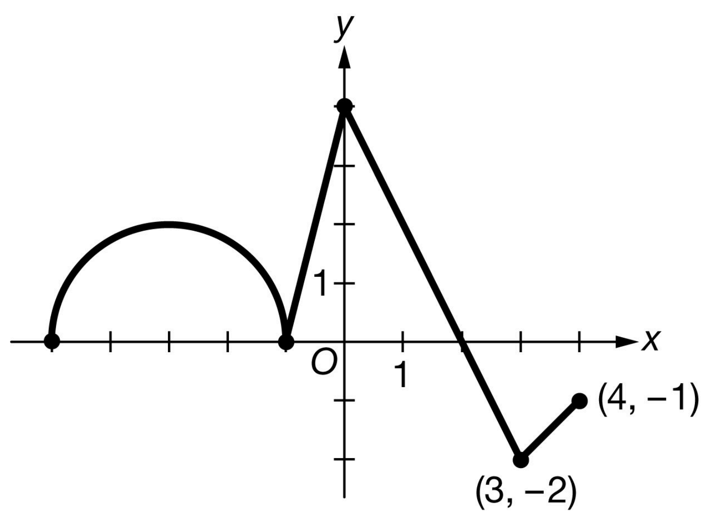
5.3a

79. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g'

Let g be a function that is defined on the closed interval $-5 \le x \le 4$ with $g(0) = 5$. The graph of g', the derivative of g, consists of a semicircle and three line segments, as shown above.

(a) On what intervals, if any, is g decreasing? Give a reason for your answer.

(b) Find the average rate of change of g on the interval $[-3, 3]$. Is there a value c, $-3 < c < 3$, for which $g'(c)$ is equal to the average rate of change of g on the interval $[-3, 3]$? Justify your answer.

(c) Find the x-coordinate of each point of inflection of the graph of g on the open interval $(-5, 4)$. Justify your answer.

(d) Let $H(x) = e^{3x} g(x)$. Find the value of $H'(0)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response does not have to say that g is continuous to earn the point.

A response with correct endpoints can use open, closed, or half-open interval notation to earn the point.

5.3a

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

g is decreasing on the interval $[2,\ 4]$ because g is continuous and $g'(x) < 0$ on $(2,\ 4]$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for declaring that the average rate of change is either $\frac{g(3)-g(-3)}{3-(-3)}$ or $\frac{\int_{-3}^{3} g'(x)\, dx}{6}$. The difference quotient and the definite integral must contain $\frac{1}{6}$ to earn the point.

The second point is earned for finding the value of the average rate of change, $\frac{\pi+5}{6}$ or equivalent.

The third point is for stating that g is continuous because g is differentiable (on $[-3,\ 3]$). Stating that g is both differentiable and continuous on the interval is not enough to earn this point.

The fourth point is for applying either the Mean Value Theorem (MVT) and concluding "yes," or for applying the Intermediate Value Theorem (IVT) and concluding "yes." Neither theorem needs to be named if the appropriate conditions are checked and the correct conclusion is reached.

If a response does not appropriately cite/apply the MVT or IVT, the response does not earn the fourth point.

0	1	2	3	4

The student response accurately includes **all four** of the criteria below.

- [ ] Difference quotient or definite integral
- [ ] Average rate of change
- [ ] Differentiable implies continuous
- [ ] Conclusion using Mean Value Theorem
  – OR –
  Conclusion using Intermediate Value Theorem

5.3a

Solution:

Average rate of change $= \frac{g(3)-g(-3)}{3-(-3)} = \frac{\int_{-3}^{3} g'(x)\,dx}{6}$

$= \frac{\pi+6-1}{6} = \frac{\pi+5}{6}$

g is differentiable on $[-3,\ 3]$. $\Rightarrow$ g is continuous on $[-3,\ 3]$.

The Mean Value Theorem guarantees that there exists a value of c in the interval $(-3,\ 3)$ such that $g'(c)$ is equal to the average rate of change of g on the interval $[-3,\ 3]$.

– OR –

g' is continuous on $[-3,\ 3]$.

The range of g' on the interval $[-3,\ 3]$ is $-2 \le g'(x) \le 4$. Because $-2 < \frac{\pi+5}{6} < 4$, there exists a value of c in the interval $(-3,\ 3)$ such that $g'(c)$ is equal to the average rate of change of g on the interval $[-3,\ 3]$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must identify all four answers.

A response that provides a correct identification and justification for just two (or three) of the four answers earns 1 of the 2 points.

The second point can be earned by correctly discussing the signs of the slopes of the graph of g'.

The second point cannot be earned by the use of vague or undefined terms such as “it” or “the graph” or “the derivative.”

The second point cannot be earned by only discussing the concavity of the graph of g.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answers
- ☐ Justification

Solution:

5.3a

The graph of g will have a point of inflection where the graph of g' changes from increasing to decreasing or from decreasing to increasing. This occurs at $x = -3,\ \ x = -1,\ \ x = 0$, and $x = 3$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for using the product rule and chain rule to correctly find $H'(x)$ or $H'(0)$.

The second point is earned for correctly using the values of $g(0)$ and $g'(0)$ in the calculation of $H'(0)$. This point can be earned without the first point if there is a single error in finding $H'(x)$.

A response of $3 \cdot 5 + 4$ is the minimal response that earns both points.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $H'(x)$
- [ ] Answer

Solution:

$H'(x) = 3e^{3x} \cdot g(x) + e^{3x} \cdot g'(x)$

$H'(0) = 3 \cdot g(0) + g'(0) = 3 \cdot 5 + 4 = 19$

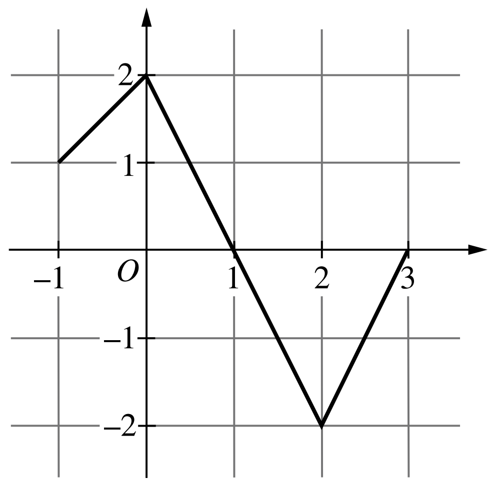
5.3a

80. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

Let f be the continuous function defined on the closed interval $[-1, 3]$ whose graph, consisting of three line segments, is shown. Let g be the function given by $g(x) = \int_0^x f(t)\,dt$.

(a) On what intervals, if any, is g decreasing? Give a reason for your answer.

(b) Find $g''\left(-\frac{1}{2}\right)$, or explain why it does not exist.

(c) What is the absolute minimum value of g on the closed interval $[-1, 3]$? Justify your answer.

(d) Let h be the function defined by $h(x) = \ln(g(x))$ for $0 < x < 2$. Find $h'\left(\frac{3}{2}\right)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $g'(x) = f(x)$ in any part of the problem.

· A response of $g''(x) = f'(x)$ or $g''\left(-\frac{1}{2}\right) = f'\left(-\frac{1}{2}\right)$ in part (b) is sufficient to earn the first point.

The interval $[1,\ 3]$ can be reported $(1, 3)$, $(1, 3]$, or $(1, 3]$.

5.3a

The second point can be earned without the first by presenting a correct interval and a statement that $f\,(x) < 0$ on this interval.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] $g'\,(x) = f\,(x)$ in any part of response
- [ ] Answer with reason

Solution:

$g'(x) = f\,(x) < 0$ on the open interval $(1, 3)$. Therefore, g is decreasing on the closed interval $[1, 3]$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An answer of 1 with no supporting work earns the point.

Any work shown must be correct in order to earn the point.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g''(x) = f'(x)$

$g''\left(-\frac{1}{2}\right) = f'\left(-\frac{1}{2}\right) = 1$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for any presented consideration of $x = -1$ and $x = 3$.

A response that finds only that $g\,(-1) = -1.5$ and declares this as the minimum earns only the second point.

5.3a

A response that evaluates $g\left(x\right)$ at any values of x in addition to $x=-1,\ \ x=1$, and $x=3$ does not earn the third point.

To earn the third point, a response must:

· have earned the first point,

· have no errors in the evaluation of $g\left(-1\right)$ and $g\left(3\right)$,

· conclude that $g\left(3\right)>g\left(-1\right)$, and

· either evaluate $g\left(1\right)$ as part of the Candidates Test or include a global argument that specifically addresses the behavior of $g\left(x\right)$ at $x=1$ (e.g., "$g'\left(x\right)=f\left(x\right)$ changes signs from positive to negative at $x=1$" or "$g'(1)=f\left(1\right)=0$ and there is a maximum at $x=1$").

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Considers $x=-1$ and $x=3$
- [ ] Answer
- [ ] Justification

Solution:

Because $g'(x)=f(x)>0$ for $-1<x<1$ and $g'(x)=f(x)<0$ for $1<x<3$, the minimum value of g must occur at either $x=-1$ or $x=3$.

$$g\left(-1\right)=\int_{0}^{-1}f\left(t\right)dt=-\int_{-1}^{0}f\left(t\right)dt=-1.5$$

$$g\left(3\right)=\int_{0}^{3}f\left(t\right)dt=-1$$

Therefore, the minimum value of g is $g\left(-1\right)=-1.5$.

Alternate solution using Candidates Test:

$$g'(x)=f(x)=0\Rightarrow x=1$$

$$g\left(-1\right)=\int_{0}^{-1}f(t)dt=-\int_{-1}^{0}f(t)dt=-1.5$$

$$g\left(1\right)=\int_{0}^{1}f\left(t\right)\,dt=1$$

5.3a

$g(3) = \int_0^3 f(t)\, dt = -1$

Therefore, the minimum value of g is $g(-1) = -1.5$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $h'(x) = \frac{1}{g(x)} \cdot g'(x)$, $h'\left(\frac{3}{2}\right) = \frac{1}{g\left(\frac{3}{2}\right)} \cdot g'\left(\frac{3}{2}\right)$, or $\frac{1}{\frac{3}{4}} \cdot (-1)$.

To earn the second point a response must use the correct value of either $g\left(\frac{3}{2}\right)$ or $g'\left(\frac{3}{2}\right)$ (or both) in an expression for $h'\left(\frac{3}{2}\right)$.

The minimal answer that earns all 3 points is $\frac{1}{\frac{3}{4}} \cdot (-1)$.

An answer of $-\frac{4}{3}$ with incorrect or without supporting work does not earn any points.

The third point is earned only for $-\frac{4}{3}$ or equivalent.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ $h'(x)$
- ☐ Correct value of $g\left(\frac{3}{2}\right)$ or $g'\left(\frac{3}{2}\right)$
- ☐ Answer

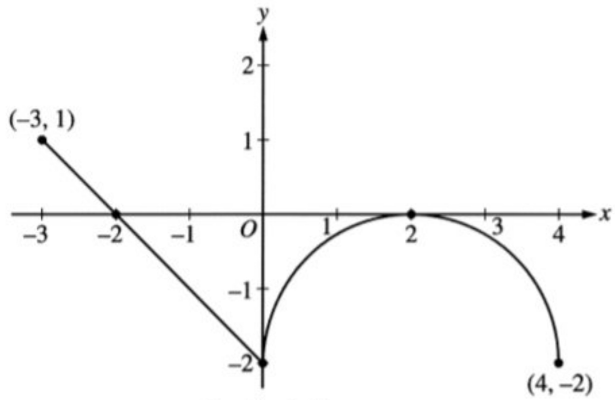
Solution:

$$h'(x) = \frac{1}{g(x)} \cdot g'(x)$$

$$h'\left(\frac{3}{2}\right) = \frac{1}{g\left(\frac{3}{2}\right)} \cdot g'\left(\frac{3}{2}\right) = \frac{1}{\frac{3}{4}} \cdot (-1) = -\frac{4}{3}$$

5.3a

81.



Graph of f'

Let f be a function defined on the closed interval $-3 \leq x \leq 4$ with $f(0) = 3$. The graph of f', the derivative of f, consists of one line segment and a semicircle, as shown above.

(a) On what intervals, if any, is f increasing? Justify your answer.

(b) Find the x-coordinate of each point of inflection of the graph of f on the open interval $-3 < x < 4$. Justify your answer.

(c) Find an equation for the line tangent to the graph of f at the point (0, 3).

(d) Find $f(-3)$ and $f(4)$. Show the work that leads to your answers.

Part A

1 point is earned for the correct interval

1 point is earned for the correct reason

The function f is increasing on [-3, -2] since $f' > 0$ for $-3 \leq x < -2$.

0	1	2

The student response earns all of the following points:

1 point is earned for the correct interval

1 point is earned for the correct reason

The function f is increasing on [-3, -2] since $f' > 0$ for $-3 \leq x < -2$.

Part B

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

5.3a

$x = 0$ and $x = 2$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$

0	1	2

The student response earns all of the following points:

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

$x = 0$ and $x = 2$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$

Part C

1 point is earned for the correct equation

$f'(0) = -2$

Tangent line is $y = -2x + 3$.

0	1

The student response earns all of the following points:

1 point is earned for the correct equation

$f'(0) = -2$

Tangent line is $y = -2x + 3$.

Part D

1 point is earned for the difference of areas of triangles, $\pm\left(\frac{1}{2} - 2\right)$

$$\begin{aligned} f(0) - f(-3) &= \int_{-3}^{0} f'(t)dt \\ &= \tfrac{1}{2}(1)(1) - \tfrac{1}{2}(2)(2) = -\tfrac{3}{2} \end{aligned}$$

5.3a

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \tfrac{3}{2} = \tfrac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm\left(8 - \dfrac{1}{2}(2)^2\pi\right)$

$$\begin{aligned} f(4) - f(0) &= \int_0^4 f'(t)dt \\ &= -\left(8 - \tfrac{1}{2}(2)^2\pi\right) = -8 + 2\pi \end{aligned}$$

1 point is earned for the correct answer of $f(4)$ using FTC

$$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$$

0	1	2	3	4 ✓

The student response earns all of the following points:

1 point is earned for the difference of areas of triangles, $\pm\left(\dfrac{1}{2} - 2\right)$

$$\begin{aligned} f(0) - f(-3) &= \int_{-3}^0 f'(t)dt \\ &= \tfrac{1}{2}(1)(1) - \tfrac{1}{2}(2)(2) = -\tfrac{3}{2} \end{aligned}$$

1 point is earned for the correct answer for $f(-3)$ using FTC

$$f(-3) = f(0) + \tfrac{3}{2} = \tfrac{9}{2}$$

1 point is earned for the area of rectangle - area of semicircle, $\pm\left(8 - \dfrac{1}{2}(2)^2\pi\right)$

$$\begin{aligned} f(4) - f(0) &= \int_0^4 f'(t)dt \\ &= -\left(8 - \tfrac{1}{2}(2)^2\pi\right) = -8 + 2\pi \end{aligned}$$

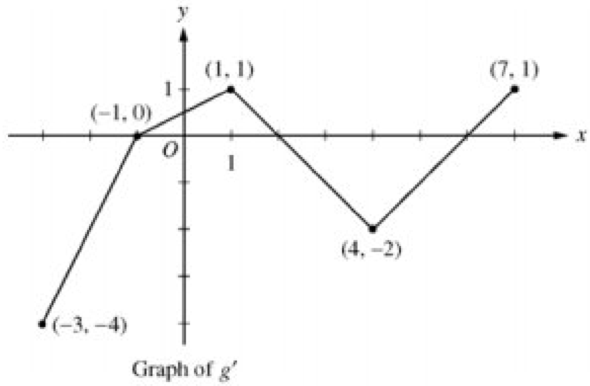
5.3a

1 point is earned for the correct answer of $f(4)$ using FTC

$f(4) = f(0) - 8 + 2\pi = -5 + 2\pi$

5.3a

82.



Let g be a continuous function with $g(2) = 5$. The graph of the piecewise-linear function g', the derivative of g, is shown above for $-3 \le x \le 7$.

(a) Find the x-coordinate of all points of inflection of the graph of $y = g(x)$ for $-3 < x < 7$. Justify your answer.

(b) Find the absolute maximum value of g on the interval $-3 \le x \le 7$. Justify your answer.

(c) Find the average rate of change of $g(x)$ on the interval $-3 \le x \le 7$.

(d) Find the average rate of change of $g'(x)$ on the interval $-3 \le x \le 7$. Does the Mean Value Theorem applied on the interval $-3 \le x \le 7$ guarantee a value of c, for $-3 < c < 7$, such that $g''(c)$ is equal to this average rate of change? Why or why not?

Part A

1 point is earned for x-values

1 point is earned for the justification

g' changes from increasing to decreasing at $x = 1$;

g' changes from decreasing to increasing at $x = 4$.

Points of inflection for the graph of $y = g(x)$ occur at $x = 1$ and $x = 4$.

0	1	2

The student response earns all of the following points:

1 point is earned for x-values

1 point is earned for the justification

g' changes from increasing to decreasing at $x = 1$;

g' changes from decreasing to increasing at $x = 4$.

5.3a

Points of inflection for the graph of $y = g(x)$ occur at $x = 1$ and $x = 4$.

Part B

1 point is earned if identifies $x = 2$ as a candidate

1 point is earned if considers endpoints

1 point is earned for maximum value and justification

The only sign change of g' from positive to negative in the interval is at $x = 2$.

$$g(-3) = 5 + \int_2^{-3} g'(x)dx = 5 + \left(-\frac{3}{2}\right) + 4 = \frac{15}{2}$$
$$g(2) = 5$$
$$g(7) = 5 + \int_2^{7} g'(x)dx = 5 + (-4) + \frac{1}{2} = \frac{3}{2}$$

The maximum value of g for $-3 \le x \le 7$ is $\frac{15}{2}$.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned if identifies $x = 2$ as a candidate

1 point is earned if considers endpoints

1 point is earned for maximum value and justification

The only sign change of g' from positive to negative in the interval is at $x = 2$.

$$g(-3) = 5 + \int_2^{-3} g'(x)dx = 5 + \left(-\frac{3}{2}\right) + 4 = \frac{15}{2}$$
$$g(2) = 5$$
$$g(7) = 5 + \int_2^{7} g'(x)dx = 5 + (-4) + \frac{1}{2} = \frac{3}{2}$$

The maximum value of g for $-3 \le x \le 7$ is $\frac{15}{2}$.

Part C

5.3a

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g(7) - g(-3)}{7 - (-3)} = \frac{\frac{3}{2} - \frac{15}{2}}{10} = -\frac{3}{5}$$

✓

0	1	2

The student response earns all of the following points:

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g(7) - g(-3)}{7 - (-3)} = \frac{\frac{3}{2} - \frac{15}{2}}{10} = -\frac{3}{5}$$

Part D

1 point is earned for average value of $g'(x)$

1 point is earned for answer "No" with reason

$$\frac{g'(7) - g'(-3)}{7 - (-3)} = \frac{1 - (-4)}{10} = \frac{1}{2}$$

No, the MVT does *not* guarantee the existence of a value c with the stated properties because g' is not differentiable for at least one point in $-3 < x < 7$.

✓

0	1	2

The student response earns all of the following points:

1 point is earned for average value of $g'(x)$

1 point is earned for answer "No" with reason

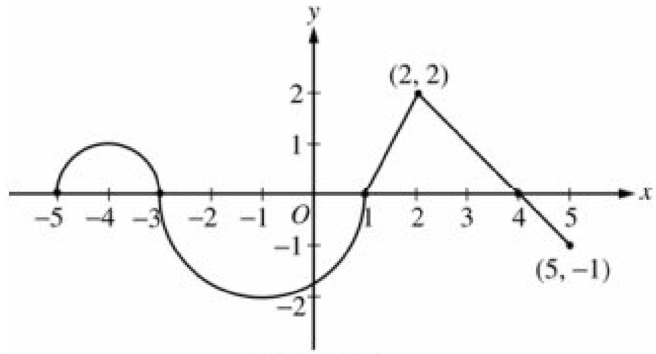
5.3a

$$\frac{g'(7) - g'(-3)}{7 - (-3)} = \frac{1 - (-4)}{10} = \frac{1}{2}$$

No, the MVT does *not* guarantee the existence of a value c with the stated properties because g' is not differentiable for at least one point in $-3 < x < 7$.

5.3a

83.



Graph of f'

Let f be a function defined on the closed interval $-5 \leq x \leq 5$ with $f(1) = 3$. The graph of f' the derivative of f, consists of two semicircles and two line segments, as shown above.

(a) For $-5 < x < 5$, find all values x at which f has a relative maximum. Justify your answer.

(b) For $-5 < x < 5$, find all values x at which the graph of f has a point of inflection. Justify your answer.

(c) Find all intervals on which the graph of f is concave up and also has positive slope. Explain your reasoning.

(d) Find the absolute minimum value of $f(x)$ over the closed interval $-5 \leq x \leq 5$. Explain your reasoning.

Part A

1 point is earned for the x-values

1 point is earned for the justification

$f'(x) = 0$ at $x = -3,\ 1,\ 4$

f' changes from positive to negative at -3 and 4.

Thus, f has a relative maximum at $x = -3$ and at $x = 4$.

0	1	2

The student response earns all of the following points:

1 point is earned for the x-values

1 point is earned for the justification

$f'(x) = 0$ at $x = -3,\ 1,\ 4$

f' changes from positive to negative at -3 and 4.

5.3a

Thus, f has a relative maximum at $x = -3$ and at $x = 4$.

Part B

1 point is earned for the x-values

1 point is earned for the justification

f' changes from increasing to decreasing, or vice versa, at $x = -4$, -1, and 2. Thus, the graph of f has points of inflection when $x = -4$, -1, and 2.

0	1	2

The student response earns all of the following points:

1 point is earned for the x-values

1 point is earned for the justification

f' changes from increasing to decreasing, or vice versa, at $x = -4$, -1, and 2. Thus, the graph of f has points of inflection when $x = -4$, -1, and 2.

Part C

1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of f is concave up with positive slope where f' is increasing and positive: $-5 < x < -4$ and $1 < x < 2$.

0	1	2

The student response earns all of the following points:

1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of f is concave up with positive slope where f' is increasing and positive: $-5 < x < -4$ and $1 < x < 2$.

Part D

1 point is earned if identifies $x = 1$ as a candidate

5.3a

1 point is earned for considering endpoint

1 point is earned for value and explanation

Candidates for the absolute minimum are where f' changes from negative to positive (at $x = 1$) and at the endpoints ($x = -5, 5$).

$$f(-5) = 3 + \int_{1}^{-5} f(x)\,dx = 3 - \frac{\pi}{2} + 2\pi > 3$$
$$f(1) = 3$$
$$f(5) = 3 + \int_{1}^{-5} f'(x)\,dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of f on [-5, 5] is $f(1) = 3$.

0	1	2	3

The student response earns all of the following points:

1 point is earned if identifies $x = 1$ as a candidate

1 point is earned for considering endpoint

1 point is earned for value and explanation

Candidates for the absolute minimum are where f' changes from negative to positive (at $x = 1$) and at the endpoints ($x = -5, 5$).

$$f(-5) = 3 + \int_{1}^{-5} f(x)\,dx = 3 - \frac{\pi}{2} + 2\pi > 3$$
$$f(1) = 3$$
$$f(5) = 3 + \int_{1}^{-5} f'(x)\,dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of f on [-5, 5] is $f(1) = 3$.

5.3a

84. Let f be the function defined by $f(x) = 3x^5 - 5x^3 + 2$.

(a) On what intervals is f increasing?

(b) On what intervals is the graph of f concave upward?

(c) Write the equation of each horizontal tangent line to the graph of f.

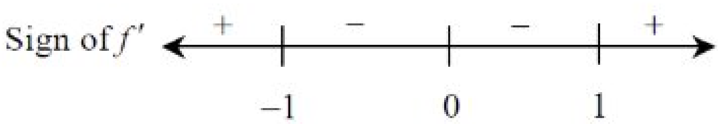
Part A

1 point is earned for $f'(x)$

1 point is earned for analyze sign of $f'(x)$ or explicitly setting students $f'(x) > 0$

1 point is earned for answer

$$f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1)$$



Answer: f is increasing on the intervals $(-\infty, -1]$ and $[1, \infty)$

0	1	2	3

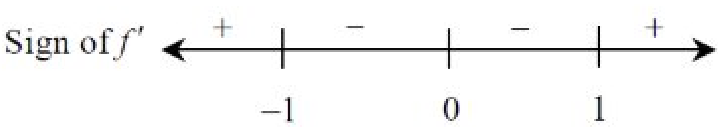
The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for analyze sign of $f'(x)$ or explicitly setting students $f'(x) > 0$

1 point is earned for answer

$$f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1)$$



Answer: f is increasing on the intervals $(-\infty, -1]$ and $[1, \infty)$

5.3a

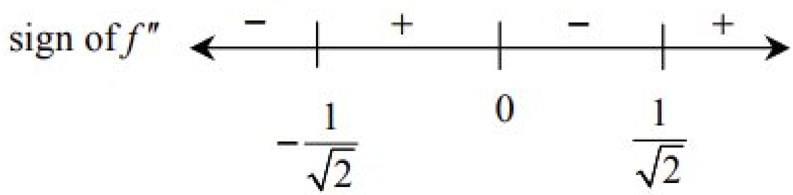
Part B

1 point is earned for $f''(x)$

1 point is earned for analyze sign of $f''(x)$ or explicitly setting students $f''(x) > 0$

1 point is earned for answer

$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$



Answer: f is concave upward on $\left(-\frac{1}{\sqrt{2}}, 0\right)$ and on $\left(\frac{1}{\sqrt{2}}, \infty\right)$

0	1	2	3

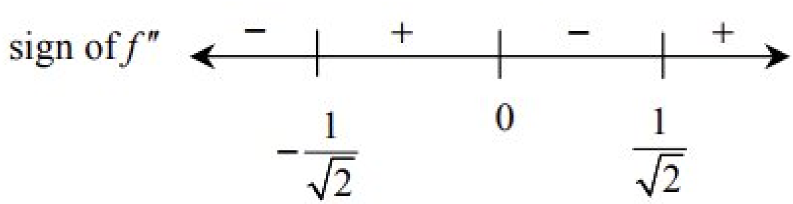
The student response earns all of the following points:

1 point is earned for $f''(x)$

1 point is earned for analyze sign of $f''(x)$ or explicitly setting students $f''(x) > 0$

1 point is earned for answer

$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$



Answer: f is concave upward on $\left(-\frac{1}{\sqrt{2}}, 0\right)$ and on $\left(\frac{1}{\sqrt{2}}, \infty\right)$

Part C

5.3a

1 point is earned for solving student's $f'(x) = 0$

1 point is earned for answer, $y =$ number

1 point is earned for consistent answer

$f'(x) = 0$ when $x = -1, 0, 1$

$x = -1 \Rightarrow f(x) = 4; y = 4$

$f(0) = 2; y = 2$

$f(1) = 0; y = 0$

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for solving student's $f'(x) = 0$

1 point is earned for answer, $y =$ number

1 point is earned for consistent answer

$f'(x) = 0$ when $x = -1, 0, 1$

$x = -1 \Rightarrow f(x) = 4; y = 4$

$f(0) = 2; y = 2$

$f(1) = 0; y = 0$

5.3a

85. Consider the function f defined by $f(x) = e^x \cos x$ with domain $[0, 2\pi]$.

(a) Find the absolute maximum and minimum values of $f(x)$.

(b) Find the intervals on which f is increasing.

(c) Find the x-coordinate of each point of inflection of the graph of f.

Part A

1 point is earned for derivative

1 point is earned for finding critical values

1 point is earned for evaluating at critical points and ends

1 point is earned for choosing absolute max

1 point is earned for absolute min

$$\begin{aligned} f'(x) &= -e^x \sin x + e^x \cos x \\ &= e^x [\cos x - \sin x] \end{aligned}$$

$f'(x) = 0$ when $\sin x = \cos x$, $x = \dfrac{\pi}{4}, \dfrac{5\pi}{4}$

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2} e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2} e^{5\pi/4}$
2π	$e^{2\pi}$

Max: $e^{2\pi}$; Min: $-\dfrac{\sqrt{2}}{2} e^{5\pi/4}$

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for derivative

5.3a

1 point is earned for finding critical values

1 point is earned for evaluating at critical points and ends

1 point is earned for choosing absolute max

1 point is earned for absolute min

$$\begin{aligned} f'(x) &= -e^x \sin x + e^x \cos x \\ &= e^x[\cos x - \sin x] \end{aligned}$$

$f'(x) = 0$ when $\sin x = \cos x$, $x = \frac{\pi}{4}, \frac{5\pi}{4}$

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}e^{5\pi/4}$
2π	$e^{2\pi}$

Max: $e^{2\pi}$; Min: $-\frac{\sqrt{2}}{2}e^{5\pi/4}$

Part B

1 point is earned for positive intervals

1 point is earned for negative intervals

1 point is **deducted** for intervals outside of domain

Increasing on $[0, \frac{\pi}{4}], [\frac{5\pi}{4}, 2\pi]$

5.3a

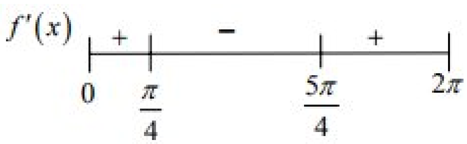
0	1	2

The student response earns all of the following points:

1 point is earned for positive intervals

1 point is earned for negative intervals

1 point is **deducted** for intervals outside of domain



Increasing on $\left[0, \frac{\pi}{4}\right], \left[\frac{5\pi}{4}, 2\pi\right]$

Part C

1 point is earned for $f''(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^x\left[-\sin x - \cos x\right] + e^x\left[\cos x - \sin x\right] \\ &= -2e^x \sin x \end{aligned}$$

$f''(x) = 0$ when $x = 0, \pi, 2\pi$

Point of inflection at $x = \pi$

0	1	2

The student response earns all of the following points:

1 point is earned for $f''(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^x\left[-\sin x - \cos x\right] + e^x\left[\cos x - \sin x\right] \\ &= -2e^x \sin x \end{aligned}$$

5.3a

$f''(x) = 0$ when $x = 0, \pi, 2\pi$

Point of inflection at $x = \pi$

86. The function f is an antiderivative of the function g defined by $g(x) = 3 - \sqrt{x^2 + x + 4\cos x}$. Which of the following is the x-coordinate of the location of a local maximum for the graph of $y = f(x)$?

(A) -3.961

(B) -2.161

(C) 1.494

(D) 3.140 ✓

Answer D

Correct. A local maximum for the graph of $y = f(x)$ occurs at a value of x where $f' = g$ changes from positive to negative. The graph of $y = g(x)$ crosses the x-axis from positive to negative at $x = 3.140$.

5.3a

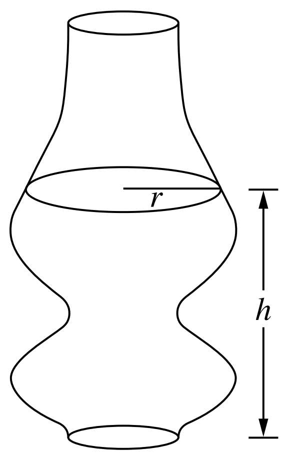
87. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



h (centimeters)	0	6.2	13	20
$r(h)$ (centimeters)	2.5	2.7	4.3	2.6

Juice is sold in 20-centimeter-tall bottles with horizontal cross sections parallel to the base that are circles, as shown in the figure. The radius of the circular cross section at height h above the base of the bottle is given by a differentiable function r, where h and $r(h)$ are measured in centimeters. Selected values of $r(h)$ are given in the table shown.

(a) Approximate $r'(3.1)$ using the average rate of change of r over the interval $0 \le h \le 6.2$. Show the computations that lead to your answer. Indicate units of measure.

(b) Interpret the meaning of $\frac{1}{20}\int_0^{20} r(h)\,dh$ in the context of the problem. Use a trapezoidal sum with the three subintervals $[0, 6.2]$, $[6.2, 13]$, and $[13, 20]$ to approximate the value of $\frac{1}{20}\int_0^{20} r(h)\,dh$.

5.3a

(c) The radius of the circular cross section at height h above the base of the bottle can also be modeled by the function f, where h and $f(h)$ are measured in centimeters. Let $f(h) = 3\sqrt{2 - 1.25\cos\left(\frac{\pi}{100}(h-20)^2\right)}$ for $0 \le h \le 20$. Based on the model, at a height of $h = 12$ centimeters, is the radius increasing or decreasing as h increases? Give a reason for your answer.

(d) Using the model f defined in part (c), find the volume of the bottle.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must provide a difference and a quotient. A response of only $\frac{2.7-2.5}{6.2}$ or $\frac{0.2}{6.2-0}$ earns the first point.

To earn the second point a response must have $\frac{\text{cm}}{\text{cm}}$ (or the equivalent) attached to a numerical value. These units do not reduce to 1 because the numerator, cm, refers to radius of the bottle and the denominator, cm, refers to height of the bottle.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Answer
- [ ] Units

Solution:

$$r'(3.1) \approx \frac{r(6.2)-r(0)}{6.2-0} = \frac{2.7-2.5}{6.2}$$

0.032 centimeter per centimeter

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must provide “average,” “radius,” and “the interval from 0 to 20” (or equivalent such as “throughout the bottle” or “from the bottom to the top”). Units are not required to earn the first point.

To earn the second point, at least five of the six factors in the trapezoidal sum must be correct.

If there is any error in the trapezoidal sum, the response does not earn the third point.

5.3a

A response of $\frac{1}{20}(6.2 \cdot \frac{2.5+2.7}{2} + 6.8 \cdot \frac{2.7+4.3}{2} + 7 \cdot \frac{4.3+2.6}{2})$ earns the second and third points, unless there is a subsequent error in simplification, in which case the response would earn only the second point.

A response that is missing the factor of $\frac{1}{2}$ in the trapezoidal sum does not earn the second and third points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Interpretation
- [ ] Form of trapezoidal sum
- [ ] Approximation

Solution:

$\frac{1}{20}\int_0^{20} r(h)\ dh$ is the average value of the radius of the juice bottle, in centimeters, over the interval $0 \leq h \leq 20$ centimeters.

$$\frac{1}{20}\int_0^{20} r(h)\ dh$$
$$\approx \frac{1}{20}\left(6.2 \cdot \frac{r(0)+r(6.2)}{2} + 6.8 \cdot \frac{r(6.2)+r(13)}{2} + 7 \cdot \frac{r(13)+r(20)}{2}\right)$$
$$= \frac{1}{20}(6.2 \cdot \frac{2.5+2.7}{2} + 6.8 \cdot \frac{2.7+4.3}{2} + 7 \cdot \frac{4.3+2.6}{2})$$
$$= \frac{1}{20} \cdot 64.07 = 3.2035$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the second point, a response does not need to provide a value for $f'(12)$; the sign of $f'(12)$ is sufficient.

Any presented value of $f'(12)$ must be correct for the digits presented, up to three decimal places.

If a response presents an incorrect sign or value for $f'\ (12)$, the response does not earn the second point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $f'(12) = -0.0006660598$. If a response presents this value, it earns the first point but is not eligible to earn the second point.

5.3a

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Considers $f'(12)$
- ☐ Answer with reason

Solution:

$f'(12) = -0.535902$

At a height of 12 centimeters, the radius is decreasing as h increases because $f'(12) < 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral with an integrand of $(f(h))^2$, regardless of any subsequent errors in presentation of $f(h)$. A response that *begins* with an error in copying $f(h)$ into an integrand of the form $(f(h))^2$ does not earn the first point, but is eligible to earn the second point for a correct answer (1007.112 or 1007.111).

An answer of 320.574π rounds and truncates to 1007.028, and therefore does not earn the second point. Similarly, an answer of 320.573π truncates to 1007.109 and rounds to 1007.110, and therefore does not earn the second point. However, answers of 320.5736π or 320.5737π do round and truncate to the correct answer and earn the second point.

Degree mode: $\pi \int_0^{20} (f\,(h))^2\,dh = 427.508$ (or 427.507) or 136.0799156π. A response that did not earn the second point in part (c) because of degree mode is eligible to earn the answer point in part (d) for this value. Furthermore, in degree mode 136.079π rounds to 427.505 and truncates to 427.504, and therefore does not earn the second point. However, in degree mode 136.080π is accurate to three decimals and is eligible for the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Integrand
- ☐ Answer

Solution:

5.3a

$$\text{Volume} = \pi \int_0^{20} (f\ (h))^2\, dh$$

$= 1007.112$ (or 1007.111) cubic centimeters

5.3a

88. For $0 \le t \le 31$, the rate of change of the number of mosquitoes on Tropical Island at time t days is modeled by $R(t) = 5\sqrt{t}\cos\left(\frac{t}{5}\right)$ mosquitoes per day. There are 1000 mosquitoes on Tropical Island at time $t = 0$.

(a) Show that the number of mosquitoes is increasing at time $t = 6$.

(b) At time $t = 6$, is the number of mosquitoes increasing at an increasing rate, or is the number of mosquitoes increasing at a decreasing rate? Give a reason for your answer.

(c) According to the model, how many mosquitoes will be on the island at time $t = 31$? Round your answer to the nearest whole number.

(d) To the nearest whole number, what is the maximum number of mosquitoes for $0 \le t \le 31$? Show the analysis that leads to your conclusion.

Part A

1 point is earned for showing that $R(6) > 0$

Since $R(6) = 4.438 > 0$, the number of mosquitoes is increasing at $t = 6$.

0	1

The student response earns all of the following points:

1 point is earned for showing that $R(6) > 0$

Since $R(6) = 4.438 > 0$, the number of mosquitoes is increasing at $t = 6$.

Part B

1 point is earned for considering $R'(6)$

1 point is earned for the answer with reason

$R'(6) = -1.913$

Since $R'(6) < 0$, the number of mosquitoes is increasing at a decreasing rate at $t = 6$.

0	1	2

The student response earns all of the following points:

1 point is earned for considering $R'(6)$

1 point is earned for the answer with reason

5.3a

$R'(6) = -1.913$

Since $R'(6) < 0$, the number of mosquitoes is increasing at a decreasing rate at $t = 6$.

Part C

1 point is earned for the integral

1 point is earned for the answer

$$1000 + \int_0^{31} R(t)dt = 964.335$$

To the nearest whole number, there are 964 mosquitoes.

0	1	2

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$1000 + \int_0^{31} R(t)dt = 964.335$$

To the nearest whole number, there are 964 mosquitoes.

Part D

2 points are earned for absolute maximum value where 1 point is earned for the integral and 1 point is earned for the answer

2 points are earned for analysis where 1 point is earned for computing interior critical points and 1 point is earned for completing analysis

$R(t) = 0$ when $t = 0$, $t = 2.5\pi$, or $t = 7.5\pi$

$R(t) > 0$ on $0 < t < 2.5\pi$

$R(t) < 0$ on $2.5\pi < t < 7.5\pi$

$R(t) > 0$ on $7.5\pi < t < 31$

The absolute maximum number of mosquitoes occurs at $t = 2.5\pi$ or at $t = 31$.

5.3a

$$1000 + \int_0^{2.5\pi} R(t)dt = 1039.357,$$

There are 964 mosquitoes at $t = 31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.

0	1	2	3	4

The student response earns all of the following points:

2 points are earned for absolute maximum value where 1 point is earned for the integral and 1 point is earned for the answer

2 points are earned for analysis where 1 point is earned for computing interior critical points and 1 point is earned for completing analysis

$R(t) = 0$ when $t = 0$, $t = 2.5\pi$, or $t = 7.5\pi$

$R(t) > 0$ on $0 < t < 2.5\pi$

$R(t) < 0$ on $2.5\pi < t < 7.5\pi$

$R(t) > 0$ on $7.5\pi < t < 31$

The absolute maximum number of mosquitoes occurs at $t = 2.5\pi$ or at $t = 31$.

$$1000 + \int_0^{2.5\pi} R(t)dt = 1039.357,$$

There are 964 mosquitoes at $t = 31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.

5.3a

89. Let f be the function given by $f(x) = e^x \sin x$ on the closed interval $[0, 2\pi]$. On which of the following intervals is f increasing?

(A) $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

(B) $\left[\frac{3\pi}{4}, \frac{7\pi}{4}\right]$

(C) $\left[0, \frac{\pi}{2}\right]$ and $\left[\frac{3\pi}{2}, 2\pi\right]$

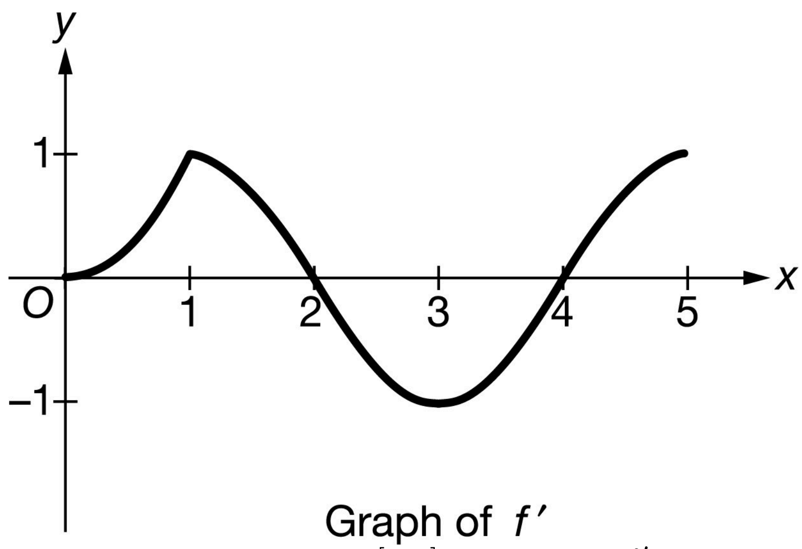
(D) $\left[0, \frac{3\pi}{4}\right]$ and $\left[\frac{7\pi}{4}, 2\pi\right]$

Answer D

Correct. By the product rule, $f'(x) = e^x \sin x + e^x \cos x = e^x (\sin x + \cos x)$.
$\sin x + \cos x = 0$ when $\sin x = -\cos x$, or $\tan x = -1$.
On the interval $[0, 2\pi]$, this happens at $x = \frac{3\pi}{4}$ and $x = \frac{7\pi}{4}$.
Check a value of x in each of the three intervals $\left[0, \frac{3\pi}{4}\right)$, $\left(\frac{3\pi}{4}, \frac{7\pi}{4}\right)$, and $\left(\frac{7\pi}{4}, 2\pi\right]$.
$x = 0$: $\sin 0 + \cos 0 = 1 > 0$
$x = \pi$: $\sin \pi + \cos \pi = -1 < 0$
$x = 2\pi$: $\sin(2\pi) + \cos(2\pi) = 1 > 0$
Therefore, $f'(x) > 0$ on the open intervals $\left(0, \frac{3\pi}{4}\right)$ and $\left(\frac{7\pi}{4}, 2\pi\right)$, and so f is increasing on the closed intervals $\left[0, \frac{3\pi}{4}\right]$ and $\left[\frac{7\pi}{4}, 2\pi\right]$.
Alternatively, sketching a graph of $y = \cos x$ and $y = -\sin x$ on the same set of axes, then looking for where the graph of $y = \cos x$ is above the graph of $y = -\sin x$ relative to the points $x = \frac{3\pi}{4}$ and $x = \frac{7\pi}{4}$ will indicate where $f'(x) > 0$.

5.3a

90.



Graph of f'

The function f is continuous on the closed interval $[0, 5]$. The graph of f', the derivative of f, is shown above.

On which of the following intervals is f increasing?

(A) $[0, 1]$ and $[2, 4]$

(B) $[0, 1]$ and $[3, 5]$

(C) $[0, 1]$ and $[4, 5]$ only

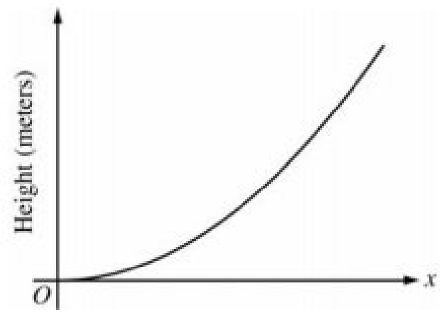
(D) $[0, 2]$ and $[4, 5]$

Answer D

Correct. The function f is increasing on closed intervals where f' is positive on the corresponding open intervals. The graph indicates that $f'(x) > 0$ on the intervals $(0, 2)$ and $(4, 5)$, so f is increasing on the intervals $[0, 2]$ and $[4, 5]$.

5.3a

91.



The figure above is the graph of a function of x, which models the height of a skateboard ramp. The function meets the following requirements.

(i) At $x = 0$, the value of the function is 0, and the slope of the graph of the function is 0.

(ii) At $x = 4$, the value of the function is 1, and the slope of the graph of the function is 1.

(iii) Between $x = 0$ and $x = 4$, the function is increasing.

(a) Let $f(x) = ax^2$, where a is a nonzero constant. Show that it is not possible to find a value for a so that f meets requirement (ii) above.

(b) Let $g(x) = cx^3 - \frac{x^2}{16}$, where c is a nonzero constant. Find the value of c so that g meets requirement (ii) above. Show the work that leads to your answer.

(c) Using the function g and your value of c from part (b), show that g does not meet requirement (iii) above.

(d) Let $h(x) = x^n/k$, where k is a nonzero constant and n is a positive integer. Find the values of k and n so that h meets requirement (ii) above. Show that h also meets requirements (i) and (iii) above.

Part A

1 point is earned for if $a = 1/16$ or $a = 1/8$

1 point is earned if shows a does not work

$f(4) = 1$ implies that $a = 1/16$ and $f'(4) = 2a(4) = 1$ implies that $a = 1/8$. Thus, f cannot satisfy (ii).

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for if $a = 1/16$ or $a = 1/8$

1 point is earned if shows a does not work

$f(4) = 1$ implies that $a = 1/16$ and $f'(4) = 2a(4) = 1$ implies that $a = 1/8$. Thus, f cannot satisfy (ii).

5.3a

Part B

1 point is earned for the value of c

$g(4) = 64c - 1 = 1$ implies that $c = \frac{1}{32}$.

When $c = \frac{1}{32}$, $g'(4) = 3c(4)^2 - \frac{2(4)}{16} = 3\left(\frac{1}{32}\right)(16) - \frac{1}{2} = 1$

0	1

The student response earns all of the following points:

1 point is earned for the value of c

$g(4) = 64c - 1 = 1$ implies that $c = \frac{1}{32}$.

When $c = \frac{1}{32}$, $g'(4) = 3c(4)^2 - \frac{2(4)}{16} = 3\left(\frac{1}{32}\right)(16) - \frac{1}{2} = 1$

Part C

1 point is earned for $g'(x)$

1 point is earned for the explanation

$$g'(x) = \frac{3}{32}x^2 - \frac{x}{8} = \frac{1}{32}x(3x - 4)$$

$g'(x) < 0$ for $0 < x < 4/3$, so g does not satisfy (iii).

0	1	2

The student response earns all of the following points:

5.3a

1 point is earned for $g'(x)$

1 point is earned for the explanation

$$g'(x) = \frac{3}{32}x^2 - \frac{x}{8} = \frac{1}{32}x(3x-4)$$

$g'(x) < 0$ for $0 < x < 4/3$, so g does not satisfy (iii).

Part D

1 point is earned for $\frac{4^n}{k} = 1$

1 point is earned for $\frac{n4^{n-1}}{k} = 1$

1 point is earned for values for k and n

1 point is earned for verification

$h(4) = \dfrac{4^n}{k} = 1$ implies that $4^n = k$.

$h'(4) = \dfrac{n4^{n-1}}{k} = \dfrac{n4^{n-1}}{4^n} = \dfrac{n}{4} = 1$ gives $n = 4$ and $k = 4^4 = 256$.

$$h(x) = \frac{x^4}{256} \Rightarrow h(0) = 0.$$
$$h'(x) = \frac{4x^3}{256} \Rightarrow h'(0) = 0 \text{ and } h'(x) > 0 \text{ for } 0 < x < 4.$$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for $\frac{4^n}{k} = 1$

1 point is earned for $\frac{n4^{n-1}}{k} = 1$

1 point is earned for values for k and n

5.3a

1 point is earned for verification

$h(4) = \dfrac{4^n}{k} = 1$ implies that $4^n = k$.

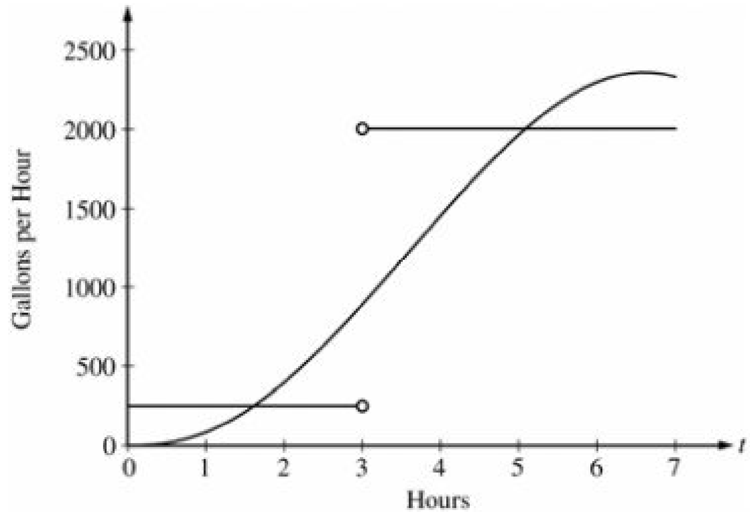
$h'(4) = \dfrac{n4^{n-1}}{k} = \dfrac{n4^{n-1}}{4^n} = \dfrac{n}{4} = 1$ gives $n = 4$ and $k = 4^4 = 256$.

$h(x) = \dfrac{x^4}{256} \Rightarrow h(0) = 0.$

$h'(x) = \dfrac{4x^3}{256} \Rightarrow h'(0) = 0$ and $h'(x) > 0$ for $0 < x < 4$.

5.3a

92.



The amount of water in a storage tank, in gallons, is modeled by a continuous function on the time interval $0 \leq t \leq 7$, where t is measured in hours. In this model, rates are given as follows:

i. The rate at which water enters the tank is $f(t) = 100t^2 \sin\left(\sqrt{t}\right)$ gallons per hour for $0 \leq t \leq 7$.

ii. The rate at which water leaves the tank is

$$g(t) = \begin{cases} 250 & \text{for } 0 \leq t < 3 \\ 2000 & \text{for } 3 < t \leq 7 \end{cases} \text{gallons per hour.}$$

The graphs of f and g, which intersect at $t = 1.617$ and $t = 5.076$, are shown in the figure above. At time $t = 0$, the amount of water in the tank is 5000 gallons.

(a) How many gallons of water enter the tank during the time interval $0 \leq t \leq 7$? Round your answer to the nearest gallon.

(b) For $0 \leq t \leq 7$, find the time intervals during which the amount of water in the tank is decreasing. Give a reason for each answer.

(c) For $0 \leq t \leq 7$, at what time t is the amount of water in the tank greatest? To the nearest gallon, compute the amount of water at this time. Justify your answer.

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^7 f(t)dt \approx 8264 \text{ gallons}$$

5.3a

0	1	2

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^7 f(t)dt \approx 8264 \text{ gallons}$$

Part B

1 point is earned for the intervals

1 point is earned for the reason

The amount of water in the tank is decreasing on the intervals $0 \le t \le 1.617$ and $3 \le t \le 5.076$ because $f(t) < g(t)$ for $0 \le t < 1.617$ and $3 < t < 5.076$.

0	1	2

The student response earns all of the following points:

1 point is earned for the intervals

1 point is earned for the reason

The amount of water in the tank is decreasing on the intervals $0 \le t \le 1.617$ and $3 \le t \le 5.076$ because $f(t) < g(t)$ for $0 \le t < 1.617$ and $3 < t < 5.076$.

Part C

1 point is earned if identifies $t = 3$ as a candidate

1 point is earned for the integrand

1 point is earned for the amount of water at $t = 3$

1 point is earned for the amount of water at $t = 7$

1 point is earned for the conclusion

5.3a

Since $f(t)$ - $g(t)$ changes sign from positive to negative only at t = 3, the candidates for the absolute maximum are t = 0, 3, and 7.

t (hours)	gallons of water
0	5000
3	$5000 + \int_0^3 f(t)\,dt - 250(3) = 5126.591$
7	$5126.591 + \int_3^7 f(t)\,dt - 2000(4) = 4513.807$

The amount of water in the tank is greatest at 3 hours. At that time, the amount of water in the tank, rounded to the nearest gallon, is 5127 gallons.

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned if identifies t = 3 as a candidate

1 point is earned for the integrand

1 point is earned for the amount of water at t = 3

1 point is earned for the amount of water at t = 7

1 point is earned for the conclusion

Since $f(t)$ - $g(t)$ changes sign from positive to negative only at t = 3, the candidates for the absolute maximum are t = 0, 3, and 7.

t (hours)	gallons of water
0	5000
3	$5000 + \int_0^3 f(t)\,dt - 250(3) = 5126.591$
7	$5126.591 + \int_3^7 f(t)\,dt - 2000(4) = 4513.807$

The amount of water in the tank is greatest at 3 hours. At that time, the amount of water in the tank, rounded to the nearest gallon, is 5127 gallons.

5.3a

93. T*raffic flow is defined as the rate at which cars pass through an* intersection, measured in cars per minute. The traffic flow at a particular intersection is modeled by the function F defined by

$$F(t) = 82 + 4\sin\left(\frac{t}{2}\right) \text{ for } 0 \le t \le 30,$$

where $F(t)$ is measured in cars per minute and t is measured in minutes.

(a) To the nearest whole number, how many cars pass through the intersection over the 30-minute period?

(b) Is the traffic flow increasing or decreasing at $t = 7$*? Give a reason for your answer.*

(c) What is the average value of the traffic flow over the time interval $10 \le t \le 15$? Indicate units of measure.

(d) What is the average rate of change of the traffic flow over the time interval $10 \le t \le 15$? Indicate units of measure.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^{30} F(t)dt = 2474 \text{ cars}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^{30} F(t)dt = 2474 \text{ cars}$$

Part B

1 point is earned for the answer with reason

$F'(7) = -1.872$ or -1.873

5.3a

Since $F'(7) < 0$, the traffic flow is decreasing at $t = 7$.

0	1

The student response earns all of the following points:

1 point is earned for the answer with reason

$F'(7) = -1.872$ or -1.873

Since $F'(7) < 0$, the traffic flow is decreasing at $t = 7$.

Part C

1 point is earned for limits

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{5}\int_{10}^{15} F(t)dt = 81.899 \text{ cars/min}$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min^2 in (d)

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for limits

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{5}\int_{10}^{15} F(t)dt = 81.899 \text{ cars/min}$$

5.3a

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min^2 in (d)

Part D

1 point is earned for the answer

$$\frac{F(15)-F(10)}{15-10} = 1.517 \text{ or } 1.518 \ \text{cars}/\text{min}^2$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min^2 in (d)

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{F(15)-F(10)}{15-10} = 1.517 \text{ or } 1.518 \ \text{cars}/\text{min}^2$$

1 point is earned for units in (c) and (d)

Units of cars/min in (c) and cars/min^2 in (d)

5.3a

94. Let $F(x) = \int_0^x \sin\left(t^2\right) dt$ for $0 \le x \le 3$.

(a) Use the trapezoidal rule with four equal subdivisions of the closed interval [0,1] to approximate $F(1)$.

(b) On what intervals is F increasing?

(c) If the average rate of change of F on the closed interval [1,3] is k, find $\int_1^3 \sin\left(t^2\right) dt$ in terms of k.

Part A

1 point is earned for correctly using $\int_0^1 \sin(t^2) dt$ or

$x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$

$F(1) = \int_0^1 \sin(t^2) dt$

1 point is earned for correctly substituting $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ into $\sin\left(x^2\right)$

$$\approx \frac{(1-0)}{4} \cdot \frac{1}{2} \cdot \left[\sin 0^2 + 2\sin\left(\frac{1}{4}\right)^2 + 2\sin\left(\frac{1}{2}\right)^2 + 2\sin\left(\frac{3}{4}\right)^2 + \sin 1^2\right]$$

1 point is earned for the correct answer using $\sin\left(x^2\right)$-values in trapezoidal rule formula on [0,1]

≈ 0.316

0	1	2	3

The student response earns all of the following points:

1 point is earned for correctly using $\int_0^1 \sin(t^2) dt$ or

$x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$

$F(1) = \int_0^1 \sin(t^2) dt$

1 point is earned for correctly substituting $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ into $\sin\left(x^2\right)$

5.3a

$$\approx \frac{(1-0)}{4} \cdot \frac{1}{2} \cdot [\sin 0^2 + 2\sin\left(\frac{1}{4}\right)^2 + 2\sin\left(\frac{1}{2}\right)^2 + 2\sin\left(\frac{3}{4}\right)^2 + \sin 1^2]$$

1 point is earned for the correct answer using $\sin\left(x^2\right)$-values in trapezoidal rule formula on [0,1]

≈ 0.316

Part B

1 point is earned for correctly using $F'(x) = \sin\left(x^2\right)$

$F'(x) = \sin\left(x^2\right)$

$F'(x) = 0$ when $x^2 = 0, \pi, 2\pi, \ldots$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$x = 0, \sqrt{\pi}, \sqrt{2\pi}$

1 point is earned for correctly analyzing sign of F' within [0,3]

1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $\left[0, \sqrt{\pi}\right]$ and on $\left[\sqrt{2\pi}, 3\right]$

0	1	2	3	4

The student response earns all of the following points:

1 point is earned for correctly using $F'(x) = \sin\left(x^2\right)$

$F'(x) = \sin\left(x^2\right)$

$F'(x) = 0$ when $x^2 = 0, \pi, 2\pi, \ldots$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$x = 0, \sqrt{\pi}, \sqrt{2\pi}$

5.3a

1 point is earned for correctly analyzing sign of F' within [0,3]

1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $\left[0, \sqrt{\pi}\right]$ and on $\left[\sqrt{2\pi}, 3\right]$

Part C

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2)dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2)dt = 2k$$

0	1	2

The student response earns all of the following points:

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2)dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2)dt = 2k$$

5.3a

95. A particle moves along the y-axis with velocity given by $v(t) = t \sin(t^2)$ for $t \geq 0$.

(a) In which direction (up or down) is the particle moving at time $t = 1.5$? Why?

(b) Find the acceleration of the particle at time $t = 1.5$. Is the velocity of the particle increasing at $t = 1.5$? Why or why not?

(c) Given that $y(t)$ is the position of the particle at time t and that $y(0) = 3$, find $y(2)$.

(d) Find the total distance traveled by the particle from $t = 0$ to $t = 2$.

Part A

1 point is earned for the answer and reason

$v(1.5) = 1.5 \sin(1.5^2) = 1.167$

Up, because $v(1.5) > 0$

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer and reason

$v(1.5) = 1.5 \sin(1.5^2) = 1.167$

Up, because $v(1.5) > 0$

Part B

1 point is earned for: $a(1.5)$

1 point is earned for the conclusion and reason

$$a(t) = v'(t) = \sin t^2 + 2t^2 \cos t^2$$
$$a(1.5) = v'(1.5) = -2.048 \text{ or } -2.049$$

No; v is decreasing at 1.5 because $v'(1.5) < 0$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: $a(1.5)$

5.3a

1 point is earned for the conclusion and reason

$$a(t) = v'(t) = \sin t^2 + 2t^2 \cos t^2$$
$$a(1.5) = v'(1.5) = -2.048 \text{ or } -2.049$$

No; v is decreasing at 1.5 because $v'(1.5) < 0$

Part C

1 point is earned for: $y(t) = \int v(t)dt$

1 point is earned for: $y\left(t\right) = -\dfrac{1}{2}\cos t^2 + C$

1 point is earned for: $y(2)$

$$\begin{aligned} y(t) &= \int v(t)dt \\ &= \int t\sin t^2 dt = -\frac{\cos t^2}{2} + C \\ y(0) &= 3 = -\tfrac{1}{2} + C \Rightarrow C = \tfrac{7}{2} \\ y(t) &= -\tfrac{1}{2}\cos t^2 + \tfrac{7}{2} \\ y(2) &= -\tfrac{1}{2}\cos 4 + \tfrac{7}{2} = 3.826 \text{ or } 3.827 \end{aligned}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $y(t) = \int v(t)dt$

1 point is earned for: $y\left(t\right) = -\dfrac{1}{2}\cos t^2 + C$

1 point is earned for: $y(2)$

5.3a

$$y(t) = \int v(t)dt$$
$$= \int t\sin t^2 dt = -\frac{\cos t^2}{2} + C$$
$$y(0) = 3 = -\frac{1}{2} + C \Rightarrow C = \frac{7}{2}$$
$$y(t) = -\frac{1}{2}\cos t^2 + \frac{7}{2}$$
$$y(2) = -\frac{1}{2}\cos 4 + \frac{7}{2} = 3.826 \text{ or } 3.827$$

Part D

1 point is earned for the limits of 0 and 2 on an integral of $v(t)$ or $|v(t)|$

or

uses $y(0)$ and $y(2)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

1 point is earned for the answer

0/1 if incorrect turning point

$$\text{distance} = \int_0^2 |v(t)|dt = 1.173$$

or

$$v(t) = t\sin t^2 = 0$$
$$t = 0 \text{ or } t = \sqrt{\pi} \approx 1.772$$
$$y(0) = 3;\ y(\sqrt{\pi}) = 4;\ y(2) = 3.826 \text{ or } 3.827$$
$$\left[y(\sqrt{\pi}) - y(0)\right] + \left[y(\sqrt{\pi}) - y(2)\right]$$
$$= 1.173 \text{ or } 1.174$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for the limits of 0 and 2 on an integral of $v(t)$ or $|v(t)|$

or

uses $y(0)$ and $y(2)$ to compute distance

1 point is earned for the handles change of direction at student's turning point

5.3a

1 point is earned for the answer

0/1 if incorrect turning point

$$\text{distance} = \int_0^2 |v(t)|dt = 1.173$$

or

$v(t) = t\sin t^2 = 0$

$t = 0 \text{ or } t = \sqrt{\pi} \approx 1.772$

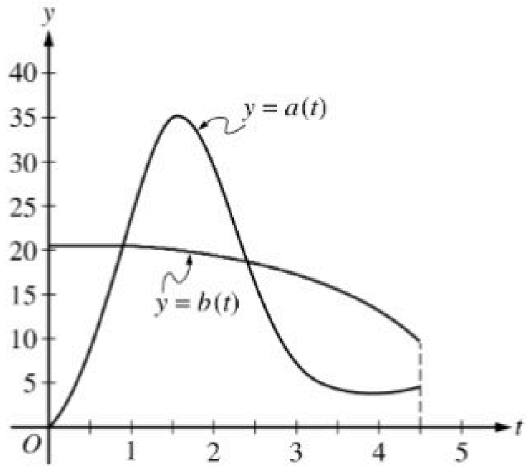
$y(0) = 3;\ y(\sqrt{\pi}) = 4;\ y(2) = 3.826 \text{ or } 3.827$

$\left[y(\sqrt{\pi}) - y(0)\right] + \left[y(\sqrt{\pi}) - y(2)\right]$

$= 1.173 \text{ or } 1.174$

5.3a

96.



During the time interval $0 \le t \le 4.5$ hours, water flows into tank A at a rate of $a(t) = (2t - 5) + 5e^{2\sin t}$ liters per hour. During the same time interval, water flows into tank B at a rate of $b(t)$ liters per hour. Both tanks are empty at time $t = 0$. The graphs of $y = a(t)$ and $y = b(t)$, shown in the figure above, intersect at $t = k$ and $t = 2.416$.

(a) How much water will be in tank A at time $t = 4.5$?

(b) During the time interval $0 \le t \le k$ hours, water flows into tank B at a constant rate of 20.5 liters per hour. What is the difference between the amount of water in tank A and the amount of water in tank B at time $t = k$?

(c) The area of the region bounded by the graphs of $y = a(t)$ and $y = b(t)$ for $k \le t \le 2.416$ is 14.470. How much water is in tank B at time $t = 2.416$?

(d) During the time interval $2.7 \le t \le 4.5$ hours, the rate at which water flows into tank B is modeled by $w(t) = 21 - \frac{30t}{(t-8)^2}$ liters per hour. Is the difference $w(t)$ - $a(t)$ increasing or decreasing at time $t = 3.5$? Show the work that leads to your answer.

Part A

1 point is earned for integral

1 point is earned for answer

$$\int_0^{4.5} a(t)dt = 66.532128$$

At time $t = 4.5$, tank A contains 66.532 liters of water.

0	1	2

5.3a

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\int_{0}^{4.5} a\left(t\right)dt = 66.532128$$

At time $t = 4.5$, tank A contains 66.532 liters of water.

Part B

1 point is earned for sets $a(k) = 20.5$

1 point is earned for integral

1 point is earned for answer

$$a\left(k\right) = 20.5 \Rightarrow k = 0.892040$$
$$\int_{0}^{k} (20.5 - a\left(t\right))dt = 10.599191$$

At time $t = k$, the difference in the amounts of water in the tanks is 10.599 liters.

0	1	2	3

The student response earns all of the following points:

1 point is earned for sets $a(k) = 20.5$

1 point is earned for integral

1 point is earned for answer

$$a\left(k\right) = 20.5 \Rightarrow k = 0.892040$$
$$\int_{0}^{k} (20.5 - a\left(t\right))dt = 10.599191$$

At time $t = k$, the difference in the amounts of water in the tanks is 10.599 liters.

Part C

5.3a

1 point is earned for $\int_{k}^{2.416} a(t)dt$

1 point is earned for answer

$$\int_{0}^{2.416} b(t)dt = \int_{0}^{k} b(t)dt + \int_{k}^{2.416} b(t)dt$$
$$\int_{0}^{k} b(t)dt = 20.5 \cdot k = 18.286826$$

On $k < t < 2.416$, tank A receives $\int_{k}^{2.416} a(t)dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B.

Therefore, $\int_{k}^{2.416} b(t)dt = \int_{k}^{2.416} a(t)dt - 14.470 = 30.027051.$

$$\int_{0}^{k} b(t)dt + \int_{k}^{2.416} b(t)dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

✓

0	1	2

The student response earns all of the following points:

1 point is earned for $\int_{k}^{2.416} a(t)dt$

1 point is earned for answer

$$\int_{0}^{2.416} b(t)dt = \int_{0}^{k} b(t)dt + \int_{k}^{2.416} b(t)dt$$
$$\int_{0}^{k} b(t)dt = 20.5 \cdot k = 18.286826$$

5.3a

On $k < t < 2.416$, tank A receives $\int_k^{2.416} a(t)dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B.

Therefore, $\int_k^{2.416} b(t)dt = \int_k^{2.416} a(t)dt - 14.470 = 30.027051.$

$$\int_0^k b(t)dt + \int_k^{2.416} b(t)dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

Part D

1 point is earned for $w'(3.5) - a'(3.5) < 0$

1 point is earned for conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

0	1	2

The student response earns all of the following points:

1 point is earned for $w'(3.5) - a'(3.5) < 0$

1 point is earned for conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

5.3a

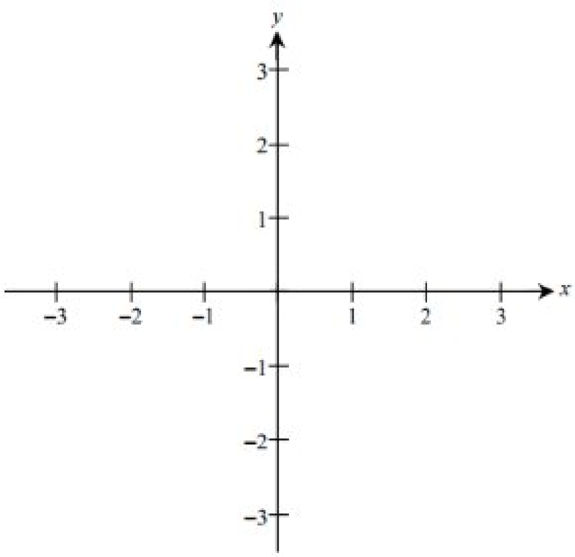
97. Let f be the function defined by $f(x) = xe^{1-x}$ for all real numbers x.

(a) Find each interval on which f is increasing.

(b) Find the range of f.

(c) Find the x-coordinate of each point of inflection of the graph of f.

(d) Using the results found in parts (a), (b), and (c), sketch the graph of f in the xy- plane provided below. (Indicate all intercepts.)



Part A

1 point is earned for derivative

1 point is earned for answer

$$f'(x) = xe^{1-x}(-1) + e^{1-x} = (1-x)e^{1-x}$$

$\boxed{f \text{ increases on } (-\infty, 1]}$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for derivative

1 point is earned for answer

5.3a

$f'(x) = xe^{1-x}(-1) + e^{1-x} = (1-x)e^{1-x}$

$\boxed{f \text{ increases on } (-\infty, 1]}$

Part B

1 point is earned for left hand endpoint

1 point is earned for right hand endpoint

Note: 0/2 for All reals without supporting work.

$f(1) = 1; \lim\limits_{x \to -\infty} f(x) = -\infty$

$\boxed{\text{Range: } (-\infty, 1]}$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for left hand endpoint

1 point is earned for right hand endpoint

Note: 0/2 for All reals without supporting work.

$f(1) = 1; \lim\limits_{x \to -\infty} f(x) = -\infty$

$\boxed{\text{Range: } (-\infty, 1]}$

Part C

1 point is earned for $f''(x)$ for student's $f'(x)$

1 point is earned for answer

$f''(x) = e^{1-x}(-1) + (1-x)e^{1-x}(-1)$
$= (x-2)e^{1-x}$

$\boxed{\text{Point of inflection at } x = 2}$

5.3a

0	1	2

The student response earns all of the following points:

1 point is earned for $f''(x)$ for student's $f'(x)$

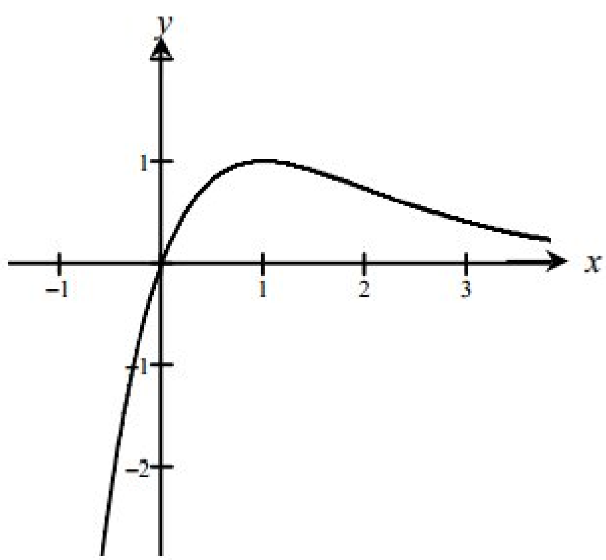
1 point is earned for answer

$$f''(x) = e^{1-x}(-1) + (1-x)e^{1-x}(-1)$$
$$= (x-2)e^{1-x}$$
$$\boxed{\text{Point of inflection at } x = 2}$$

Part D

3 points are earned for answer without mistake

1 point is **deducted** for each error: intercept, domain, increasing, decreasing, range, inflection points.



0	1	2	3

The student response earns all of the following points:

3 points are earned for answer without mistake

5.3a

1 point is **deducted** for each error: intercept, domain, increasing, decreasing, range, inflection points.